

IENC Feature Catalogue

Edition 2.6 corr 3

2026-05-29

Introduction

The IENC Feature Catalogue is based on features, attributes and enumerations of following Data Dictionaries

Data Dictionary Name	Token	Owner	Place of Publishment
HYDRO Data Dictionary	HYDRO	IHO	http://registry.iho.int
IENC Data Dictionary	IENC	IEHG	http://registry.iho.int

Used Abbreviations

Feature Types

G	Geo
M	Meta
C	Cartographic
O	Collection
I	Information

Feature Primitives

P	Point
L	Line
A	Area
N	None

Use of Feature Attribute Bindings

O	Optional
M	Mandatory
C	Conditional mandatory

Type of Attribute Use

F	Feature
N	National
S	Spatial
C	Cartographic

Type of Attribute Value

E	Enumeration
L	List
F	Float
I	Integer
T	Text
S	Structured text

Feature	Accuracy of data
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Acronym: M_ACCY Code: 300

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area within which the best estimate of the overall accuracy of the data is uniform. The overall accuracy takes into account for example the source accuracy, chart scale, digitising accuracy, etc.

Attribute Bindings:

acronym	usage	constraints
POSACC	O	unit = "m" decimal digits = "2"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
CATTEV	M	value list = "4,5,6"
INFORM	O	
NINFOM	O	
SOUACC	O	unit = "m" decimal digits = "2"
TXTDSC	O	
NTXTDS	O	
VERACC	O	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"

Feature	Administration Area (Named)
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Acronym: ADMARE Code: 1
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
 Definition: A defined (and possibly named) administration area.

Attribute Bindings:

acronym	usage	constraints
JRSDTN	M	value list = "1,2,3"
NATION	M	format = "cc"
CONDTN	C	value list = "3"
OBJNAM	M	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Aggregation
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Acronym: C_AGGR Code: 400
 Type: O
 Primitive: N

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Used to identify an aggregation of two or more objects. This aggregation may be named.

Attribute Bindings:

acronym	usage	constraints
NOBJNM	O	
unlocd	C	
OBJNAM	M	
INFORM	O	
NINFOM	O	
NTXTDS	O	
PICREP	O	
SCAMIN	O	min = "1"
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
locmrn	C	

Feature	Airport/airfield
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Acronym: AIRARE Code: 2

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area containing at least one runway, used for landing, take-off, and movement of aircraft.

Attribute Bindings:

acronym	usage	constraints
CATAIR	O	value list = "1,2,4,6"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Association
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Acronym: C ASSO Code: 401
Type: O
Primitive: N

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Used to identify an association between two or more objects. The association may be named.

Attribute Bindings:

acronym	usage	constraints
NOBJNM	O	
OBJNAM	O	
INFORM	O	
NINFOM	O	
NTXTDS	O	
PICREP	O	
SCAMIN	M	min = "1"
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Beacon, cardinal
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Acronym: BCNCAR Code: 5

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A beacon is a prominent, specially constructed object forming a conspicuous mark as a fixed aid to navigation for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A cardinal beacon is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
PICREP	O	
CONDTN	O	value list = "1,2,3,5"
HEIGHT	O	unit = "m" decimal digits = "2"
VERLEN	O	unit = "m" decimal digits = "1"
BCNSHP	M	value list = "1,2,3,4,5,6,7"
CATCAM	M	value list = "1,2,3,4"
COLOUR	M	value list = "2,6"
COLPAT	M	value list = "1"
MARSYS	C	value list = "1,2"
INFORM	O	
NINFOM	O	
OBJNAM	O	
NOBJNM	O	
CONRAD	O	value list = "3"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
DATSTA	O	format = "ccyymmdd"

acronym	usage	constraints
DATEND	O	format = "ccyymmdd"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Beacon, isolated danger
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Acronym: BCNISD Code: 6
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A beacon, is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). An isolated danger beacon is a beacon erected on an isolated danger of limited extent, which has navigable water all around it. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BCNSHP	M	value list = "1,2,3,4,5"
COLOUR	M	value list = "2,3"
COLPAT	M	value list = "1"
CONRAD	O	value list = "3"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Beacon, lateral
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Acronym: BCNLAT Code: 7
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A beacon, is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A lateral beacon, is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BCNSHP	M	value list = "1"
CATLAM	M	value list = "1,2"
COLOUR	M	value list = "1,2,3,4"
COLPAT	C	value list = "1"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Beacon, safe water
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Acronym: BCNSAW Code: 8
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A safe water beacon is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A safe water beacon may be used to indicate that there is navigable water around the mark. (UKHO NP735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
PICREP	O	
CONDTN	O	value list = "1,2,3,5"
HEIGHT	O	unit = "m" decimal digits = "2"
VERLEN	O	unit = "m" decimal digits = "1"
BCNSHP	M	value list = "1,2,3,4,5,6,7"
COLOUR	M	value list = "1,3"
COLPAT	M	value list = "2"
MARSYS	C	value list = "1,2"
CONRAD	O	value list = "3"
INFORM	O	
NINFOM	O	
OBJNAM	O	
NOBJNM	O	
TXTDSC	O	
NTXTDS	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"

acronym	usage	constraints
PEREND	O	format = "ccyymmdd"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17,18"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Beacon, special purpose/general
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Acronym: BCNSPP Code: 9

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A beacon, is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A special purpose beacon is primarily used to indicate an area or features, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners. (UKHO NP 735, 5th Edition). Beacon in general: A beacon whose appearance or purpose is not adequately known.

Attribute Bindings:

acronym	usage	constraints
MARSYS	C	value list = "1,2"
CONDTN	O	value list = "1,2,3,5"
COLPAT	O	value list = "1,2,3,4,5,6"
HEIGHT	O	unit = "m" decimal digits = "2"
VERLEN	O	unit = "m" decimal digits = "1"
BCNSHP	M	value list = "1,2,3,4,5,6,7"
CATSPM	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
CONRAD	O	value list = "3"
OBJNAM	O	
NOBJNM	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
INFORM	O	

acronym	usage	constraints
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
NTXTDS	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"

Feature	Building, single
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Acronym: BUISGL Code: 12

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A relatively permanent structure, roofed and usually walled. It is designed for some particular use which it may be important to indicate. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CONVIS	O	value list = "1,2"
FUNCTN	C	value list = "2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42"
CONDTN	O	value list = "1,2,3,4,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Built-up area
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Acronym: BUAARE Code: 13

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area containing a concentration of buildings and the supporting road or rail infrastructure.

Attribute Bindings:

acronym	usage	constraints
CATBUA	O	value list = "1,2,3,4,5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
unlocd	C	

Feature	Buoy, cardinal
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Acronym: BOYCAR Code: 14

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A buoy, is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary S-32 5th Edition, 565). A cardinal buoy, is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "1,2,3,4,5,6,8"
CATCAM	M	value list = "1,2,3,4"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
CONRAD	O	value list = "3"
MARSYS	C	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
NOBJNM	C	
OBJNAM	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	

acronym	usage	constraints
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
mmsico	O	format = "xxxxxxxx"
STATUS	O	value list = "5"

Feature	Buoy, isolated danger
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Acronym: BOYISD Code: 16
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary S-32 5th Edition, 565). An isolated danger buoy is a buoy moored on or above an isolated danger of limited extent, which has navigable water all around it. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "4,5"
COLOUR	M	value list = "2,3"
COLPAT	M	value list = "1"
CONRAD	O	value list = "3"
MARSYS	C	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
NOBJNM	C	
OBJNAM	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
mmsico	O	format = "xxxxxxxxx"
STATUS	O	value list = "5"

Feature	Buoy, lateral
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Acronym: BOYLAT Code: 17
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A buoy, is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A lateral buoy, is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "1,2,3,4,5,6,8"
CATLAM	M	value list = "1,2,3,4"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
CONRAD	O	value list = "3"
MARSYS	C	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
NOBJNM	C	
OBJNAM	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
typatn	O	value list = "1,2,3"
mmsico	O	format = "xxxxxxxx"
STATUS	O	value list = "5"

Feature	Buoy, safe water
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Acronym: BOYSAW Code: 18

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A buoy, is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A safe water buoy, is used to indicate that there is navigable water around the mark. (UKHO NP735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "1,2,3,4,5,6,8"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	M	value list = "1,2,3,4,5,6"
CONRAD	O	value list = "3"
MARSYS	C	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
mmsico	O	format = "xxxxxxxx"
STATUS	O	value list = "5"

Feature	Buoy, special purpose/general
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Acronym: BOYSPP Code: 19

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A buoy, is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A special purpose buoy, is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners. (UKHO NP 735, 5th Edition). Buoy in general: A buoy, whose appearance or purpose is not adequately known.

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "1,2,3,4,5,6,8"
CATSPM	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
CONRAD	O	value list = "3"
MARSYS	C	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	

acronym	usage	constraints
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
typatn	O	value list = "1,2,3"
mmsico	O	format = "xxxxxxxx"
STATUS	O	value list = "5"

Feature	Cable area
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Acronym: CBLARE Code: 20
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
 Definition: An area which contains one or more submarine cables.

Attribute Bindings:

acronym	usage	constraints
CATCBL	O	value list = "1,3,4,5,6"
STATUS	C	value list = "5,18"
RESTRN	O	value list = "1,38"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"

Feature	Cable, submarine
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Acronym: CBLSUB Code: 22
 Type: G
 Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An assembly of wires or fibres, or a wire rope or chain which has been laid underwater or buried beneath the seabed (Hydrographic Service, Royal Australian Navy)

Attribute Bindings:

acronym	usage	constraints
CATCBL	O	value list = "1,3,4,5,6"
STATUS	C	value list = "4,5,18"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Canal
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Acronym: CANALS Code: 23
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An artificial waterway with no flow, or a controlled flow, used for navigation, or for draining or irrigating land (ditch). (United States Geological Survey, Jan.89)

Attribute Bindings:

acronym	usage	constraints
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Cargo Transhipment Area
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Acronym: CTSARE Code: 25
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area designated for the transfer of cargo from one vessel to another. (adapted from IHO Dictionary, S-32, 5th Edition, 5593).

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
DATEND	O	format = "ccyymmdd"
DATSTA	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
INFORM	O	
NINFOM	O	
STATUS	O	value list = "2,4,5"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Causeway
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Acronym: CAUSWY Code: 26
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A raised way across low or wet ground or water. (IHO Dictionary, S-32, 5th Edition, 662)

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
CONDTN	O	value list = "1,2,3,5"
WATLEV	M	value list = "2,4"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Caution area
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Acronym: CTNARE Code: 27
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Generally, an area where the mariner has to be made aware of circumstances influencing the safety of navigation.

Attribute Bindings:

acronym	usage	constraints
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
NOBJNM	O	
OBJNAM	O	
INFORM	M	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Coastguard station
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Acronym: CGUSTA Code: 29
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Watch keeping stations at which a watch is kept either continuously, or at certain times only. (IHO Chart Specifications, M-4)

Attribute Bindings:

acronym	usage	constraints
COMCHA	M	format = "[XXXX];[XXXX];....]"
OBJNAM	O	
NOBJNM	O	
STATUS	C	value list = "1,2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18"
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
PICREP	O	

Feature	Coastline
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Acronym: COALNE Code: 30
 Type: G
 Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The line where shore and water meet. Although the terminology of coasts and shores is rather confused, shoreline and coastline are generally used as synonyms. (IHO Dictionary, S-32, 5th Edition, 858,4695)

Attribute Bindings:

acronym	usage	constraints
CATCOA	O	value list = "1,2,3,4,5,6,7,8,9,10,11"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	decimal digits = "2" unit = "m"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Conveyor
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Acronym: CONVYR Code: 34
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A mechanical apparatus for moving bulk material or people from place to place (as by a moving belt or chain of receptacles).

Attribute Bindings:

acronym	usage	constraints
CATCON	M	value list = "1,2"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PRODC	O	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
VERCLR	O	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "4,12"

Feature	Coverage
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Acronym: M_COVR Code: 302
Type: M
Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A geographical area that describes the coverage and extent of spatial objects.

Attribute Bindings:

acronym	usage	constraints
CATCOV	M	value list = "1,2"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Crane
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Acronym: CRANES Code: 35

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A machine for lifting, shifting and lowering objects or materials by means of a swinging boom or with a lifting apparatus supported on an overhead track. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CATCRN	O	value list = "2,3,4,5"
VERCLR	O	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Dam
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Acronym: DAMCON Code: 38
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding. (IHO Dictionary, S-32, 5th Edition, 1196)

Attribute Bindings:

acronym	usage	constraints
CATDAM	M	value list = "1,2,3"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
NATCON	O	value list = "1,2,3,4,5,6,7,8,9"
CONDTN	O	value list = "1,2,3,5"
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
OBJNAM	O	
NOBJNM	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Daymark
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Acronym: DAYMAR Code: 39

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid. (IHO Dictionary, S-32, 5th Edition, 1248)

Attribute Bindings:

acronym	usage	constraints
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
TOPSHP	M	value list = "1-33"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Depth area
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Acronym: DEPARE Code: 42
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A depth area is a water area whose depth is within a defined range of values.

Attribute Bindings:

acronym	usage	constraints
DRVAL1	M	unit = "m" decimal digits = "2"
DRVAL2	M	unit = "m" decimal digits = "2"
INFORM	C	
NINFOM	O	
QUASOU	C	value list = "1,2,8,10,11"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Depth contour
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Acronym: DEPCNT Code: 43

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A line connecting points of equal water depth which is sometimes significantly displaced outside of soundings, symbols and other chart detail for clarity as well as generalization. Depth contours, therefore, often represent an approximate location of the line of equal depth as related to the surveyed line delineated on the source. Also referred to as depth curve. (IHO Dictionary, S-32, 5th Edition, 1314, 1315)

Attribute Bindings:

acronym	usage	constraints
VALDCO	M	unit = "m" decimal digits = "2"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,ccccc,c..."
PICREP	O	
INFORM	O	
NINFOM	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Dredged area
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Acronym: DRGARE Code: 46
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area of the bottom of a body of water which has been deepened by dredging. (IHO Dictionary, S-32, 5th Edition, 1462)

Attribute Bindings:

acronym	usage	constraints
DRVAL1	O	unit = "m" decimal digits = "2"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Dry dock
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Acronym: DRYDOC Code: 47

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An artificial basin fitted with a gate or caisson, into which vessels can be floated and the water pumped out to expose the vessel's bottom. Also called graving dock. (IHO Dictionary, S-32, 5th Edition, 1426)

Attribute Bindings:

acronym	usage	constraints
DRVAL1	O	unit = "m" decimal digits = "2"
HORCLR	O	unit = "m" decimal digits = "2"
HORLEN	O	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Dumping ground
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Acronym: DMPGRD Code: 48
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-09-08

Definition: A sea area where dredged material or other potentially more harmful material, e.g. explosives, chemical waste, is deliberately deposited. (Derived from IHO Chart Specifications, M-4).

Attribute Bindings:

acronym	usage	constraints
CATDPG	M	value list = "2,4,5"
RESTRN	C	value list = "1,3,5,7,8,24"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Dyke
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Acronym: DYKCON Code: 49
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A dyke (or dike) is an artificial embankment to contain or hold back water.(IHO Dictionary, S-32, 5th Edition, 1361)

Attribute Bindings:

acronym	usage	constraints
HEIGHT	O	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Fairway
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Acronym: FAIRWY Code: 51

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: That part of a river, harbour and so on, where the main navigable channel for vessels of larger size lies. It is also the usual course followed by vessels entering or leaving harbours, called 'ship channel'.
(International Maritime Dictionary, 2nd Ed.)

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Fence/wall
---------	------------

Acronym: FNCLNE Code: 52
 Type: G
 Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A natural or man-made barrier used as an enclosure or boundary or for protection. (adapted from Digital Geographic Information Working Group, Oct.1987)

Attribute Bindings:

acronym	usage	constraints
CATFNC	M	value list = "1,4"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Ferry route
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Acronym: FERYRT Code: 53

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A route in a body of water where a ferry crosses from one shoreline to another. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CATFRY	M	value list = "1,2"
STATUS	C	value list = "2,3,4,5,8,9,12,14,16,17"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
unlocd	C	

Feature	Fishing facility
---------	------------------

Acronym: FSHFAC Code: 55
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A structure in shallow water for fishing purposes which can be an obstruction to ships in general. The position of these structures may vary frequently over time.

Attribute Bindings:

acronym	usage	constraints
CATFIF	O	value list = "1,2,3,4"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
STATUS	C	value list = "2,4,5"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
VERLEN	O	unit = "m" decimal digits = "1"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Floating dock
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Acronym: FLODOC Code: 57

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A form of dry dock consisting of a floating structure of one or more sections which can be partly submerged by controlled flooding to receive a vessel, then raised by pumping out the water so that the vessel's bottom can be exposed. (IHO Dictionary, S-32, 5th Edition, 1427)

Attribute Bindings:

acronym	usage	constraints
DRVAL1	O	unit = "m" decimal digits = "2"
HORCLR	O	unit = "m" decimal digits = "2"
HORLEN	O	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
sdrlev	O	
sdrval	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "1,4,6,7,8,12"

Feature	Fog signal
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Acronym: FOGSIG Code: 58

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal. (IHO Dictionary, S-32, 5th Edition, 1890)

Attribute Bindings:

acronym	usage	constraints
CATFOG	M	value list = "1,2,3,4,5,6,7,8,9,10"
SIGFRQ	O	unit = "Hz" min = "0"
SIGGEN	O	value list = "1,2"
SIGGRP	C	format = "(c)(c)..."
SIGPER	C	unit = "s" decimal digits = "2"
SIGSEQ	C	format = "LL.L + (EE.E)"
VALMXR	O	unit = "nm" decimal digits = "1"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Fortified Structure
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Acronym: FORSTC Code: 59
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A structure for the military defence of a site.

Attribute Bindings:

acronym	usage	constraints
CATFOR	M	value list = "1,2,3,4,5,6"
CONDTN	O	value list = "2"
CONVIS	O	value list = "1,2"
NATCON	O	value list = "1,2,6,7"
OBJNAM	O	
NOBJNM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Free port area
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Acronym: FRPARE Code: 60
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A port where certain import and export duties are waived (unless goods pass into the country) to facilitate reshipment to other countries. (IHO Dictionary, S-32, 5th Edition, 1927)

Attribute Bindings:

acronym	usage	constraints
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Gate
---------	------

Acronym: GATCON Code: 61
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A structure that may be swung, drawn, or lowered to block an entrance or passageway. (United States Geological Survey, Jan.89)

Attribute Bindings:

acronym	usage	constraints
CATGAT	M	value list = "2,4"
HORCLR	M	unit = "m" decimal digits = "2"
VERCLR	O	unit = "m" decimal digits = "2"
VERDAT	O	value list = "4,23,30"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
unlocd	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
locmrn	C	

acronym	usage	constraints
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Gridiron
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Acronym: GRIDRN Code: 62

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A structure in the intertidal zone serving as a support for vessels at low stages of the tide to permit work on the exposed portion of the vessel's hull. Also called careening grid. (adapted from IHO Dictionary, S-32, 5th Edition, 649)

Attribute Bindings:

acronym	usage	constraints
HORLEN	O	unit = "m" decimal digits = "1"
HORWID	O	unit = "m" decimal digits = "1"
VERLEN	O	unit = "m" decimal digits = "1"
NATCON	O	value list = "2,6"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
STATUS	O	value list = "4,8,14"
CONDTN	O	value list = "1,2,5"
WATLEV	O	value list = "3,4"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
PICREP	O	
TXTDSC	O	
NTXTDS	O	

Feature	Harbour facility
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Acronym: HRBFAC Code: 64
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A harbour installation with a service or commercial operation of public interest.

Attribute Bindings:

acronym	usage	constraints
CATHAF	M	value list = "5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	M	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
unlocd	C	
locmrn	C	
STATUS	O	value list = "5"

Feature	Hulk
---------	------

Acronym: HULKES Code: 65
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
 Definition: A permanently moored ship.

Attribute Bindings:

acronym	usage	constraints
CATHLK	M	value list = "1,2,3,4,5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Lake
---------	------

Acronym: LAKARE Code: 69
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A large body of water entirely surrounded by land. (IHO Dictionary, S-32, 5th Edition, 2629)

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Land area
---------	-----------

Acronym: LNDARE Code: 71
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The solid portion of the Earth's surface, as opposed to sea, water. (IHO Dictionary, S-32, 5th Edition, 2635)

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Land region
---------	-------------

Acronym: LNDRGN Code: 73
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area of natural scenery on land. It is defined by its geographical characteristics and may be known by its proper name.

Attribute Bindings:

acronym	usage	constraints
CONDTN	C	value list = "1,2,3,5"
OBJNAM	M	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
CATLND	O	value list = "2,9,11,12"

Feature	Landmark
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Acronym: LNDMRK Code: 74

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A prominent object at a fixed location which can be used in determining a location or a direction. (adapted from IHO Dictionary, S-32, 5th Edition, 2643).

Attribute Bindings:

acronym	usage	constraints
CATLMK	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21"
CONVIS	M	value list = "1"
FUNCTN	C	value list = "2-42"
CONDTN	O	value list = "1,2,3,4,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
VERLEN	O	unit = "defined in hunits" decimal digits = "1"
COLOUR	O	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"

Feature	Light
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Acronym: LIGHTS Code: 75

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A luminous or lighted aid to navigation. (adapted from IHO Dictionary, S-32, 5th Edition, 2766)

Attribute Bindings:

acronym	usage	constraints
CATLIT	C	value list = "1,4,5,6,12,13,14,15"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
EXCLIT	C	value list = "1,2,3,4"
LITCHR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,25,26,27,28,29"
LITVIS	C	value list = "3,4,7,8"
MLTYLT	C	min = "2"
ORIENT	C	unit = "deg" decimal digits = "2"
SECTR1	C	unit = "deg" decimal digits = "2"
SECTR2	C	unit = "deg" decimal digits = "2"
SIGGRP	C	format = "(c)(c)..."
SIGPER	C	unit = "s" decimal digits = "2"
SIGSEQ	C	format = "LL.L + (EE.E)"
STATUS	C	value list = "2,3,4,5,8,9,12,14,16,17"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	C	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	

acronym	usage	constraints
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HEIGHT	O	unit = "defined in hunits" decimal digits = "1"
VALNMR	O	

Feature	Magnetic Variation
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Acronym: MAGVAR Code: 81

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The angle between the magnetic and geographic (true) north at a location, expressed in degrees east or west from the direction of true north

Attribute Bindings:

acronym	usage	constraints
NTXTDS	O	
VALMAG	M	format = "sxx.xx"
VALACM	M	format = "sxx.xx"
RYRMGV	M	format = "ccyy"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"

Feature	Marine farm/culture
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Acronym: MARCUL Code: 82
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-08-12

Definition: An assemblage of cages, nets, rafts and floats or posts where fish, including shellfish, are artificially cultivated.

Attribute Bindings:

acronym	usage	constraints
CATMFA	M	value list = "1,2,3,4"
EXPSOU	C	value list = "1,2,3"
VALSOU	C	unit = "m" decimal digits = "2"
QUASOU	O	value list = "1,2,3,4,6,7,8,9"
SOUACC	C	unit = "m" decimal digits = "2"
WATLEV	C	value list = "1,2,3,4,5,7"
STATUS	C	value list = "2,4,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
PICREP	O	

acronym	usage	constraints
TXTDSC	O	
NTXTDS	O	

Feature	Military practice area
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Acronym: MIPARE Code: 83

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area within which naval, military or aerial exercises are carried out. Also called an exercise area.

Attribute Bindings:

acronym	usage	constraints
RESTRN	O	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,,2715,16,17,18,19,20,21,22,23,24,25,26"
CATMPA	O	value list = "2,3,4,5,6"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
NTXTDS	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "2,4,5"

Feature	Mooring/Warping facility
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Acronym: MORFAC Code: 84

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The equipment or structure used to secure a vessel (adapted from IHO Dictionary, S-32, 5th Edition, 3322)

Attribute Bindings:

acronym	usage	constraints
CATMOR	M	value list = "1,2,3,4,5,7"
NATCON	O	value list = "1,2,3,4,5,6,7,8,9"
WATLEV	O	value list = "1,2,3,4,5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
COLOUR	O	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
BOYSHP	C	value list = "2,3,7"

acronym	usage	constraints
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Nautical Publication Information
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Acronym: M_NPUB Code: 305
Type: M
Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Used to relate additional nautical information or publications to the data.

Attribute Bindings:

acronym	usage	constraints
TXTDSC	M	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Navigation line
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Acronym: NAVLNE Code: 85

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A navigation line is a straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing. (Service Hydrographique et Océanographique de la Marine, France)

Attribute Bindings:

acronym	usage	constraints
CATNAV	C	value list = "1,2,3"
ORIENT	M	unit = "deg" decimal digits = "2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	New Object
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Acronym:	NEWOBJ	Code:	163
Type:	G		
Primitive:	P		

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2012-01-01

Definition: A new feature specified by the IMO and that affects safety of navigation which cannot adequately be encoded by any existing object class for use in an S-57 data set.

Attribute Bindings:

acronym	usage	constraints
CLSNAM	M	
CLSDEF	M	
SYMINS	M	
DATEND	O	format = "ccyymmdd"
DATSTA	O	format = "ccyymmdd"
INFORM	M	
NINFOM	O	
OBJNAM	O	
NOBJNM	O	
TXTDSC	O	
NTXTDS	O	
PEREND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,ccccc,c..."
PICREP	O	
mmsico	O	format = "xxxxxxxxx"
STATUS	O	value list = "5"

Feature	Obstruction
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Acronym: OBSTRN Code: 86

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: In marine navigation, anything that hinders or prevents movement, particularly anything that endangers or prevents passage of a vessel. The term is usually used to refer to an isolated danger to navigation... (IHO Dictionary, S-32, 5th Edition, 3503)

Attribute Bindings:

acronym	usage	constraints
CATOBS	O	value list = "1,2,3,4,5,6,7,8,9,10,11"
NATSUR	C	value list = "9"
VALSOU	C	unit = "m" decimal digits = "2"
WATLEV	C	value list = "1,2,3,4,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
EXPSOU	C	value list = "1,2,3"
NATCON	O	value list = "1,2,3,4,6,7,8"
QUASOU	O	value list = "2,6,7,8"
HORACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Offshore platform
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Acronym: OFSPLF Code: 87

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A permanent offshore structure, either fixed or floating, used in the production of oil or natural gas. (IHO Dictionary, S-32, 5th Edition, 3895)

Attribute Bindings:

acronym	usage	constraints
INFORM	O	
NINFOM	O	
CATOPF	M	value list = "1,2,3,4,5,6,7,9"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	O	value list = "1,2,3,4,5,6"
CONDTN	O	value list = "1,2,5"
CONRAD	O	value list = "1,2,3"
CONVIS	O	value list = "1,2"
HEIGHT	O	unit = "m" decimal digits = "2"
NATCON	O	value list = "2,7,8"
OBJNAM	O	
NOBJNM	O	
PRODCT	O	value list = "1,2,18,19"
STATUS	O	value list = "2,4,7,12,16,17"
VERLEN	O	unit = "m" decimal digits = "1"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
DATSTA	O	format = "ccyymmdd"

acronym	usage	constraints
DATEND	O	format = "ccyymmdd"
PICREP	O	
TXTDSC	O	
NTXTDS	O	

Feature	Offshore production area
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Acronym: OSPARE Code: 88
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area off or away from the shore within which there are production facilities.

Attribute Bindings:

acronym	usage	constraints
INFORM	O	
NINFOM	O	
CATPRA	M	value list = "4,9"
CONDTN	O	value list = "1,2,3,4,5"
CONRAD	O	value list = "1,2,3"
CONVIS	O	value list = "1,2"
NATCON	O	value list = "2,7,8"
OBJNAM	O	
NOBJNM	O	
RESTRN	M	value list = "1,2,7,8,14"
STATUS	O	value list = "2,4,7,12,16,17"
VERLEN	O	unit = "m" decimal digits = "1"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PICREP	O	
TXTDSC	O	
NTXTDS	O	

acronym	usage	constraints
HEIGHT	O	unit = "m" decimal digits = "2"

Feature	Oil barrier
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Acronym: OILBAR Code: 89
 Type: G
 Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A construction to dam oil flow on water.

Attribute Bindings:

acronym	usage	constraints
CATOLB	O	value list = "1,2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Pile
---------	------

Acronym: PILPNT Code: 90

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A long heavy timber or section of steel, wood, concrete, etc.. forced into the earth which may serve as a support, as for a pier, or a free standing pole within a marine environment. (Adapted from IHO Dictionary, S-32, 5th Edition, 3840)

Attribute Bindings:

acronym	usage	constraints
CONDTN	O	value list = "1,2,3,5"
OBJNAM	C	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
VERLEN	O	unit = "defined in hunits" decimal digits = "1"
HEIGHT	O	unit = "defined in hunits" decimal digits = "1"
COLOUR	O	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
HORACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "1,4,6,7,8,12,14"

Feature	Pilot boarding place
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Acronym: PILBOP Code: 91
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The meeting place to which the pilot comes out. (IHO Chart Specifications, M-4)

Attribute Bindings:

acronym	usage	constraints
CATPIL	O	value list = "1,2,3"
COMCHA	O	format = "[[XXXX];[XXXX];....]"
PILDST	O	
NPLDST	O	
OBJNAM	O	
NOBJNM	O	
STATUS	C	value list = "1,2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"

Feature	Pipeline area
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Acronym: PIPARE Code: 92

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area containing one or more pipelines.

Attribute Bindings:

acronym	usage	constraints
CATPIP	O	value list = "2,3,4,6"
PRODC	O	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
RESTRN	M	value list = "1,38"
STATUS	C	value list = "18"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Pipeline, submarine/on land
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Acronym: PIPSOL Code: 94

Type: G

Primitive: P,L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A pipeline is a string of interconnected pipes used for the transport of matter, nowadays mainly oil or gas. (IHO Dictionary, S-32, 5th Edition, 3857). A submarine or land pipeline is a pipeline lying on or buried under the seabed or the land.

Attribute Bindings:

acronym	usage	constraints
CATPIP	O	value list = "2,3,4,6"
PRODC	O	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
STATUS	C	value list = "4,18"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Pontoon
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Acronym: PONTON Code: 95

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A floating structure, usually rectangular in shape which serves as landing, pier head or bridge support. (IHO Dictionary, S-32, 5th Edition, 3947)

Attribute Bindings:

acronym	usage	constraints
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"

Feature	Production/storage area
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Acronym: PRDARE Code: 97
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area on land for the exploitation or storage of natural resources.

Attribute Bindings:

acronym	usage	constraints
CATPRA	O	value list = "1,2,3,4,5,6,7,8,9,10"
PRODCT	O	value list = "1,2,4,5,6,7,14,15,17,21,22"
CONVIS	O	value list = "1,2"
STATUS	O	value list = "2,12,16,17"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
SCAMIN	M	min = "1"
NTXTDS	O	

Feature	Pylon/bridge support
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Acronym: PYLONS Code: 98
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A vertical construction consisting, for example, of a steel framework or pre-stressed concrete to carry cables, a bridge, etc.

Attribute Bindings:

acronym	usage	constraints
CATPYL	M	value list = "1,2,3,4,5"
WATLEV	M	value list = "1,2,3,4,5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Quality of data
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Acronym: M_QUAL Code: 308

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area within which a uniform assessment of the quality of the data exists.

Attribute Bindings:

acronym	usage	constraints
CATZOC	O	value list = "1,2,3,4,5,6"
DRVAL1	O	unit = "m" decimal digits = "2"
POSACC	O	unit = "m" decimal digits = "2"
SOUACC	O	unit = "m" decimal digits = "2"
SUREND	O	format = "ccyymmdd"
SURSTA	O	format = "ccyymmdd"
TECSOU	C	value list = "1,2,3,4,5,6,8,9,10,11,12,13,15,16,17"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Radar line
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Acronym: RADLNE Code: 99

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A track along which ships may be guided by coastal radar stations in the event of bad visibility. Also known as a radar guided track. (IHO Dictionary, S-32, 5th Edition, 4146).

Attribute Bindings:

acronym	usage	constraints
ORIENT	M	unit = "deg" decimal digits = "2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Radar station
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Acronym: RADSTA Code: 102

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A station with a transmitter emitting pulses of ultra-high frequency radio waves which are reflected by solid objects and are detected upon their return to the sending station. (International Maritime Dictionary, 2nd Ed.)

Attribute Bindings:

acronym	usage	constraints
CATRAS	M	value list = "1"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Radar transponder beacon
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Acronym: RTPBCN Code: 103

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A transponder beacon, transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder. Also called racon. (IHO Dictionary, S-32, 5th Edition, 4137)

Attribute Bindings:

acronym	usage	constraints
CATRTB	M	value list = "1,2,3"
RADWAL	O	format = "xx.x-b,xx.x-b,...] xx.x = value in meter, b = band"
SIGGRP	O	format = "(c)(c)..."
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

acronym	usage	constraints
STATUS	O	value list = "5"

Feature	Railway
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Acronym: RAILWY Code: 106
 Type: G
 Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A rail or set of parallel rails on which a train or tram runs. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Recommended track
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Acronym: RECTRC Code: 109

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A track recommended to all or only certain vessels. (IHO Dictionary, S-32, 5th Edition, 5576)

Attribute Bindings:

acronym	usage	constraints
CATTRK	M	value list = "1,2"
ORIENT	M	unit = "deg" decimal digits = "2"
DRVAL1	O	unit = "m" decimal digits = "2"
DRVAL2	O	unit = "m" decimal digits = "2"
TRAFIC	M	value list = "1,2,3,4"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Recommended traffic lane part
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Acronym: RCTLPT Code: 110
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A recommended traffic lane part is an area of a recommended direction of traffic control area within which traffic flow is generally along one bearing. (IHO Definition)

Attribute Bindings:

acronym	usage	constraints
ORIENT	M	unit = "deg" decimal digits = "2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
TXTDSC	O	
NTXTDS	O	
PICREP	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
SCAMIN	M	min = "1"
STATUS	O	value list = "5"

Feature	Rescue Station
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Acronym: RSCSTA Code: 111
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-09-17

Definition: A place at which life saving equipment is held. (IHO Chart Specifications, M-4)

Attribute Bindings:

acronym	usage	constraints
catrsc	M	value list = "1,2,4,5,6,7,8,9"
STATUS	O	value list = "2,4,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
NTXTDS	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
PICREP	O	

Feature	Restricted area
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Acronym: RESARE Code: 112

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A specified area designated by an appropriate authority within which navigation is restricted in accordance with certain specified conditions. (adapted from IHO Dictionary, S-32, 5th Edition, 4366)

Attribute Bindings:

acronym	usage	constraints
RESTRN	M	value list = "1,7,8,38"
CATREA	C	value list = "1,12"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,ccccc,c..."
CATTEV	O	value list = "4,5,6"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
NTXTDS	O	

acronym	usage	constraints
STATUS	O	value list = "5"

Feature	River
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Acronym: RIVERS Code: 114
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A relatively large natural stream of water. (IHO Dictionary, S-32, 5th Edition, 4405)

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Road
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Acronym: ROADWY Code: 116
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A road is an open way for the passage of vehicles. (United States Geological Survey, Jan.89)

Attribute Bindings:

acronym	usage	constraints
CATROD	M	value list = "1,2,3,4"
NATCON	O	value list = "4,5"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Runway
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Acronym: RUNWAY Code: 117

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A defined rectangular area, on a land aerodrome, prepared for the landing and take-off run of aircraft along its length.

Attribute Bindings:

acronym	usage	constraints
CATRUN	O	value list = "1,2"
CONVIS	O	value list = "1,2"
NATCON	O	value list = "4,5"
OBJNAM	O	
NOBJNM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
STATUS	O	value list = "5"

Feature	Sand waves
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Acronym: SNDWAV Code: 118

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A large mobile wave-like sediment feature in shallow water and composed of sand. The wavelength may reach 100 metres, the amplitude may be up to 20 metres.

Attribute Bindings:

acronym	usage	constraints
VERLEN	O	unit = "m" decimal digits = "1"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
SORDAT	O	format = "ccyymmdd"
SORIND	O	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Sea area/named water area
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Acronym: SEAARE Code: 119
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A geographically defined part of the sea or other navigable waters. It may be specified within its limits by its proper name.

Attribute Bindings:

acronym	usage	constraints
CATSEA	O	value list = "5,12,13,51,52,53,54,57,58,59"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	M	
NOBJNM	O	
INFORM	C	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	C	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
NTXTDS	O	

Feature	Seabed Area
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Acronym: SBDARE Code: 121
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area of the sea where the nature of bottom is homogeneous. The nature of bottom includes the material of which it is composed and its physical characteristics. Also called character (or characteristics) of the bottom, or quality of the bottom. (IHO Dictionary, S-32, 5th Edition, 515).

Attribute Bindings:

acronym	usage	constraints
NATQUA	C	value list = "1,2,3,4"
NATSUR	C	value list = "1,2,3,4,5,6,7,8,9,18"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Shoreline construction
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Acronym: SLCONS Code: 122
 Type: G
 Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A fixed (not afloat) artificial structure between the water and the land, i.e. a man-made coastline.

Attribute Bindings:

acronym	usage	constraints
CATSLC	M	value list = "1,2,4,5,6,7,8,9,10,11,12,13,14,15,16"
NATCON	C	value list = "1,2,3,4,5,6,7,8,9"
STATUS	C	value list = "2,3,4,8,9,12,14,16,17"
WATLEV	O	value list = "1,2,3,4,5"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	C	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
unlocd	C	
locmrn	C	
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Silo/tank
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Acronym: SILTNK Code: 125
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An enclosed container, used for storage (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CATSIL	O	value list = "1,2,3,4"
PRODC	O	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
CONDN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Slope topline
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Acronym: SLOTOP Code: 126

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The upper marking of a slope, e.g. the ridge line or the separation line between two different gradients.

Attribute Bindings:

acronym	usage	constraints
CATSLO	M	value list = "1,2,3,6"
NATSUR	O	value list = "1,2,3,4,5,6,7,8,9,11,14,17,18"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
NTXTDS	O	

Feature	Sloping ground
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Acronym: SLOGRD Code: 127
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An inclined surface (adapted from IHO Dictionary, S-32, 5th Edition, 4776).

Attribute Bindings:

acronym	usage	constraints
CATSLO	M	value list = "1,2,3,6"
NATSUR	O	value list = "1,2,3,4,5,6,7,8,9,11,14,17,18"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Small craft facility
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Acronym: SMCFAC Code: 128
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A place at which a service generally of interest to small craft or pleasure boats is available.

Attribute Bindings:

acronym	usage	constraints
CATSCF	M	value list = "1-33"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Sounding
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Acronym: SOUNDG Code: 129

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A measured water depth or spot which has been reduced to a vertical datum (may be a drying height).

Attribute Bindings:

acronym	usage	constraints
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
TECSOU	O	value list = "1,2,3,4,5,6,8,9,10,11,12,13,15,16,17"
SOUACC	O	unit = "m" decimal digits = "2"

Feature	Survey reliability
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Acronym: M_SREL Code: 310

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: An area within which a uniform assessment of the reliability of source survey information exists.

Attribute Bindings:

acronym	usage	constraints
QUAPOS	C	value list = "4,10"
QUASOU	M	value list = "1,2,8,10,11"
SURATH	M	
SUREND	M	format = "ccyymmdd,ccyymm"
SURSTA	M	format = "ccyymmdd,ccyymm"
SURTYP	C	value list = "2"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Tidal stream - flood/ebb
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Acronym: TS_FEB Code: 160
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The tidal stream (or tidal current) is a horizontal movement of water associated with the rise and fall of the tide caused by tide-producing forces. (Adapted from IHO Dictionary, S-32, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
CAT_TS	M	value list = "1,2,3"
CURVEL	O	unit = "knot (kt)" decimal digits = "1"
ORIENT	O	unit = "deg" decimal digits = "2"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
OBJNAM	O	
NOBJNM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Tideway
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Acronym: TIDEWY Code: 143
 Type: G
 Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A natural course in intertidal areas where water flows during the ebb or flow. A channel through which a tidal current runs. (IHO Dictionary, S-32, 5th Edition, 5502)

Attribute Bindings:

acronym	usage	constraints
PICREP	O	
OBJNAM	O	
NOBJNM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	

Feature	Topmark
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Acronym: TOPMAR Code: 144
 Type: G
 Primitive: P

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A characteristic shape secured at the top of a buoy, or beacon, to aid in its identification. (IHO Dictionary, S-32, 5th Edition, 5548)

Attribute Bindings:

acronym	usage	constraints
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
TOPSHP	M	value list = "1-33"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	C	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Traffic separation scheme boundary
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Acronym: TSSBND Code: 146

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The outer limit of a traffic lane part or a traffic separation scheme roundabout. (S-57 Edition 3.1, Appendix A – Chapter 1, Page 1.185, November 2000).

Attribute Bindings:

acronym	usage	constraints
CATTSS	O	value list = "1,2,"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "1,2,3,4,6,7,8,9,11,12,13,14,15,16,17,18"

Feature	Traffic separation scheme crossing
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Acronym: TSSCRS Code: 147
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A defined area where traffic lanes cross. (S- 57 Edition 3.1, Appendix A – Chapter 1, Page 1.186, November 2000).

Attribute Bindings:

acronym	usage	constraints
CATTSS	O	value list = "1,2"
RESTRN	C	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,16,17,18,19,20,21,22,23,24,25,27"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
STATUS	C	value list = "1,3,6,9"
SCAMIN	M	min = "1"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	O	format = "ccyymmdd"
SORIND	O	format = "cc,cc,cccc,c..."

Feature	Traffic separation scheme lane part
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Acronym: TSSLPT Code: 148
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A traffic separation scheme lane part is an area of a traffic lane in which the direction of flow of traffic is generally along one bearing. (Adapted from S-57 Edition 3.1, Appendix A – Chapter 1, Page 1.187, November 2000).

Attribute Bindings:

acronym	usage	constraints
CATTSS	O	value list = "1,2"
ORIENT	C	unit = "deg" decimal digits = "2"
RESTRN	C	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,16,17,18,19,20,21,22,23,24,25,27"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SCAMIN	M	min = "1"
SORDAT	O	format = "ccyymmdd"
SORIND	O	format = "cc,cc,cccc,c..."

Feature	Traffic separation zone
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Acronym: TSEZNE Code: 150
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-08-11

Definition: A traffic separation scheme is a scheme which aims to reduce the risk of collision in congested and/or converging areas by separating traffic moving in opposite, or nearly opposite, directions. (IHO Dictionary, S-32, 5th Edition, 5585). A traffic separation zone is a zone separating the lanes in which ships are proceeding in opposite or nearly opposite directions; or separating traffic lanes designated for particular classes of ships proceeding in the same direction (IMO Ships Routeing, 6th Edition).

Attribute Bindings:

acronym	usage	constraints
CATTSS	M	value list = "1,2"
STATUS	C	value list = "3,9"
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
TXTDSC	C	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
OBJNAM	O	
NOBJNM	O	
PICREP	O	
NTXTDS	O	

Feature	Tunnel
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Acronym: TUNNEL Code: 151

Type: G

Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A passage that is open to the atmosphere at both ends, buried under the sea bed or laid over the sea floor or bored under the ground or through mountains.

Attribute Bindings:

acronym	usage	constraints
BURDEP	O	unit = "m" decimal digits = "1"
HORCLR	C	unit = "m" decimal digits = "2"
VERCLR	C	unit = "m" decimal digits = "2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
PEREND	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
locmrn	C	
NTXTDS	O	
unlocd	O	

acronym	usage	constraints
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Two-way route part
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Acronym: TWRTPT Code: 152

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A two-way route is a route within defined limits inside which two-way traffic is established, aimed at providing safe passage of ships through waters where navigation is difficult or dangerous. (IHO Dictionary, S-32, 5th Edition, 5712). A two-way route part is an area of a two-way route within which traffic flow is generally along one bearing (and possibly its reciprocal).

Attribute Bindings:

acronym	usage	constraints
ORIENT	M	unit = "deg" decimal digits = "2"
TRAFIC	M	value list = "1,2,3,4"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "1,3,6,9"

Feature	Underwater/awash rock
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Acronym: UWTROC Code: 153

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A concreted mass of stony material or coral which dries, is awash or is below the water surface.

Attribute Bindings:

acronym	usage	constraints
VALSOU	M	unit = "m" decimal digits = "2"
WATLEV	M	value list = "1,2,3,4,5"
NATSUR	O	value list = "5,9,11,14,18"
QUASOU	O	value list = "1,2,8,10,11"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
EXPSOU	C	value list = "1,2,3"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Unsurveyed area
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Acronym: UNSARE Code: 154
 Type: G
 Primitive: A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-07-29

Definition: An area for which no bathymetric survey information is available.

Attribute Bindings:

acronym	usage	constraints
QUASOU	C	value list = "2,8"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
NTXTDS	O	
TXTDSC	O	
PICREP	O	

Feature	Vegetation
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Acronym: VEGATN Code: 155
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
 Definition: Collections of, or individual plants.

Attribute Bindings:

acronym	usage	constraints
CATVEG	M	value list = "6,11,13"
CONVIS	O	value list = "1,2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Water turbulence
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Acronym: WATTUR Code: 156
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The disturbance of water caused by the interaction of any combination of waves, currents, tidal streams, wind, shoal patches and obstructions.

Attribute Bindings:

acronym	usage	constraints
CATWAT	M	value list = "6"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
NTXTDS	O	
PICREP	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Weed/Kelp
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Acronym: WEDKLP Code: 158
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Seaweed is the general name for marine plants of the Algae class which grow in long narrow ribbons.
 (International Maritime Dictionary, 2nd Ed.)
 Kelp is one of an order (laminariales) of usually large, blade-shaped or vine-like brown algae. (IHO
 Dictionary, S-32, 5th Edition, 2611)

Attribute Bindings:

acronym	usage	constraints
CATWED	M	value list = "1,2,3,4"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
TXTDSC	O	
NTXTDS	O	
PICREP	O	

Feature	Wreck
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Acronym: WRECKS Code: 159
 Type: G
 Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The ruined remains of a stranded or sunken vessel which has been rendered useless. (IHO Dictionary, S-32, 5th Edition, 6027)

Attribute Bindings:

acronym	usage	constraints
CATWRK	M	value list = "1,2,3,4,5"
VALSOU	C	unit = "m" decimal digits = "2"
WATLEV	O	value list = "1,2,3,4,5"
QUASOU	O	value list = "1,2,8,10,11"
TECSOU	O	value list = "1,2,3,4,5,6,8,9,10,11,12,13,15,16,17"
STATUS	O	value list = "12,16,17,18"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
EXPSOU	C	value list = "1,2,3"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
CATTEV	O	value list = "4,5,6"

Feature	Anchor berth
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Acronym: achbrt

Code: 17000

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A designated area of water where a single vessel, sea plane, etc. may anchor.

Attribute Bindings:

acronym	usage	constraints
catach	O	value list = "1,2,3,4,5,6,7,9,10,11,12,13"
clsdng	O	value list = "1,2,3,4,5"
restrn	O	value list = "1,2,7,8,13,14,27,28,29,30,31,32,33,34,35,36,38"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
NATSUR	O	value list = "1,2,3,4,5,6,7,8,9,11,14,17,18"
unlocd	C	
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

acronym	usage	constraints
locmrn	C	

Feature	Anchorage area
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Acronym: achare

Code: 17001

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area in which vessels anchor or may anchor. (IHO Dictionary, S-32, 5th Edition, 130)

Attribute Bindings:

acronym	usage	constraints
catach	O	value list = "1,2,3,4,5,6,7,9,10,11,12,13"
clsdng	O	value list = "1,2,3,4,5"
restrn	O	value list = "1,2,7,8,13,14,27,28,29,30,31,32,33,34,35,36,38"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
NATSUR	O	value list = "1,2,3,4,5,6,7,8,9,11,14,17,18"
unlocd	C	
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

acronym	usage	constraints
locmrn	C	

Feature	Beacon, lateral
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Acronym: bcnlal

Code: 17028

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A beacon, is a prominent specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey (IHO Dictionary, S-32, 5th Edition, 420). A lateral beacon, is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage. (UKHO NP 735, 5th Edition)

Attribute Bindings:

acronym	usage	constraints
BCNSHP	M	value list = "1,5"
catlam	M	value list = "1-27"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
dirimp	C	value list = "1,2,3,4"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"

acronym	usage	constraints
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
typatn	O	value list = "1,2,3"
mmsico	O	format = "xxxxxxxxx"
STATUS	O	value list = "5"

Feature	Berth
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Acronym: berths

Code: 17010

Type: G

Primitive: P,L,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A named or numbered place where a vessel is moored at a wharf. (IHO Dictionary, S-32, 5th Edition, 470)

Attribute Bindings:

acronym	usage	constraints
catbrt	O	value list = "1,2,3,4,5,6,7,8,9,10"
clsdng	O	value list = "1,2,3,4,5"
DRVAL1	O	unit = "m" decimal digits = "2"
QUASOU	C	value list = "1,2,8,10,11"
SOUACC	C	unit = "m" decimal digits = "2"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
trshgd	O	value list = "1,2,3,4,5,6,7,8,9,10"
unlocd	C	
verdat	C	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"

acronym	usage	constraints
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	

Feature	Bridge
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Acronym: bridge Code: 17011

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A structure erected over a depression or an obstacle such as a body of water, railroad, etc. to provide a roadway for vehicles, pedestrians or to carry utility services. (IHO Dictionary, S-32, 5th Edition, 544)

Attribute Bindings:

acronym	usage	constraints
CATBRG	M	value list = "1,3,4,5,7,9,10,11,12,13"
HORCLR	O	unit = "m" decimal digits = "2"
VERCCL	C	unit = "m" decimal digits = "2"
VERCLR	C	unit = "m" decimal digits = "2"
VERCOP	C	unit = "m" decimal digits = "2"
hunits	C	value list = "1,2,3,4,5,6"
unlocd	C	
verdat	C	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
wtwdis	C	decimal digits = "3"
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	C	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"

acronym	usage	constraints
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
elevwl	O	unit = "m" decimal digits = "2"
reflev	O	value list = "1,2,3,4,5,6,7,8,9,10"
locmrn	C	
refgag	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Bridge Arch Aggregation
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Acronym: c_brga Code: 18003
Type: O
Primitive: N

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2012-08-03

Definition: A collection of the bridge arch elements comprising an entire bridge arch.

Attribute Bindings:

acronym	usage	constraints
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
PICREP	O	
TXTDSC	O	
NTXTDS	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Bunker station
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Acronym: bunsta

Code: 17054

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A station, at which a ship is able to bunker fuel, water or ballast or to obtain electrical power supply.

Attribute Bindings:

acronym	usage	constraints
bunves	M	value list = "1,2"
catbun	O	value list = "1,2,3,4,5,6,7,8"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
catvol	C	value list = "1,2"

acronym	usage	constraints
catfrq	C	value list = "1,2"
amoamp	C	
allcon	O	
catplg	O	
shrnum	O	
HORACC	O	unit = "m" decimal digits = "1"
VERACC	O	unit = "m" decimal digits = "1"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Buoy, lateral
---------	---------------

Acronym: boylat

Code: 17029

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A buoy is a floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes. (IHO Dictionary, S-32, 5th Edition, 565). A lateral buoy is used to indicate the right-hand or left-hand side of a channel limit in the waterway.

Attribute Bindings:

acronym	usage	constraints
BOYSHP	M	value list = "1,2,3,4,5,6,8"
catlam	M	value list = "1-27"
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
CONRAD	O	value list = "3"
marsys	C	value list = "1,2,9,10,11,12,13,15"
OBJNAM	C	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
typatn	O	value list = "1,2,3"
mmsico	O	format = "xxxxxxxx"
STATUS	O	value list = "5"

Feature	Cable, overhead
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Acronym: cblohd

Code: 17012

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An overhead cable is an assembly of wires or fibres, or a wire rope or chain, which is supported by structures such as poles or pylons and passing over or nearby navigable waters. (Hydrographic Service, Royal Australian Navy).

Attribute Bindings:

acronym	usage	constraints
catchl	M	value list = "1,3,4,5,6,7"
VERCLR	M	unit = "m" decimal digits = "1"
hunits	O	value list = "1,2,3,4,5,6"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
wtwdis	O	decimal digits = "3"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
locmrn	C	
refgag	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Checkpoint
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Acronym: chkpnt

Code: 17027

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An official place to register, declare or check goods and people.

Attribute Bindings:

acronym	usage	constraints
catchp	M	value list = "1,2"
NATION	M	format = "cc"
unlocd	C	
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
STATUS	O	value list = "5"

Feature	Communication area
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Acronym: comare

Code: 17055

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Indicates the coverage of an area, in which a vessel has to report or may request information.

Attribute Bindings:

acronym	usage	constraints
catcom	O	value list = "1,2,3,4,5,6,7,8,9"
COMCHA	M	format = "[XXXX];[XXXX];....]"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	

Feature	Conveyor
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Acronym: convyr

Code: 17034

Type: G

Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A mechanical apparatus for moving bulk material or people from place to place (as by a moving belt or chain of receptacles).

Attribute Bindings:

acronym	usage	constraints
CATCON	M	value list = "1,2"
PRODC	O	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
VERCLR	O	unit = "m" decimal digits = "2"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXDSC	O	
DATSTA	O	format = "ccymmdd"
DATEND	O	format = "ccymmdd"
SORDAT	C	format = "ccymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Crane
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Acronym: cranes

Code: 17030

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A machine for lifting, shifting and lowering objects or materials by means of a swinging boom or with a lifting apparatus supported on an overhead track. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
CATCRN	O	value list = "2,3,4,5"
VERCLR	O	unit = "m" decimal digits = "2"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
locmrn	C	
unlocd	C	
HORACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Current, non-gravitational
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Acronym: curent

Code: 17019

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Currents (non-gravitational) include either singly or in combination: ocean currents (wind and/or density driven), inter-oceanic equalising currents, currents of navigable rivers, river outflow effects offshore and other non-tidal flows.

Attribute Bindings:

acronym	usage	constraints
curvhw	C	unit = "km/h" decimal digits = "1"
curvlw	C	unit = "km/h" decimal digits = "1"
curvmw	C	unit = "km/h" decimal digits = "1"
curvow	C	unit = "km/h" decimal digits = "1"
dirimp	C	value list = "1,2,3,4"
hignam	C	
lownam	C	
meanam	C	
othnam	C	
ORIENT	C	unit = "deg" decimal digits = "2"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"

acronym	usage	constraints
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Daymark
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Acronym: daymar

Code: 17035

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2009-09-11

Definition: The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid. (IHO Dictionary, S-32, 5th Edition, 1248)

Attribute Bindings:

acronym	usage	constraints
COLOUR	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13"
COLPAT	C	value list = "1,2,3,4,5,6"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
TOPSHP	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33"
ORIENT	C	unit = "deg" decimal digits = "2"
dirimp	C	value list = "1,2,3,4"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	

acronym	usage	constraints
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Depth area
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Acronym: depare

Code: 17003

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A depth area is a water area whose depth is within a defined range of values.

Attribute Bindings:

acronym	usage	constraints
DRVAL1	M	unit = "m" decimal digits = "2"
DRVAL2	M	unit = "m" decimal digits = "2"
eleva1	C	unit = "m" decimal digits = "2"
eleva2	C	unit = "m" decimal digits = "2"
QUASOU	C	value list = "1,2,8,10,11"
hunits	M	value list = "1,2,3,4,5,6"
wtwdis	M	decimal digits = "3"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Distance mark
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Acronym: disarm

Code: 17004

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A distance mark indicates the distance measured from an origin and consists of a distinct location without special installation, used to serve as a reference along the waterway. (Adapted from S-57 Standard). Due to the natural or historic changes in the waterway the distance can deviate from the real distance to the origin.

Attribute Bindings:

acronym	usage	constraints
CATDIS	M	value list = "1,2,3,4"
hunits	M	value list = "1,2,3,4,5,6"
unlocd	C	
wtwdis	M	decimal digits = " 1"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
STATUS	O	value list = "5"

Feature	Exceptional navigation structure
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Acronym: excnst

Code: 17070

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An exceptional navigational construction like aqueduct, lift-lock, etc.

Attribute Bindings:

acronym	usage	constraints
catexs	M	value list = "1,2,3,4,5"
DRVAL1	M	unit = "m" decimal digits = "2"
unlocd	C	
verdat	C	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
hunits	M	value list = "1,2,3,4,5,6"
wtwdis	M	decimal digits = "3"
CONDTN	O	value list = "1,2,3,5"
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
sdrlev	O	

acronym	usage	constraints
sdrval	O	unit = "m" decimal digits = "2"
locmrn	C	
OBJNAM	O	
NOBJNM	O	
horclw	C	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Ferry route
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Acronym: feryrt

Code: 17013

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A route in a body of water where a ferry crosses from one shoreline to another. (Digital Geographic Information Working Group, Oct.87)

Attribute Bindings:

acronym	usage	constraints
catfry	M	value list = "4"
NOBJNM	O	
OBJNAM	O	
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
INFORM	O	
NINFOM	O	
NTXTDS	O	
PICREP	O	
SCAMIN	M	min = "1"
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
locmrn	C	
unlocd	C	

Feature	Floating dock
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Acronym: flodoc

Code: 17025

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A form of dry dock consisting of a floating structure of one or more sections which can be partly submerged by controlled flooding to receive a vessel, then raised by pumping out the water so that the vessel's bottom can be exposed. (IHO Dictionary, S-32, 5th Edition, 1427)

Attribute Bindings:

acronym	usage	constraints
DRVAL1	O	unit = "m" decimal digits = "2"
HORCLR	O	unit = "m" decimal digits = "2"
HORLEN	O	unit = "m" decimal digits = "2"
horcll	C	unit = "m" decimal digits = "2"
horclw	C	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"

acronym	usage	constraints
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
sdrlev	O	
sdrval	O	unit = "m" decimal digits = "2"
vcrlev	O	
vcrval	O	unit = "m" decimal digits = "2"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "1,4,6,7,8,12"

Feature	Gate
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Acronym: gatcon

Code: 17031

Type: G

Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A structure that may be swung, drawn, or lowered to block an entrance or passageway. (United States Geological Survey, Jan.89)

Attribute Bindings:

acronym	usage	constraints
CATGAT	M	value list = "2,4"
HORCLR	M	unit = "m" decimal digits = "2"
VERCLR	O	unit = "m" decimal digits = "2"
hunits	C	value list = "1,2,3,4,5,6"
wtwdis	O	decimal digits = "3"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
unlocd	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vcrlev	O	

acronym	usage	constraints
vcrval	O	unit = "m" decimal digits = "2"
locmrn	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Harbour area (administrative)
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Acronym: hrbare

Code: 17014

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The term "harbour" applies only to the area of water with the works necessary for its formation, protections and maintenance (International Maritime Dictionary, 2d. Edition). A harbour area not only covers the area of water but also the area of land which supplies the harbour installations.

Attribute Bindings:

acronym	usage	constraints
cathbr	O	value list = "1,2,3,4,5"
unlocd	C	
CONDTN	C	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	

acronym	usage	constraints
STATUS	O	value list = "5"

Feature	Harbour basin
---------	---------------

Acronym: hrbbsn

Code: 17056

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An enclosed area of water surrounded by quay walls constructed to provide means for the transfer of cargos from and to ships (International Maritime Dictionary, 2d. Edition).

Attribute Bindings:

acronym	usage	constraints
HORLEN	O	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	

acronym	usage	constraints
STATUS	O	value list = "5"

Feature	Harbour facility
---------	------------------

Acronym: hrbfac

Code: 17015

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A harbour installation with a service or commercial operation of public interest.

Attribute Bindings:

acronym	usage	constraints
cathaf	M	value list = "4,6,9,12,13,16,17"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
unlocd	C	
STATUS	O	value list = "5"

Feature	Hulk
---------	------

Acronym: hulkes

Code: 17020

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A permanently moored ship.

Attribute Bindings:

acronym	usage	constraints
cathlk	O	value list = "1,2,3,4,5,6"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Lock basin
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Acronym: lokbsn

Code: 17016

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A lock basin is a wet dock in a waterway, permitting a ship to pass from one level to another. (adapted from IHO Dictionary, S-32, 5th Edition, 2881)

Attribute Bindings:

acronym	usage	constraints
horcfl	C	unit = "m" decimal digits = "2"
horclw	C	unit = "m" decimal digits = "2"
HORLEN	O	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
locmrn	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Lock basin part
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Acronym: lkbspt

Code: 17058

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A lock basin is divided into several lock basin parts, if this lock basin has one ground level but several gates.

Attribute Bindings:

acronym	usage	constraints
horcll	C	unit = "m" decimal digits = "2"
horclw	C	unit = "m" decimal digits = "2"
HORLEN	O	unit = "m" decimal digits = "2"
HORWID	O	unit = "m" decimal digits = "2"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"
locmrn	C	

Feature	Maximum permitted ship dimensions
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Acronym: lg_sdm

Code: 18001

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Waterway or waterway section for which a juridical regulation with respect to the maximum permitted vessel dimensions exists.

Attribute Bindings:

acronym	usage	constraints
lg_rel	O	value list = "1,2,3,4"
lg_des	O	
lc_csi	O	value list = "1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32"
lc_cse	O	value list = "1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32"
lc_asl	O	value list = "1,2,3,5,6,7,8,9,10"
lc_ase	O	value list = "1,2,3,5,6,7,8,9,10"
lc_cci	O	value list = "1,2,4,5,6,7,8,9"
lc_cce	O	value list = "1,2,4,5,6,7,8,9"
lg_bme	O	unit = "m" decimal digits = "2"
lg_lgs	O	unit = "m" decimal digits = "2"
lg_drt	O	unit = "m" decimal digits = "2"
lg_wdp	O	decimal digits = "1"
lg_wdu	O	value list = "1,2,3"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
lg_pbr	O	

Feature	Maximum permitted vessel speed
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Acronym: lg_vsp

Code: 18002

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Waterway or waterway section for which a juridical regulation with respect to the maximum permitted vessel speed exists.

Attribute Bindings:

acronym	usage	constraints
lg_rel	O	value list = "1,2,3,4"
lg_des	O	
lc_csi	O	value list = "1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32"
lc_cse	O	value list = "1,2,3,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32"
lc_asl	O	value list = "1,2,3,5,6,7,8,9,10"
lc_ase	O	value list = "1,2,3,5,6,7,8,9,10"
lc_cci	O	value list = "1,2,4,5,6,7,8,9"
lc_cce	O	value list = "1,2,4,5,6,7,8,9"
lg_wdu	O	value list = "1,2,3"
lg_spd	O	unit = "km/h" decimal digits = "2"
lg_spr	O	value list = "1,2,3"
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
lg_pbr	O	

Feature	Navigational system of marks
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Acronym: m_nsys

Code: 17018

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area within which a specific system of navigational marks applies and/or a common direction of buoyage.

Attribute Bindings:

acronym	usage	constraints
marsys	M	value list = "1,2,9,10,11,12,13,15"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
ORIENT	C	unit = "deg" decimal digits = "2"

Feature	Notice mark
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Acronym: notmrk

Code: 17050

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A signboard used to indicate prohibitions, regulations, restrictions, recommendations and general information which apply to a waterway or a section of a waterway

Attribute Bindings:

acronym	usage	constraints
addmrk	O	value list = "1,2,3,4,5"
catnmk	M	value list = "1 - 127"
dirimp	O	value list = "1,2,3,4,5"
disipd	O	unit = "m" decimal digits = "1"
disipu	O	unit = "m" decimal digits = "1"
disbk1	O	unit = "m" decimal digits = "1"
disbk2	O	unit = "m" decimal digits = "1"
fnctnm	M	value list = "1,2,3,4,5"
marsys	O	value list = "1,2,9,10,11,12,13,15"
ORIENT	C	unit = "deg" decimal digits = "2"
STATUS	O	value list = "2,3,4,5,8,9,12,14,16,17"
bnkwtw	O	value list = "1,2"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	

acronym	usage	constraints
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
mmsico	O	format = "xxxxxxxxx"

Feature	Pipeline, overhead
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Acronym: pipohd

Code: 17024

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A pipeline is a string of interconnected pipes used for the transport of matter, nowadays mainly oil or gas. (IHO Dictionary, S-32, 5th Edition, 3857). An overhead pipeline is a pipeline supported by pylons and passing over or nearby navigable waters.

Attribute Bindings:

acronym	usage	constraints
CATPIP	M	value list = "2,3,4,6"
PRODC	M	value list = "1,2,3,4,5,6,7,8,14,15,17,21,22"
VERCLR	M	unit = "m" decimal digits = "2"
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
hunits	O	value list = "1,2,3,4,5,6"
wtwdis	O	decimal digits = "3"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"

acronym	usage	constraints
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
vrlev	O	
vrval	O	unit = "m" decimal digits = "2"
locmrn	C	
refgag	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Pontoon
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Acronym: ponton

Code: 17021

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A floating structure, usually rectangular in shape which serves as landing, pier head or bridge support. (IHO Dictionary, S-32, 5th Edition, 3947)

Attribute Bindings:

acronym	usage	constraints
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"

acronym	usage	constraints
CATTEV	O	value list = "4,5,6"
STATUS	O	value list = "5"

Feature	Port area
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Acronym: prtare

Code: 17059

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Apart from harbours a port includes a city or borough with accommodation and facilities for landing passengers and goods and some amount of overseas trade. A port may possess a harbour but a harbour is not necessarily a port (International Maritime Dictionary, 2d. Edition).

Attribute Bindings:

acronym	usage	constraints
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
locmrn	C	
STATUS	O	value list = "5"

Feature	Radio calling-in point
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Acronym: rdocal

Code: 17017

Type: G

Primitive: P,L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Also called radio reporting points, they have been established in certain busy waterways and port approaches to assist traffic control. On passing these points or crossing a defined line vessels are required to report on VHF to a Traffic Control Centre. (adapted from IHO Chart Specifications, M-4)

Attribute Bindings:

acronym	usage	constraints
catcom	M	value list = "1,2,3,4,5,6,7,8"
COMCHA	M	format = "[XXXX];[XXXX];....]"
ORIENT	M	unit = "deg" decimal digits = "2"
TRAFIC	M	value list = "1,2,3,4"
OBJNAM	O	
NOBJNM	O	
unlocd	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	C	
DATSTA	O	format = "ccymmdd"
DATEND	O	format = "ccymmdd"
PERSTA	O	format = "ccymmdd"
PEREND	O	format = "ccymmdd"
SORDAT	C	format = "ccymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

acronym	usage	constraints
locmrn	C	
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Refuse dump
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Acronym: refdmp

Code: 17062

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: At a refuse dump ships are able to unload their refuse like waste oil or black water.

Attribute Bindings:

acronym	usage	constraints
catrfd	O	value list = "1,2,3,4"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
locmrn	C	
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Restricted area
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Acronym: resare

Code: 17005

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A specified area designated by an appropriate authority within which navigation is restricted in accordance with certain specified conditions. (adapted from IHO Dictionary, S-32, 5th Edition, 4366)

Attribute Bindings:

acronym	usage	constraints
CATREA	O	value list = "1,4,5,6,7,8,9,10,12,14,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33"
restrn	M	value list = "1,2,7,8,13,14,27,28,29,30,31,32,33,34,35,36,37,38,40,41"
NATSUR	O	value list = "1,2,3,4,5,6,7,8,9,11,14,17,18"
OBJNAM	O	
NOBJNM	O	
INFORM	C	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	C	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Sensor
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Acronym: sensor

Code: 18004

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2011-12-19

Definition: A device that responds to a physical stimulus (as heat, light, sound, pressure, magnetism or a particular motion) and transmits a resulting impulse (as for measurement or operating a control).

Attribute Bindings:

acronym	usage	constraints
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
catsen	M	value list = "1,2,3"
fnctsn	M	value list = "1"
SCAMIN	M	min = "1"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
COMCHA	C	format = "[[XXXX];[XXXX];....]"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
TXTDSC	O	
PICREP	O	
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Shoreline Construction
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Acronym: slcons

Code: 17032

Type: G

Primitive: L,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A fixed (not afloat) artificial structure between the water and the land, i.e. a man-made coastline.

Attribute Bindings:

acronym	usage	constraints
catslc	M	value list = "2,7,8,9,18,19,20"
NATCON	O	value list = "1,2,3,4,5,6,7,8,9"
watlev	C	value list = "1,2,3,4,8,9"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Signal station, traffic
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Acronym: sistat

Code: 17007

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A signal station is a place on shore from which signals are made to ships at sea. (IHO Dictionary, S-32, 5th Edition, 4742). Traffic signal stations regulate the movement of traffic. (IHO Chart Specifications, M-4)

Attribute Bindings:

acronym	usage	constraints
catsit	M	value list = "2,6,8,10"
dirimp	O	value list = "1,2,3,4"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
unlocd	C	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
locmrn	C	
NTXTDS	O	

acronym	usage	constraints
STATUS	O	value list = "5"

Feature	Signal station, warning
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Acronym: sistaw

Code: 17008

Type: G

Primitive: P

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A signal station is a place on shore from which signals are made to ships at sea. (IHO Dictionary, S-32, 5th Edition, 4742)

Attribute Bindings:

acronym	usage	constraints
catsiw	M	value list = "1,15,16,18"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	C	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

Feature	Sounding datum
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Acronym: m_sdat

Code: 17022

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area of uniform sounding datum.

Attribute Bindings:

acronym	usage	constraints
verdat	M	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Terminal
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Acronym: termnl

Code: 17064

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A terminal covers that area on shore which provides buildings and constructions for the transfer of cargo or passengers from and to ships.

Attribute Bindings:

acronym	usage	constraints
cathaf	M	value list = "1,3,7,8,10,11"
trshgd	O	value list = "1,2,3,4,5,6,7,8,9,10"
unlocd	C	
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"

acronym	usage	constraints
locmrn	C	

Feature	Time Schedule - in general
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Acronym: tisdge Code: 17068

Type: O

Primitive: N

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A schedule listing events and the times at which they will take place (www.wordiq.com/dictionary).

Attribute Bindings:

acronym	usage	constraints
aptref	O	
cattab	M	value list = "1,2"
dirimp	O	value list = "1,2,3,4"
schref	M	
shptyp	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15"
useshp	M	value list = "1,2,3"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"

Feature	Turning basin
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Acronym: trnbsn

Code: 17065

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area of water or enlargement of a channel used for turning vessels (International Maritime Dictionary, 2d Edition).

Attribute Bindings:

acronym	usage	constraints
HORCLR	O	unit = "m" decimal digits = "2"
unlocd	C	
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
STATUS	O	value list = "5"
locmrn	C	

Feature	Underwater rock / awash rock
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Acronym: uwtroc

Code: 17033

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A concreted mass of stony material or coral which dries, is awash or is below the water surface.

Attribute Bindings:

acronym	usage	constraints
VALSOU	M	unit = "m" decimal digits = "2"
watlev	M	value list = "1,2,3,4,8,9"
NATSUR	O	value list = "5,9,11,14,18"
QUASOU	O	value list = "1,2,8,10,11"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
NTXTDS	O	
EXPSOU	C	value list = "1,2,3"
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Feature	Vehicle transfer
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Acronym: vehtrf

Code: 17069

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A place where vehicles can be loaded or unloaded from the inland vessel with onboard or on-shore facilities.

Attribute Bindings:

acronym	usage	constraints
catvtr	M	value list = "1,2,3,4,5,6"
HEIGHT	M	unit = "m" decimal digits = "2"
unlocd	C	
verdat	O	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
CONDTN	O	value list = "1,2,3,5"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
vcrlev	O	

acronym	usage	constraints
vcrval	O	unit = "m" decimal digits = "2"
NTXTDS	O	
STATUS	O	value list = "5"
locmrn	C	

Feature	Vertical datum
---------	----------------

Acronym: m_vdat

Code: 17023

Type: M

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area of uniform vertical datum.

Attribute Bindings:

acronym	usage	constraints
verdat	M	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."

Feature	Waterway area
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Acronym: wtware

Code: 17066

Type: G

Primitive: A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: An area in which uniform general information of the waterway exists.

Attribute Bindings:

acronym	usage	constraints
catccl	M	value list = "1,2,3,4,5,6,7,8,9,10,11,12"
dirimp	M	value list = "1,2,3,4"
unlocd	C	
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"
NTXTDS	O	
locmrn	C	

Feature	Waterway axis
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Acronym: wtwaxs

Code: 17051

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The waterway axis can be defined by e.g: - the middle line of a fairway, (Definition of fairway: That part of a river, harbour; etc. where the main navigable channel for vessels of larger size lies. It is also the usual course followed by vessels entering or leaving harbours, called 'ship channel'. (International Maritime Dictionary, 2nd Ed.). - the middle line of a waterway (Definition of waterway: The waterway covers the entire area of a river or canal).

Attribute Bindings:

acronym	usage	constraints
catccl	O	value list = "1,2,3,4,5,6,7,8,9,10,11,12"
OBJNAM	M	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"
NTXTDS	O	

Feature	Waterway gauge
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Acronym: wtwgag

Code: 17067

Type: G

Primitive: P,A

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A waterway gauge is an instrument for measuring water levels

Attribute Bindings:

acronym	usage	constraints
catgag	O	value list = "1,2,3,4,5"
disipd	O	unit = "m" decimal digits = "1"
disipu	O	unit = "m" decimal digits = "1"
ELEVAT	O	unit = "m" decimal digits = "2"
higwat	O	unit = "m" decimal digits = "2"
hignam	O	
lowwat	O	unit = "m" decimal digits = "2"
lownam	O	
meawat	O	unit = "m" decimal digits = "2"
meanam	O	
othwat	O	unit = "m" decimal digits = "2"
othnam	O	
reflev	O	value list = "1,2,3,4,5,6,7,8,9,10"
sdrlev	O	
unlocd	C	
vcrlev	O	
verdat	O	value list = "12,23,24,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
hunits	O	value list = "1,2,3,4,5,6"
wtwdis	O	decimal digits = "3"
CONDTN	O	value list = "1,2,3,5"

acronym	usage	constraints
OBJNAM	C	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"
sdrval	O	unit = "m" decimal digits = "2"
vcrval	O	unit = "m" decimal digits = "2"
NTXTDS	O	
locmrn	C	

Feature	Waterway profile
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Acronym: wtwprf

Code: 17052

Type: G

Primitive: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: A waterway profile is an imaginary (i.e., physically non-existent) line across the waterway.

Attribute Bindings:

acronym	usage	constraints
hunits	M	value list = "1,2,3,5,6"
wtwdis	M	decimal digits = "3"
HEIGHT	C	unit = "m" decimal digits = "2"
reflev	C	value list = "1,2,3,4,5,6,7,8,9,10"
verdat	C	value list = "12,23,24,30,31,32,33,34,35,36,37,38,39,40,41,42,43,45"
OBJNAM	O	
NOBJNM	O	
INFORM	O	
NINFOM	O	
SCAMIN	M	min = "1"
PICREP	O	
TXTDSC	O	
DATSTA	O	format = "ccyymmdd"
DATEND	O	format = "ccyymmdd"
PERSTA	O	format = "ccyymmdd"
PEREND	O	format = "ccyymmdd"
SORDAT	C	format = "ccyymmdd"
SORIND	C	format = "cc,cc,cccc,c..."
STATUS	O	value list = "5"
sdrlev	O	

acronym	usage	constraints
sdrval	O	unit = "m" decimal digits = "2"
NTXTDS	O	
HORACC	O	unit = "m" decimal digits = "2"
VERACC	O	unit = "m" decimal digits = "2"
CATTEV	O	value list = "4,5,6"

Attribute	Beacon shape
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Acronym: BCNSHP Code: 2

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: BCNSHP_1 Date accepted: 2000-11-01 Name: stake, pole, perch, post Definition: an elongated wood or metal pole, embedded in the bottom to serve as a navigational aid or a support for a navigational aid. (adapted from IHO Dictionary S-32, 5th Edition, 4960)
2	DD Name: HYDRO Code: BCNSHP_2 Date accepted: 2000-11-01 Name: withy Definition: a tree without roots stuck or spoiled into the bottom of the sea to serve as a navigational aid.
3	DD Name: HYDRO Code: BCNSHP_3 Date accepted: 2000-11-01 Name: beacon tower Definition: a solid structure of the order of 10 metres in height used as a navigational aid.
4	DD Name: HYDRO Code: BCNSHP_4 Date accepted: 2000-11-01 Name: lattice beacon Definition: a structure consisting of strips of metal or wood crossed or interlaced to form a structure to serve as an aid to navigation or as a support for an aid to navigation.
5	DD Name: HYDRO Code: BCNSHP_5 Date accepted: 2000-11-01 Name: pile beacon Definition: a long heavy timber(s) or section(s) of steel, wood, concrete, etc., forced into the seabed to serve as an aid to navigation or as a support for an aid to navigation. (Adapted from IHO Dictionary, S-32, 5th Edition, 3840 and Navigation Dictionary, US National Oceanic and Atmospheric Administration - NOAA, 1969)

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO	Code: BCNSHP_6	Date accepted: 2000-11-01
	Name: cairn		
	Definition: a mound of stones, usually conical or pyramidal, raised specifically for maritime navigation. (adapted from IHO Dictionary, S-32, 5th Edition, 601).		
7	DD Name: HYDRO	Code: BCNSHP_7	Date accepted: 2000-11-01
	Name: buoyant beacon		
	Definition: a tall spar-like beacon fitted with a permanently submerged buoyancy chamber, the lower end of the body is secured to seabed sinker either by a flexible joint or by a cable under tension. (IHO Specifications, M-4, 459.1)		

Attribute	Buoy shape
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Acronym: BOYSHP Code: 4

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: BOYSHP_1 Date accepted: 2000-11-01 Name: conical (nun, ogival) Definition: the upper part of the body above the water-line, or the greater part of the superstructure, has approximately the shape or the appearance of a pointed cone with the point upwards.
2	DD Name: HYDRO Code: BOYSHP_2 Date accepted: 2000-11-01 Name: can (cylindrical) Definition: the upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a cylinder, or a truncated cone that approximates to a cylinder, with a flat end uppermost.
3	DD Name: HYDRO Code: BOYSHP_3 Date accepted: 2000-11-01 Name: spherical Definition: the upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a part of a sphere.
4	DD Name: HYDRO Code: BOYSHP_4 Date accepted: 2000-11-01 Name: pillar Definition: the upper part of the body above the water-line, or the greater part of the superstructure is a narrow vertical structure, pillar or lattice tower.
5	DD Name: HYDRO Code: BOYSHP_5 Date accepted: 2000-11-01 Name: spar (spindle) Definition: the upper part of the body above the water-line, or the greater part of the superstructure, has the form of a pole, or of a very long cylinder, floating upright.

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO	Code: BOYSHP_6	Date accepted: 2000-11-01
	Name: barrel (tun)		
	Definition: the upper part of the body above the water-line, or the greater part of the superstructure, has the form of a barrel or cylinder floating horizontally.		
7	DD Name: HYDRO	Code: BOYSHP_7	Date accepted: 2000-11-01
	Name: super-buoy		
	Definition: a very large buoy, generally more than 5m in diameter		
8	DD Name: HYDRO	Code: BOYSHP_8	Date accepted: 2000-11-01
	Name: ice buoy		
	Definition: a specially constructed shuttle shaped buoy which is used in ice conditions.		

Attribute	Buried depth
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Acronym: BURDEP Code: 5
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
Definition: The depth below the sea bed to which an object is buried.

Attribute	Category of airport/airfield
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Acronym: CATAIR Code: 7

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | <p>DD Name: HYDRO Code: CATAIR_1 Date accepted: 2000-11-01</p> <p>Name: military aeroplane airport</p> <p>Definition: a large military airfield usually equipped with a control tower, hangars and accommodation for the receiving and discharging of passengers or cargo. (adapted from The Macquarie Dictionary, 1988)</p> |
| 2 | <p>DD Name: HYDRO Code: CATAIR_2 Date accepted: 2000-11-01</p> <p>Name: civil aeroplane airport</p> <p>Definition: a large airfield usually equipped with a control tower, hangars and accommodation for the receiving and discharging of passengers or cargo. (The Macquarie Dictionary, 1988)</p> |
| 4 | <p>DD Name: HYDRO Code: CATAIR_4 Date accepted: 2000-11-01</p> <p>Name: civil heliport</p> <p>Definition: a landing place for helicopters, often the roof of a building. (The Macquarie Dictionary, 1988)</p> |
| 6 | <p>DD Name: HYDRO Code: CATAIR_6 Date accepted: 2000-11-01</p> <p>Name: small planes airfield</p> <p>Definition: an area of land set aside for the take-off and landing of small aeroplanes.</p> |

Attribute	Category of bridge
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Acronym: CATBRG Code: 9

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATBRG_1 Date accepted: 2000-11-01

Name: fixed bridge

Definition: a bridge having permanent horizontal and vertical alignment. (McGraw-Hill Dictionary of Scientific and Technical Terms, 3rd Edition, 1984)

3 DD Name: HYDRO Code: CATBRG_3 Date accepted: 2000-11-01

Name: swing bridge

Definition: a movable bridge (or span thereof) which rotates in a horizontal plane about a vertical pivot to allow the passage of vessels. (adapted from McGraw-Hill Encyclopaedia of Science and Technology, 7th Edition, 1992)

4 DD Name: HYDRO Code: CATBRG_4 Date accepted: 2000-11-01

Name: lifting bridge

Definition: a movable bridge (or span thereof) which is capable of being lifted vertically to allow vessels to pass beneath. (adapted from IHO Dictionary, S-32, 5th Edition, 547)

5 DD Name: HYDRO Code: CATBRG_5 Date accepted: 2000-11-01

Name: bascule bridge

Definition: a counterpoise bridge rotated in a vertical plane about an axis at one or both ends. Also called a balance. (IHO Dictionary, S-32, 5th Edition, 545)

7 DD Name: HYDRO Code: CATBRG_7 Date accepted: 2000-11-01

Name: draw bridge

Definition: a general name for bridges of which part or the entire span of the bridge may be raised or

Value	Data Dictionary (DD) Reference		
	drawn aside to allow ships to pass through. IHO Dictionary, S-32, 5th Edition, 546)		
9	DD Name: HYDRO	Code: CATBRG_9	Date accepted: 2000-11-01
	Name: footbridge		
	Definition: a bridge structure used only for pedestrian traffic. (McGraw-Hill Dictionary of Scientific and Technical Terms, 3rd Edition, 1984)		
11	DD Name: HYDRO	Code: CATBRG_11	Date accepted: 2000-11-01
	Name: aqueduct		
	Definition: a bridge supporting an artificially elevated channel, for the conveyance of water. (adapted from The New Shorter English Dictionary, 1993)		
12	DD Name: HYDRO	Code: CATBRG_12	Date accepted: 2000-11-01
	Name: suspension bridge		
	Definition: a fixed bridge consisting of either a roadway or a truss suspended from two or more cables which pass over towers and are anchored by backstays to a firm foundation. (McGraw-Hill Encyclopedia of Science and Technology, 7th Edition, 1992)		
13	DD Name: IENC	Code: CATBRG_13	Date accepted: 2000-11-01
	Name: bridge arch		
	Definition: an arched bridge with known clearances		

Attribute	Category of built-up area
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Acronym: CATBUA Code: 10

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATBUA_1 Date accepted: 2000-11-01 Name: urban area Definition: an area predominantly occupied by man-made structures used for residential, commercial, and industrial purposes. (Nautical Chart Manual, US Department of Commerce, 1992)
2	DD Name: HYDRO Code: CATBUA_2 Date accepted: 2000-11-01 Name: settlement Definition: a small collection of dwellings in a remote area.
3	DD Name: HYDRO Code: CATBUA_3 Date accepted: 2000-11-01 Name: village Definition: a collection of houses in a rural district, usually smaller than a town.
4	DD Name: HYDRO Code: CATBUA_4 Date accepted: 2000-11-01 Name: town Definition: any considerable collection of dwellings and other buildings larger than a village, but not incorporated as a city.
5	DD Name: HYDRO Code: CATBUA_5 Date accepted: 2000-11-01 Name: city Definition: a major town inhabited by a large permanent community with all essential services.

Attribute	Category of cable
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Acronym: CATCBL Code: 11

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATCBL_1 Date accepted: 2000-11-01

Name: power line

Definition: a cable used for the supply of electricity.

3 DD Name: HYDRO Code: CATCBL_3 Date accepted: 2000-11-01

Name: transmission line

Definition: multiple un-insulated cables usually supported by steel lattice towers. Such features are generally more prominent than normal power lines.

4 DD Name: HYDRO Code: CATCBL_4 Date accepted: 2000-11-01

Name: telephone

Definition: a cable used for the transmission of telephone signals.

5 DD Name: HYDRO Code: CATCBL_5 Date accepted: 2000-11-01

Name: telegraph

Definition: a cable used for the transmission of telegraph signals.

6 DD Name: HYDRO Code: CATCBL_6 Date accepted: 2000-11-01

Name: mooring cable/chain

Definition: a cable or chain used to secure a mooring buoy or other floating structure.

Attribute	Category of cardinal mark
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Acronym: CATCAM Code: 13

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATCAM_1 Date accepted: 2000-11-01

Name: north cardinal mark

Definition: Quadrant bounded by the true bearing NW-NE taken from the point of interest it should be passed to the north side of the mark.

2 DD Name: HYDRO Code: CATCAM_2 Date accepted: 2000-11-01

Name: east cardinal mark

Definition: Quadrant bounded by the true bearing NE-SE taken from the point of interest it should be passed to the east side of the mark.

3 DD Name: HYDRO Code: CATCAM_3 Date accepted: 2000-11-01

Name: south cardinal mark

Definition: Quadrant bounded by the true bearing SE-SW taken from the point of interest it should be passed to the south side of the mark.

4 DD Name: HYDRO Code: CATCAM_4 Date accepted: 2000-11-01

Name: west cardinal mark

Definition: Quadrant bounded by the true bearing SW-NW taken from the point of interest it should be passed to the west side of the mark.

Attribute	Category of coastline
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Acronym: CATCOA Code: 15

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATCOA_1 Date accepted: 2000-11-01 Name: steep coast Definition: a coast backed by rock or earth cliffs, gives a good radar return and is useful for visual identification from a considerable distance off, where cliffs alternate with low lying coast along the shoreline. (IHO Chart Specifications, M-4)
2	DD Name: HYDRO Code: CATCOA_2 Date accepted: 2000-11-01 Name: flat coast Definition: a level coast with no obvious topographic features.
3	DD Name: HYDRO Code: CATCOA_3 Date accepted: 2000-11-01 Name: sandy shore Definition: a shoreline area made up of sand, i.e. loose material consisting of small but easily distinguishable, separate grains, between 0.0625 and 2.000 millimetres in diameter. (adapted from IHO Dictionary, S-32, 5th Edition, 4497)
4	DD Name: HYDRO Code: CATCOA_4 Date accepted: 2000-11-01 Name: stony shore Definition: a shoreline area made up of rock and rock fragments ranging in size from pebbles and gravel to boulders or large rock masses. (adapted from IHO Dictionary, S-32, 5th Edition, 5059)
5	DD Name: HYDRO Code: CATCOA_5 Date accepted: 2000-11-01 Name: shingly shore Definition: a shoreline area made up of rounded, often flat waterworn rock fragments larger than

Value	Data Dictionary (DD) Reference		
	approximately 16 millimetres. (adapted from IHO Dictionary, S-32, 5th Edition, 4683)		
6	DD Name: HYDRO	Code: CATCOA_6	Date accepted: 2000-11-01
	Name: glacier (seaward end)		
	Definition: projecting seaward extension of glacier, usually afloat. Also called glacier tongue. (IHO Hydrographic Dictionary, S-32, 5th Edition, 2043)		
7	DD Name: HYDRO	Code: CATCOA_7	Date accepted: 2000-11-01
	Name: mangrove		
	Definition: one of several genera of tropical trees or shrubs which produce many prop roots and grow along low lying coasts into shallow water. (IHO Hydrographic Dictionary, S-32, 5th Edition, 3064)		
8	DD Name: HYDRO	Code: CATCOA_8	Date accepted: 2000-11-01
	Name: marshy shore		
	Definition: a shoreline area made up of spongy land saturated with water. It may have a shallow covering of water, usually with a considerable amount of vegetation appearing above the surface. (adapted from IHO Dictionary, S-32, 5th Edition, 5240)		
9	DD Name: HYDRO	Code: CATCOA_9	Date accepted: 2000-11-01
	Name: coral reef		
	Definition: a reef, often of large extent, composed chiefly of coral and its derivatives. (IHO Dictionary, S-32, 5th Edition, 1063)		
10	DD Name: HYDRO	Code: CATCOA_10	Date accepted: 2000-11-01
	Name: ice coast		
	Definition: a vertical cliff forming the seaward edge of an ice shelf, ranging in height from 2m to 50m or more above sea level.		
11	DD Name: HYDRO	Code: CATCOA_11	Date accepted: 2000-11-01
	Name: shelly shore		
	Definition: a shoreline area made up of shells i.e. made up of the hard outside covering of marine animals.		

Attribute	Category of conveyor
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Acronym: CATCON Code: 17
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATCON_2 Date accepted: 2000-11-01

Name: belt conveyor

Definition: a conveyor along which material or people are transported by means of a moving belt.

1 DD Name: HYDRO Code: CATCON_1 Date accepted: 2000-11-01

Name: aerial cableway (telepheric)

Definition: a transportation system consisting of load cables strung between pylons on which carrier units (for example: cars or buckets intended to transport people, material, and/or equipment) are suspended.

Attribute	Category of coverage
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Acronym: CATCOV Code: 18
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATCOV_1 Date accepted: 2000-11-01

Name: coverage available

Definition: continuous coverage of spatial objects is available within this area.

2 DD Name: HYDRO Code: CATCOV_2 Date accepted: 2000-11-01

Name: no coverage available

Definition: an area containing no spatial objects.

Attribute	Category of crane
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Acronym: CATCRN Code: 19

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATCRN_2 Date accepted: 2000-11-01

Name: container crane/gantry

Definition: a high speed, shore-based crane used in the lift-on/lift-off operation of specially constructed containers. (adapted from Nautical Chart Manual, US Department of Commerce, Coast and Geodetic Survey, 7th Edition)

3 DD Name: HYDRO Code: CATCRN_3 Date accepted: 2000-11-01

Name: sheerlegs

Definition: a tripodal structure used in dockyards and harbours for stepping masts or lifting loads in to and out of vessels.

4 DD Name: HYDRO Code: CATCRN_4 Date accepted: 2000-11-01

Name: travelling crane

Definition: a crane mounted on rails (track) that can move (usually parallel to the wharf face) in order to load and unload cargo vessels. (Canadian Hydrographic Service)

5 DD Name: HYDRO Code: CATCRN_5 Date accepted: 2000-11-01

Name: A-frame

Definition: a type of crane shaped like the letter "A". They are often positioned on river banks or the coastline and are used for lifting logs from logging trucks and depositing them in the water. (Canadian Hydrographic Service)

Attribute	Category of dam
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Acronym: CATDAM Code: 20

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATDAM_1 Date accepted: 2000-11-01

Name: weir

Definition: a dam erected across a river to raise the level of the water. A fence of stakes set in a river or along the shore as a trap for fish.

2 DD Name: HYDRO Code: CATDAM_2 Date accepted: 2000-11-01

Name: dam

Definition: a barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding. (IHO Dictionary, S-32, 5th Edition, 1196)

3 DD Name: HYDRO Code: CATDAM_3 Date accepted: 2000-11-01

Name: flood barrage

Definition: an opening dam across a channel which, when required, is closed to control flood waters. (IHO Chart Specifications, M-4 326.7)

Attribute	Category of distance mark
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Acronym: CATDIS Code: 21

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | <p>DD Name: HYDRO Code: CATDIS_1 Date accepted: 2000-11-01</p> <p>Name: distance mark not physically installed</p> <p>Definition: a point at which a distance from an origin along a feature is given for information, but at which no specific marker exists.</p> |
| 2 | <p>DD Name: HYDRO Code: CATDIS_2 Date accepted: 2000-11-01</p> <p>Name: visible mark, pole</p> <p>Definition: a point at which a distance from an origin along a feature is given for information and which is marked by a pole.</p> |
| 3 | <p>DD Name: HYDRO Code: CATDIS_3 Date accepted: 2000-11-01</p> <p>Name: visible mark, board</p> <p>Definition: a point at which a distance from an origin along a feature is given for information and which is marked by a board.</p> |
| 4 | <p>DD Name: HYDRO Code: CATDIS_4 Date accepted: 2000-11-01</p> <p>Name: visible mark, unknown shape</p> <p>Definition: a point at which a distance from an origin along a feature is given for information and which is physically marked, but the shape of the mark is not known or not given.</p> |

Attribute	Category of dumping ground
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Acronym: CATDPG Code: 23

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-09-08

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATDPG_2 Date accepted: 2010-09-08

Name: chemical waste dumping ground

Definition: an area at sea where chemical waste is dumped.

4 DD Name: HYDRO Code: CATDPG_4 Date accepted: 2010-09-08

Name: explosives dumping ground

Definition: an area at sea where explosives are dumped.

5 DD Name: HYDRO Code: CATDPG_5 Date accepted: 2010-09-08

Name: spoil ground

Definition: an area at sea where dredged material is deposited. Also called dumping ground. (IHO Dictionary, S-32, 5th Edition, 4930)

Attribute	Category of fence/wall
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Acronym: CATFNC Code: 24
 Use Type: F
 Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATFNC_1 Date accepted: 2000-11-01

Name: fence

Definition: a man made barrier used as an enclosure or boundary or for protection. (Digital Geographic Information Working Group -DGIWG, Oct. 1987)

4 DD Name: HYDRO Code: CATFNC_4 Date accepted: 2000-11-01

Name: wall

Definition: a fence constructed from masonry or stone.

Attribute	Category of ferry
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Acronym: CATFRY Code: 25
 Use Type: F
 Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | DD Name: HYDRO Code: CATFRY_1 Date accepted: 2000-11-01
Name: 'free-moving' ferry
Definition: a ferry which may have routes that vary with weather, tide and traffic. (adapted from M-4) |
| 2 | DD Name: HYDRO Code: CATFRY_2 Date accepted: 2000-11-01
Name: cable ferry
Definition: a ferry that follows a fixed route guided by a cable. (adapted from IHO Specifications, M-4) |

Attribute	Category of fishing facility
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Acronym: CATFIF Code: 26

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of fishing facility

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATFIF_1 Date accepted: 2000-11-01 Name: fishing stake Definition: a pole or stake placed in shallow water to outline a fishing ground or to catch fish
2	DD Name: HYDRO Code: CATFIF_2 Date accepted: 2000-11-01 Name: fish trap Definition: a structure (usually portable) for catching fish
3	DD Name: HYDRO Code: CATFIF_3 Date accepted: 2000-11-01 Name: fish weir Definition: a fence of stakes or stones set in a river or along the shore to trap fish
4	DD Name: HYDRO Code: CATFIF_4 Date accepted: 2000-11-01 Name: tunny net Definition: a net built at sea for catching tunny.

Attribute	Category of fog signal
-----------	------------------------

Acronym: CATFOG Code: 27

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATFOG_1 Date accepted: 2000-11-01 Name: explosive Definition: a signal produced by the firing of explosive charges. (Admiralty List of Lights and Fog Signals)
2	DD Name: HYDRO Code: CATFOG_2 Date accepted: 2000-11-01 Name: diaphone Definition: a diaphone uses compressed air and generally emits a powerful low-pitched sound, which often concludes with a brief sound of suddenly lowered pitch, termed the 'grunt'. (Admiralty List of Lights and Fog Signals)
3	DD Name: HYDRO Code: CATFOG_3 Date accepted: 2000-11-01 Name: siren Definition: a siren uses compressed air and exists in a variety of types which differ considerably in their sound and power. (Admiralty List of Lights and Fog Signals)
4	DD Name: HYDRO Code: CATFOG_4 Date accepted: 2000-11-01 Name: nautophone Definition: a horn having a diaphragm oscillated by electricity (IHO Dictionary, S-32, 5th Edition, 3371).
5	DD Name: HYDRO Code: CATFOG_5 Date accepted: 2000-11-01 Name: reed Definition: a reed uses compressed air and emits a weak, high pitched sound. (Admiralty List of Lights and Fog Signals)

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO	Code: CATFOG_6	Date accepted: 2000-11-01
	Name: tyfon		
	Definition: a diaphragm horn which operates under the influence of compressed air or steam (IHO Dictionary, S-32, 5th Edition, 5717).		
7	DD Name: HYDRO	Code: CATFOG_7	Date accepted: 2000-11-01
	Name: bell		
	Definition: a ringing sound with a short range. The apparatus may be operated automatically, by hand or by wave action. (IHO Chart Specifications, M-4, 452.5)		
8	DD Name: HYDRO	Code: CATFOG_8	Date accepted: 2000-11-01
	Name: whistle		
	Definition: a distinctive sound made by a jet of air passing through an orifice. The apparatus may be operated automatically, by hand or by air being forced up a tube by waves acting on a buoy. (IHO Chart Specifications, M-4, 452.6)		
9	DD Name: HYDRO	Code: CATFOG_9	Date accepted: 2000-11-01
	Name: gong		
	Definition: a sound produced by vibration of a disc when struck. The apparatus may be operated automatically, by hand or by wave action. (IHO Chart Specifications, M-4, 452.7)		
10	DD Name: HYDRO	Code: CATFOG_10	Date accepted: 2000-11-01
	Name: horn		
	Definition: a horn uses compressed air or electricity to vibrate a diaphragm and exists in a variety of types which differ greatly in their sound and power. (Admiralty List of Lights and Fog Signals)		

Attribute	Category of fortified structure
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Acronym: CATFOR Code: 28

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of fortified structure

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATFOR_1 Date accepted: 2000-11-01 Name: castle Definition: a large fortified building or structure
2	DD Name: HYDRO Code: CATFOR_2 Date accepted: 2000-11-01 Name: fort Definition: a fortified enclosure, building, or position able to be defended against an enemy
3	DD Name: HYDRO Code: CATFOR_3 Date accepted: 2000-11-01 Name: battery Definition: a fortified structure on which artillery is mounted
4	DD Name: HYDRO Code: CATFOR_4 Date accepted: 2000-11-01 Name: blockhouse Definition: a concrete structure strengthened to give protection against enemy fire, with apertures to allow defensive gunfire
5	DD Name: HYDRO Code: CATFOR_5 Date accepted: 2000-11-01 Name: martello tower Definition: a round fort for coastal defence
6	DD Name: HYDRO Code: CATFOR_6 Date accepted: 2000-11-01 Name: redoubt

Value Data Dictionary (DD) Reference

Definition: an outwork or fieldwork usually square or polygonal and without flanking defences.

Attribute	Category of gate
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Acronym: CATGAT Code: 29
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATGAT_2 Date accepted: 2000-11-01

Name: flood barrage gate

Definition: an opening gate used to control flood water.

4 DD Name: HYDRO Code: CATGAT_4 Date accepted: 2000-11-01

Name: lock gate

Definition: lock gates are the massive hinged doors at each end of a lock. (adapted from IHO Dictionary, S-32, 5th Edition, 2882)

Attribute	Category of harbour facility
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Acronym: CATHAF Code: 30
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

5 DD Name: HYDRO Code: CATHAF_5 Date accepted: 2000-11-01

Name: yacht harbour/marina

Definition: a harbour with facilities for small boats and yachts (IHO Dictionary, S-32, 5th Edition, 3095).

Attribute	Category of hulk
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Acronym: CATHLK Code: 31

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATHLK_1 Date accepted: 2000-11-01 Name: floating restaurant Definition: a permanently moored floating structure, such as an old ship, used as a restaurant.
2	DD Name: HYDRO Code: CATHLK_2 Date accepted: 2000-11-01 Name: historic ship Definition: a ship of historical interest permanently moored as a tourist attraction.
3	DD Name: HYDRO Code: CATHLK_3 Date accepted: 2000-11-01 Name: museum Definition: a permanently moored floating structure, such as an old ship, used as a museum.
4	DD Name: HYDRO Code: CATHLK_4 Date accepted: 2000-11-01 Name: accommodation Definition: a permanently moored floating structure, such as an old ship, used for accommodation.
5	DD Name: HYDRO Code: CATHLK_5 Date accepted: 2000-11-01 Name: floating breakwater Definition: a permanently moored floating structure, often constructed from old ships, used as a breakwater.

Attribute	Category of land region
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Acronym: CATLND Code: 34

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of land region

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATLND_2 Date accepted: 2000-11-01

Name: marsh

Definition: an area of wet, often spongy ground that is subject to frequent flooding or tidal inundations, but not considered to be continually under water. It is characterized by the growth of non woody plants and by the lack of trees

9 DD Name: HYDRO Code: CATLND_9 Date accepted: 2000-11-01

Name: agricultural land

Definition: areas used for cultivation of the soil, the breeding of livestock etc. and general farming.

11 DD Name: HYDRO Code: CATLND_11 Date accepted: 2000-11-01

Name: parkland

Definition: a piece of ground kept for ornament and/or recreation or maintained in its natural state as a public property or area.

12 DD Name: HYDRO Code: CATLND_12 Date accepted: 2000-11-01

Name: swamp

Definition: an area of spongy land saturated with water. It may have a shallow covering of water, usually with a considerable amount of vegetation appearing above the surface.

Attribute	Category of landmark
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Acronym: CATLMK Code: 35

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | DD Name: HYDRO Code: CATLMK_1 Date accepted: 2000-11-01
Name: cairn
Definition: a mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying. (IHO Dictionary, S-32, 5th Edition, 601) |
| 2 | DD Name: HYDRO Code: CATLMK_2 Date accepted: 2000-11-01
Name: cemetery
Definition: an area of land for burying the dead. |
| 3 | DD Name: HYDRO Code: CATLMK_3 Date accepted: 2000-11-01
Name: chimney
Definition: a vertical structure containing a passage or flue for discharging smoke and gases. (Digital Geographic Information Standard - DIGEST) |
| 4 | DD Name: HYDRO Code: CATLMK_4 Date accepted: 2000-11-01
Name: dish aerial
Definition: a parabolic aerial for the receipt and transmission of high frequency radio signals. (IHO Dictionary, S-32, 5th Edition, 1400) |
| 5 | DD Name: HYDRO Code: CATLMK_5 Date accepted: 2000-11-01
Name: flagstaff (flagpole)
Definition: a staff or pole on which flags are raised. (Digital Geographic Information Standard - DIGEST 1.28) |

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO	Code: CATLMK_6	Date accepted: 2000-11-01
	Name: flare stack		
	Definition: a tall structure used for burning-off waste oil or gas. (IHO Dictionary, S-32, 5th Edition, 1836). Normally showing a flame and located at refineries (IHO Chart specifications, M-4).		
7	DD Name: HYDRO	Code: CATLMK_7	Date accepted: 2000-11-01
	Name: mast		
	Definition: a straight vertical piece of timber or a hollow cylinder. (adapted from Digital Geographic Information Standard - DIGEST)		
8	DD Name: HYDRO	Code: CATLMK_8	Date accepted: 2000-11-01
	Name: wind sock		
	Definition: a tapered fabric sleeve mounted so as to catch and swing with the wind, thus indicating the wind direction. (Navigation dictionary, US National Oceanic and Atmospheric Administration - NOAA, 1969)		
9	DD Name: HYDRO	Code: CATLMK_9	Date accepted: 2000-11-01
	Name: monument		
	Definition: a structure erected or maintained as a memorial to a person or event. (Digital Geographic Information Standard - DIGEST)		
10	DD Name: HYDRO	Code: CATLMK_10	Date accepted: 2000-11-01
	Name: column (pillar)		
	Definition: a cylindrical or slightly tapering body of considerably greater length than diameter erected vertically. (Oxford English Dictionary)		
11	DD Name: HYDRO	Code: CATLMK_11	Date accepted: 2000-11-01
	Name: memorial plaque		
	Definition: a slab of metal, usually ornamented, erected as a memorial to a person or event.		
12	DD Name: HYDRO	Code: CATLMK_12	Date accepted: 2000-11-01
	Name: obelisk		
	Definition: a tapering shaft usually of stone or concrete, square or rectangular in section, with a pyramidal apex. (Adapted from Oxford English Dictionary)		
13	DD Name: HYDRO	Code: CATLMK_13	Date accepted: 2000-11-01
	Name: statue		
	Definition: a representation of a human, animal or fantasy figure in marble, bronze, etc.		
14	DD Name: HYDRO	Code: CATLMK_14	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	cross	
	Definition:	a monument, or other structure in form of a cross. (Funk and Wagnalls Dictionary)	
15	DD Name: HYDRO	Code: CATLMK_15	Date accepted: 2000-11-01
	Name:	dome	
	Definition:	a landmark comprising a hemispherical or spheroidal shaped structure (adapted from the Macquarie Dictionary).	
16	DD Name: HYDRO	Code: CATLMK_16	Date accepted: 2000-11-01
	Name:	radar scanner	
	Definition:	a device used for directing a radar beam through a search pattern (adapted from Navigation Dictionary, US National Oceanic and Atmospheric Administration - NOAA, 1969)	
17	DD Name: HYDRO	Code: CATLMK_17	Date accepted: 2000-11-01
	Name:	tower	
	Definition:	a relatively tall structure which may be used for observation, support, storage or communication etc. (Digital Geographic Information Working Group -DGIWG, Oct. 1987)	
18	DD Name: HYDRO	Code: CATLMK_18	Date accepted: 2000-11-01
	Name:	windmill	
	Definition:	a wind driven system of vanes attached to a tower like structure (excluding wind-generated power plants). (Digital Geographic Information Standard - DIGEST)	
19	DD Name: HYDRO	Code: CATLMK_19	Date accepted: 2000-11-01
	Name:	windmotor	
	Definition:	a modern structure for the use of windpower. (IHO Chart Specifications, M-4)	
20	DD Name: HYDRO	Code: CATLMK_20	Date accepted: 2000-11-01
	Name:	spire/minaret	
	Definition:	a tall conical or pyramid-shaped structure often built on the roof or tower of a building, especially a church or mosque. (adapted from The New Shorter Oxford English Dictionary, 1993)	
21	DD Name: HYDRO	Code: CATLMK_21	Date accepted: 2000-11-01
	Name:	large rock or boulder on land	
	Definition:	an isolated rocky formation or a single large stone (IHO Dictionary, S-32, 5th Edition).	

Attribute	Category of lateral mark
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Acronym: CATLAM Code: 36

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATLAM_1 Date accepted: 2000-11-01 Name: port-hand lateral mark Definition: indicates the port boundary of a navigational channel or suggested route when proceeding in the 'conventional direction of buoyage'.
2	DD Name: HYDRO Code: CATLAM_2 Date accepted: 2000-11-01 Name: starboard-hand lateral mark Definition: indicates the starboard boundary of a navigational channel or suggested route when proceeding in the 'conventional direction of buoyage'.
3	DD Name: HYDRO Code: CATLAM_3 Date accepted: 2000-11-01 Name: preferred channel to starboard lateral mark Definition: at a point where a channel divides, when proceeding in the 'conventional direction of buoyage', the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
4	DD Name: HYDRO Code: CATLAM_4 Date accepted: 2000-11-01 Name: preferred channel to port lateral mark Definition: at a point where a channel divides, when proceeding in the 'conventional direction of buoyage', the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.

Attribute	Category of lights
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Acronym: CATLIT Code: 37

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATLIT_1 Date accepted: 2000-11-01 Name: directional function Definition: a light illuminating a sector of very narrow angle and intended to mark a direction to follow. (IHO Dictionary, S-32, 5th Edition, 2778)
4	DD Name: HYDRO Code: CATLIT_4 Date accepted: 2000-11-01 Name: leading light Definition: a light associated with other lights so as to form a leading line to be followed. (adapted from IHO Dictionary, S-32, 5th Edition, 2794)
5	DD Name: HYDRO Code: CATLIT_5 Date accepted: 2000-11-01 Name: aero light Definition: an aero light is established for aeronautical navigation and may be of higher power than marine lights and visible from well offshore
6	DD Name: HYDRO Code: CATLIT_6 Date accepted: 2000-11-01 Name: air obstruction light Definition: a light marking an obstacle which constitutes a danger to air navigation
12	DD Name: HYDRO Code: CATLIT_12 Date accepted: 2000-11-01 Name: front Definition: terms used with leading lights to describe the position of the light on the lead as viewed from seaward.

Value	Data Dictionary (DD) Reference		
13	DD Name: HYDRO	Code: CATLIT_13	Date accepted: 2000-11-01
	Name: rear		
	Definition: terms used with leading lights to describe the position of the light on the lead as viewed from seaward.		
14	DD Name: HYDRO	Code: CATLIT_14	Date accepted: 2000-11-01
	Name: lower		
	Definition: terms used with leading lights to describe the position of the light on the lead as viewed from seaward.		
15	DD Name: HYDRO	Code: CATLIT_15	Date accepted: 2000-11-01
	Name: upper		
	Definition: terms used with leading lights to describe the position of the light on the lead as viewed from seaward.		

Attribute	Category of marine farm/culture
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Acronym: CATMFA Code: 38

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-08-12

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATMFA_1 Date accepted: 2010-08-12

Name: crustaceans

Definition: hard shelled animals, for example crabs or lobsters

2 DD Name: HYDRO Code: CATMFA_2 Date accepted: 2010-08-12

Name: oysters/mussels

Definition: edible bivalve molluscs

3 DD Name: HYDRO Code: CATMFA_3 Date accepted: 2010-08-12

Name: fish

Definition: vertebrate cold blooded animal with gills, living in water.

4 DD Name: HYDRO Code: CATMFA_4 Date accepted: 2010-08-12

Name: seaweed

Definition: the general name for marine plants of the Algae class which grow in long narrow ribbons.
(International Maritime Dictionary, 2nd Ed.)

Attribute	Category of military practice area
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Acronym: CATMPA Code: 39

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of military practice area

Enumerations:

Value	Data Dictionary (DD) Reference
2	DD Name: HYDRO Code: CATMPA_2 Date accepted: 2000-11-01 Name: torpedo exercise area Definition: an area within which exercises are carried out with torpedos
3	DD Name: HYDRO Code: CATMPA_3 Date accepted: 2000-11-01 Name: submarine exercise area Definition: an area within which submarine exercises are carried out.
4	DD Name: HYDRO Code: CATMPA_4 Date accepted: 2000-11-01 Name: firing danger area Definition: areas for bombing and missile exercises.
5	DD Name: HYDRO Code: CATMPA_5 Date accepted: 2000-11-01 Name: mine-laying practice area Definition: an area within which mine laying exercises are carried out.
6	DD Name: HYDRO Code: CATMPA_6 Date accepted: 2000-11-01 Name: small arms firing range Definition: an area for shooting pistols, rifles and machine guns etc. at a target.

Attribute	Category of mooring/warping facility
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Acronym: CATMOR Code: 40

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATMOR_1 Date accepted: 2000-11-01 Name: dolphin Definition: a post or group of posts, which may support a deck, used for mooring or warping a vessel. (IHO Dictionary, S-32, 5th Edition, 1433)
3	DD Name: HYDRO Code: CATMOR_3 Date accepted: 2000-11-01 Name: bollard Definition: small shaped post, mounted on a wharf or dolphin used to secure ship=s lines.
5	DD Name: HYDRO Code: CATMOR_5 Date accepted: 2000-11-01 Name: post or pile Definition: a long heavy timber or section of steel, wood, concrete, etc., forced into the seabed to serve as a mooring facility. (IHO Dictionary, S-32, 5th Edition, 3840)
7	DD Name: HYDRO Code: CATMOR_7 Date accepted: 2000-11-01 Name: mooring buoy Definition: a buoy secured to the bottom by permanent moorings with means for mooring a vessel by use of its anchor chain or mooring lines. (IHO Dictionary, S-32, 5th Edition, 575)
4	DD Name: HYDRO Code: CATMOR_4 Date accepted: 2000-11-01 Name: tie-up wall Definition: a tie-up wall is a section of wall designated for tying-up vessels awaiting transit. Bollards and mooring devices are available for both large and small ships.

Value	Data Dictionary (DD) Reference
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2	DD Name: HYDRO Code: CATMOR_2 Date accepted: 2000-11-01 Name: deviation dolphin Definition: a post or group of posts, which a vessel may swing around for compass adjustment.
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Attribute	Category of navigation line
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Acronym: CATNAV Code: 41

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATNAV_1 Date accepted: 2000-11-01

Name: clearing line

Definition: a straight line that marks the boundary between a safe and a dangerous area or that passes clear of a navigational danger. (adapted from IHO Dictionary, S-32, 5th Edition, 826)

2 DD Name: HYDRO Code: CATNAV_2 Date accepted: 2000-11-01

Name: transit line

Definition: a line passing through one or more fixed marks.

3 DD Name: HYDRO Code: CATNAV_3 Date accepted: 2000-11-01

Name: leading line bearing a recommended track

Definition: a line passing through one or more clearly defined objects, along the path of which a vessel can approach safely up to a certain distance off. (Adapted from IHO Dictionary, S-32, 5th Edition, 2696)

Attribute	Category of obstruction
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Acronym: CATOBS Code: 42

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | DD Name: HYDRO Code: CATOBS_1 Date accepted: 2000-11-01 |
| | Name: snag/stump |
| | Definition: a tree, branch or broken pile embedded in the ocean floor, river or lake bottom and not visible on the surface, forming thereby a hazard to vessels. (IHO Dictionary, S-32, 5th Edition, 4794) |
| 2 | DD Name: HYDRO Code: CATOBS_2 Date accepted: 2000-11-01 |
| | Name: wellhead |
| | Definition: a submarine structure projecting some distance above the seabed and capping a temporarily abandoned or suspended oil or gas well. (IHO Dictionary, S-32, 5th Edition, 5976) |
| 3 | DD Name: HYDRO Code: CATOBS_3 Date accepted: 2000-11-01 |
| | Name: diffuser |
| | Definition: a structure on an outfall through which liquids are discharged. The structure will usually project above the level of the outfall and can be an obstruction to navigation. |
| 4 | DD Name: HYDRO Code: CATOBS_4 Date accepted: 2000-11-01 |
| | Name: crib |
| | Definition: a permanent structure set in the water, framed with wooden beams and filled with rocks or boulders. They are used to anchor log booms or support other constructions, e.g. submerged outfalls, diffusers etc.. They may always be dry, submerged or cover and uncover. |
| 5 | DD Name: HYDRO Code: CATOBS_5 Date accepted: 2000-11-01 |
| | Name: fish haven |
| | Definition: areas established by private interests, usually sport fishermen, to simulate natural reefs and |

Value	Data Dictionary (DD) Reference		
	wrecks that attract fish. The reefs are constructed by dumping assorted junk in areas which may be of very small extent or may stretch a considerable distance along a depth contour. Also called fishery reefs.		
6	DD Name: HYDRO	Code: CATOBS_6	Date accepted: 2000-11-01
	Name: foul area		
	Definition: an area of numerous unidentified dangers to navigation. The area serves as a warning to the mariner that all dangers are not identified individually and that navigation through the area may be hazardous. Commonly used to encode areas behind danger lines on navigation charts. (adapted from IHO Dictionary, S-32, 5th Edition, 1915)		
7	DD Name: HYDRO	Code: CATOBS_7	Date accepted: 2000-11-01
	Name: foul ground		
	Definition: areas over which it is safe to navigate but which should be avoided for anchoring, taking the ground or ground fishing. (IHO Chart Specifications, M-4, 442.8)		
8	DD Name: HYDRO	Code: CATOBS_8	Date accepted: 2000-11-01
	Name: ice boom		
	Definition: floating barriers, anchored to the bottom, used to deflect the path of floating ice in order to prevent the obstruction of locks, intakes, etc., and to prevent damage to bridge piers and other structures. (Canadian Hydrographic Service, Chart specifications).		
9	DD Name: HYDRO	Code: CATOBS_9	Date accepted: 2000-11-01
	Name: ground tackle		
	Definition: equipment such as anchors, concrete blocks, chains and cables, etc., used to position floating structures such as trot and mooring buoys etc.		
10	DD Name: HYDRO	Code: CATOBS_10	Date accepted: 2000-11-01
	Name: boom		
	Definition: a floating barrier used to protect a river or harbour mouth or to create a sheltered area for storage purposes. (IHO Dictionary, S-32, 5th Edition, 505).		
11	DD Name: HYDRO	Code: CATOBS_11	Date accepted: 2011-02-23
	Name: fishing net		
	Definition: a piece of open-meshed material made of twine, cord, or something similar, used for catching fish		

Attribute	Category of offshore platform
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Acronym: CATOFP Code: 43

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of offshore platform

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATOFP_1 Date accepted: 2000-11-01 Name: oil derrick/rig Definition: a temporary mobile structure, either fixed or floating, used in the exploration stages of oil and gas fields. (IHO Dictionary, S-32, 5th Edition)
2	DD Name: HYDRO Code: CATOFP_2 Date accepted: 2000-11-01 Name: production platform Definition: a term used to indicate a permanent offshore structure equipped to control the flow of oil or gas. It does not include entirely submarine structures. (IHO Dictionary, S-32, 5th Edition, 4037)
3	DD Name: HYDRO Code: CATOFP_3 Date accepted: 2000-11-01 Name: observation/research platform Definition: a platform from which one=s surroundings or events can be observed, noted or recorded such as for scientific study. (adapted from IHO Dictionary, S-32, 5th Edition, 3493/3500)
4	DD Name: HYDRO Code: CATOFP_4 Date accepted: 2000-11-01 Name: articulated loading platform (ALP) Definition: a metal lattice tower, buoyant at one end and attached at the other by a universal joint to a concrete filled base on the sea bed. The platform may be fitted with a helicopter platform, emergency accommodation and hawser/hose retrieval. (adapted from United Kingdom Hydrographic Office CSDO 607.2 (12), May 1994)
5	DD Name: HYDRO Code: CATOFP_5 Date accepted: 2000-11-01 Name: single anchor leg mooring

Value	Data Dictionary (DD) Reference		
	Definition: a rigid frame or tube with a buoyancy device at its upper end , secured at its lower end to a universal joint on a large steel or concrete base resting on the sea bed, and at its upper end to a mooring buoy by a chain or wire. (adapted from United Kingdom Hydrographic Office CSDO 607.2 (12), May 1994)		
6	DD Name: HYDRO	Code: CATOFP_6	Date accepted: 2000-11-01
	Name: mooring tower		
	Definition: a platform secured to the sea bed and surmounted by a turntable to which ships moor. (adapted from United Kingdom Hydrographic Office CSDO 607.2 (12), May 1994)		
7	DD Name: HYDRO	Code: CATOFP_7	Date accepted: 2000-11-01
	Name: artificial island		
	Definition: a man-made structure usually built for the exploration or exploitation of marine resources, marine scientific research, tidal observations, etc. (adapted from IHO Dictionary, S-32, 5th Edition, 240)		
9	DD Name: HYDRO	Code: CATOFP_9	Date accepted: 2000-11-01
	Name: accommodation platform		
	Definition: a platform used primarily for eating, sleeping and recreation purposes.		

Attribute	Category of oil barrier
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Acronym: CATOLB Code: 44

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATOLB_1 Date accepted: 2000-11-01

Name: oil retention (high pressure pipe)

Definition: a pipe with holes from which air blows. When the air bubbles reach the surface they form a barrier which prevents the spread of oil. (Kort- og Matrikelstyrelsen, Denmark)

2 DD Name: HYDRO Code: CATOLB_2 Date accepted: 2000-11-01

Name: floating oil barrier

Definition: a floating tube shaped structure, with a curtain (2 metre) hanging under it, below the surface, which prevents the spread of oil. (Kort- og Matrikelstyrelsen, Denmark)

Attribute	Category of pilot boarding place
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Acronym: CATPIL Code: 46
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of pilot boarding place.

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATPIL_1 Date accepted: 2000-11-01

Name: boarding by pilot-cruising vessel

Definition: pilot boards from a cruising vessel.

2 DD Name: HYDRO Code: CATPIL_2 Date accepted: 2000-11-01

Name: boarding by helicopter

Definition: pilot boards by helicopter which comes out from the shore.

3 DD Name: HYDRO Code: CATPIL_3 Date accepted: 2000-11-01

Name: pilot comes out from shore

Definition: pilot boards from a vessel which comes out from the shore on request.

Attribute	Category of pipeline/pipe
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Acronym: CATPIP Code: 47

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- 2 DD Name: HYDRO Code: CATPIP_2 Date accepted: 2000-11-01
 Name: outfall pipe
 Definition: a pipe (generally a sewer or drainage pipe) discharging in to the sea or a river.
- 3 DD Name: HYDRO Code: CATPIP_3 Date accepted: 2000-11-01
 Name: intake pipe
 Definition: a pipe taking water from a river or other body of water, to drive a mill or supply a canal, waterworks, etc. (IHO Dictionary, S-32, 5th Edition, 2468)
- 4 DD Name: HYDRO Code: CATPIP_4 Date accepted: 2000-11-01
 Name: sewer
 Definition: a pipe in a sewage system for carrying water or sewage to a disposal area.
- 6 DD Name: HYDRO Code: CATPIP_6 Date accepted: 2000-11-01
 Name: supply pipe
 Definition: a pipe used for supplying of gas or liquid product.

Attribute	Category of production area
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Acronym: CATPRA Code: 48

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

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|---|--|
| 1 | DD Name: HYDRO Code: CATPRA_1 Date accepted: 2000-11-01
Name: quarry
Definition: an excavation in solid rock from which building stone, limestone, etc. is removed. |
| 2 | DD Name: HYDRO Code: CATPRA_2 Date accepted: 2000-11-01
Name: mine
Definition: an excavation in the earth for the purpose of extracting earth materials. |
| 3 | DD Name: HYDRO Code: CATPRA_3 Date accepted: 2000-11-01
Name: stockpile
Definition: a reserve stock of material, equipment or other supplies. |
| 4 | DD Name: HYDRO Code: CATPRA_4 Date accepted: 2000-11-01
Name: power station area
Definition: a stationary plant containing apparatus for large-scale conversion of some form of energy (hydraulic, steam, chemical, nuclear, etc.) into electrical energy. |
| 5 | DD Name: HYDRO Code: CATPRA_5 Date accepted: 2000-11-01
Name: refinery area
Definition: a system of process units used to convert crude petroleum into fuels, lubricants and other petroleum-derived products. |
| 6 | DD Name: HYDRO Code: CATPRA_6 Date accepted: 2000-11-01 |

Value	Data Dictionary (DD) Reference		
	Name:	timber yard	
	Definition:	a storage area for wood used for building, carpentry or joinery.	
7	DD Name: HYDRO	Code: CATPRA_7	Date accepted: 2000-11-01
	Name:	factory area	
	Definition:	a group of buildings where goods are manufactured.	
8	DD Name: HYDRO	Code: CATPRA_8	Date accepted: 2000-11-01
	Name:	tank farm	
	Definition:	an area in which a number of large-capacity storage tanks are located, generally used for crude oil or petroleum products.	
9	DD Name: HYDRO	Code: CATPRA_9	Date accepted: 2000-11-01
	Name:	wind farm	
	Definition:	an area in which numerous wind motors are located.	
10	DD Name: HYDRO	Code: CATPRA_10	Date accepted: 2000-11-01
	Name:	slag heap/spoil heap	
	Definition:	hill of refuse from a mine, industrial plant etc. on land (adapted from Concise Oxford Dictionary).	

Attribute	Category of pylon
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Acronym: CATPYL Code: 49

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATPYL_1 Date accepted: 2000-11-01 Name: power transmission pylon/pole Definition: a vertical construction consisting, for example, of a steel framework or of pre-stressed concrete, to support a power transmission cable or line. (adapted from Digital Geographic Information Standard - DIGEST FACC 1.2)
2	DD Name: HYDRO Code: CATPYL_2 Date accepted: 2000-11-01 Name: telephone/telegraph pylon/pole Definition: a pylon or pole used to support a telephone or telegraph line. (Digital Geographic Information Standard - DIGEST FACC 1.2)
3	DD Name: HYDRO Code: CATPYL_3 Date accepted: 2000-11-01 Name: aerial cableway/sky pylon Definition: a tower or pylon supporting steel cables which convey cars, buckets, or other suspended carrier units. (adapted from Digital Geographic Information Standard - DIGEST FACC 1.2)
4	DD Name: HYDRO Code: CATPYL_4 Date accepted: 2000-11-01 Name: bridge pylon/tower Definition: a tower, abutment or pylon from which a bridge deck is suspended. (adapted from Digital Geographic Information Standard - DIGEST FACC 1.2)
5	DD Name: HYDRO Code: CATPYL_5 Date accepted: 2000-11-01 Name: bridge pier Definition: a support in the form of a pillar or pier for the spans of a bridge. (adapted from Digital

Value	Data Dictionary (DD) Reference Geographic Information Standard - DIGEST FACC 1.2)
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Attribute	Category of radar station
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Acronym: CATRAS Code: 51
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATRAS_1 Date accepted: 2000-11-01

Name: radar surveillance station

Definition: a radar station established for traffic surveillance. (IHO Dictionary, S-32, 5th Edition, 4144)

Attribute	Category of radar transponder beacon
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Acronym: CATRTB Code: 52

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- 1 DD Name: HYDRO Code: CATRTB_1 Date accepted: 2000-11-01
Name: ramark, radar beacon transmitting continuously
Definition: a radar marker beacon which continuously transmits a signal appearing as a radial line on a radar screen, the line indicating the direction of the beacon. Ramarks are intended primarily for marine use. The name 'ramark' is derived from the words radar marker. (IHO Dictionary, S-32, 5th Edition, 4208)
- 2 DD Name: HYDRO Code: CATRTB_2 Date accepted: 2000-11-01
Name: racon, radar transponder beacon
Definition: a radar beacon which returns a coded signal which provides identification of the beacon, as well as range and bearing. The range and bearing are indicated by the location of the first character received on the radar screen. The name 'racon' is derived from the words radar beacon. (IHO Dictionary, S-32, 5th Edition, 4132)
- 3 DD Name: HYDRO Code: CATRTB_3 Date accepted: 2000-11-01
Name: leading racon/radar transponder beacon
Definition: a radar beacon that may be used (in conjunction with at least one other radar beacon) to indicate a leading line.

Attribute	Category of recommended track
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Acronym: CATTRK Code: 54

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATTRK_1 Date accepted: 2000-11-01

Name: based on a system of fixed marks

Definition: a straight route (known as a recommended track, range or leading line), which comprises at least two structures (usually beacons or daymarks) and/or natural features, which may carry lights and/or top-marks. The structures/features are positioned so that when observed to be in line, a vessel can follow a known bearing with safety. (adapted from International Association of Lighthouse Authorities - IALA Aids to Navigation Guide, 1990)

2 DD Name: HYDRO Code: CATTRK_2 Date accepted: 2000-11-01

Name: not based on a system of fixed marks

Definition: a route (known as a recommended track or preferred route) which is not based on a series of structures or features in line.

Attribute	Category of restricted area
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Acronym: CATREA Code: 56

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
4	DD Name: HYDRO Code: CATREA_4 Date accepted: 2000-11-01 Name: nature reserve Definition: a tract of land managed so as to preserve it's flora, fauna, physical features, etc.
5	DD Name: HYDRO Code: CATREA_5 Date accepted: 2000-11-01 Name: bird sanctuary Definition: a place where birds are bred and protected.
9	DD Name: HYDRO Code: CATREA_9 Date accepted: 2000-11-01 Name: military area Definition: an area controlled by the military in which restrictions may apply. (Hydrographic Service, Royal Australian Navy)
12	DD Name: HYDRO Code: CATREA_12 Date accepted: 2000-11-01 Name: navigational aid safety zone Definition: an area around a navigational aid which vessels are prohibited from entering.
19	DD Name: HYDRO Code: CATREA_19 Date accepted: 2000-11-01 Name: waiting area Definition: an area reserved for vessels waiting to enter a harbour.
22	DD Name: HYDRO Code: CATREA_22 Date accepted: 2000-11-01 Name: fish sanctuary

Value	Data Dictionary (DD) Reference		
	Definition:	a place where fish are protected.	
23	DD Name: HYDRO	Code: CATREA_23	Date accepted: 2000-11-01
	Name:	ecological reserve	
	Definition:	a tract of land managed so as to preserve the relation of plants and living creatures to each other and to their surroundings.	
25	DD Name: HYDRO	Code: CATREA_25	Date accepted: 2000-11-01
	Name:	swinging area	
	Definition:	an area where vessels turn. (Service Hydrographique et Océanographique de la Marine, France).	
26	DD Name: HYDRO	Code: CATREA_26	Date accepted: 2000-11-01
	Name:	water skiing area	
	Definition:	an area within which people may water ski and therefore vessel movement may be restricted.	
6	DD Name: HYDRO	Code: CATREA_6	Date accepted: 2000-11-01
	Name:	game reserve	
	Definition:	a place where wild animals or birds hunted for sport or food are kept undisturbed for private use	
7	DD Name: HYDRO	Code: CATREA_7	Date accepted: 2000-11-01
	Name:	seal sanctuary	
	Definition:	a place where seals are protected.	
8	DD Name: HYDRO	Code: CATREA_8	Date accepted: 2000-11-01
	Name:	degaussing range	
	Definition:	an area, usually about two cables diameter, within which ships' magnetic fields may be measured; sensing instruments and cables are installed on the sea bed in the range and there are cables leading from the range to a control position ashore.	
10	DD Name: HYDRO	Code: CATREA_10	Date accepted: 2000-11-01
	Name:	historic wreck area	
	Definition:	an area around certain wrecks of historical importance to protect the wrecks from unauthorized interference by diving, salvage or deposition (including anchoring).	
14	DD Name: HYDRO	Code: CATREA_14	Date accepted: 2000-11-01
	Name:	minefield	
	Definition:	an area laid and maintained with explosive mines for defence or practice purposes.	

Value	Data Dictionary (DD) Reference		
18	DD Name: HYDRO	Code: CATREA_18	Date accepted: 2000-11-01
	Name: swimming area		
	Definition: an area in which people may swim and therefore vessel movement may be restricted.		
20	DD Name: HYDRO	Code: CATREA_20	Date accepted: 2000-11-01
	Name: research area		
	Definition: an area where marine research takes place.		
21	DD Name: HYDRO	Code: CATREA_21	Date accepted: 2000-11-01
	Name: dredging area		
	Definition: an area where dredging is taking place.		
24	DD Name: HYDRO	Code: CATREA_24	Date accepted: 2000-11-01
	Name: no wake area		
	Definition: an area in which a vessels' speed must be reduced in order to reduce the size of the wake it produces.		
27	DD Name: HYDRO	Code: CATREA_27	Date accepted: 2022-11-23
	Name: environmentally sensitive area		
	Definition: a generic term which may be used to describe a wide range of areas considered sensitive for a variety of environmental reasons.		
28	DD Name: HYDRO	Code: CATREA_28	Date accepted: 2022-11-23
	Name: particularly sensitive sea area		
	Definition: an area that needs special protection through action by IMO because of its significance for regional ecological, socio-economic, or scientific reasons and because it may be vulnerable to damage by international shipping activities.		
29	DD Name: HYDRO	Code: CATREA_29	Date accepted: 2022-11-23
	Name: disengagement area		
	Definition: an area near a fairway where vessels can go to clear the way or make an about turn and possibly return to a waiting area when nautical conditions impose it.		
30	DD Name: HYDRO	Code: CATREA_30	Date accepted: 2022-11-23
	Name: port security area		
	Definition: an area in which defence, law and treaty enforcement, and counter-terrorism activities that fall within the port and maritime domain apply.		
31	DD Name: HYDRO	Code: CATREA_31	Date accepted: 2022-11-23

Value	Data Dictionary (DD) Reference		
	Name:	coral sanctuary	
	Definition:	a place where coral is protected.	
32	DD Name: HYDRO	Code: CATREA_32	Date accepted: 2022-11-23
	Name:	recreation area	
	Definition:	an area within which recreational activities regularly take place and therefore vessel movement may be restricted.	
1	DD Name: HYDRO	Code: CATREA_1	Date accepted: 2000-11-01
	Name:	offshore safety zone	
	Definition:	the area around an offshore installation within which vessels are prohibited from entering without permission; special regulations protect installations within a safety zone and vessels of all nationalities are required to respect the zone	
33	DD Name: IENC	Code: CATREA_33	Date accepted: 2017-11-06
	Name:	ship pollution emission control	
	Definition:	an area set up to regulate ship pollution emissions	

Attribute	Category of road
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Acronym: CATROD Code: 57
 Use Type: F
 Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATROD_1 Date accepted: 2000-11-01 Name: motorway Definition: a main road with separate carriageways and limited access, specially constructed and controlled for fast motor traffic.
2	DD Name: HYDRO Code: CATROD_2 Date accepted: 2000-11-01 Name: major road Definition: a hard surfaced (metalled) road; a main through route.
3	DD Name: HYDRO Code: CATROD_3 Date accepted: 2000-11-01 Name: minor road Definition: a secondary road for local traffic.
4	DD Name: HYDRO Code: CATROD_4 Date accepted: 2000-11-01 Name: track/path Definition: track - a rough path or way formed by use. path - a way or track laid down for walking or made by continual treading.

Attribute	Category of runway
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Acronym: CATRUN Code: 58
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
Definition: Category of runway

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATRUN_1 Date accepted: 2000-11-01 Name: aeroplane runway Definition: a level stretch of land where aeroplanes take off and land
2	DD Name: HYDRO Code: CATRUN_2 Date accepted: 2000-11-01 Name: helicopter landing pad Definition: a site on which helicopters may land and take off

Attribute	Category of sea area
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Acronym: CATSEA Code: 59

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

13	DD Name: HYDRO Code: CATSEA_13 Date accepted: 2000-11-01
	Name: shoal
	Definition: an offshore hazard to surface navigation that is composed of unconsolidated material.
51	DD Name: HYDRO Code: CATSEA_51 Date accepted: 2000-11-01
	Name: canal
	Definition: an artificial water course used for navigation.
53	DD Name: HYDRO Code: CATSEA_53 Date accepted: 2000-11-01
	Name: river
	Definition: a relatively large natural stream of water.
52	DD Name: HYDRO Code: CATSEA_52 Date accepted: 2000-11-01
	Name: lake
	Definition: a large body of water entirely surrounded by land.
12	DD Name: HYDRO Code: CATSEA_12 Date accepted: 2000-11-01
	Name: narrows
	Definition: a navigable narrow part of a bay, strait, river, etc.
54	DD Name: HYDRO Code: CATSEA_54 Date accepted: 2000-11-01
	Name: reach

Value	Data Dictionary (DD) Reference
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	Definition: a straight section of a river, specially a navigable river between two bends or an arm of the sea extending into the land
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5	DD Name: HYDRO	Code: CATSEA_5	Date accepted: 2000-11-01
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	Name: bay
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	Definition: an indentation in the coastline
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57	DD Name: IENC	Code: CATSEA_57	Date accepted: 2014-12-10
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	Name: chute
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	Definition: an inclined plane, sloping channel, or passage down or through which things may pass
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58	DD Name: IENC	Code: CATSEA_58	Date accepted: 2014-12-10
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	Name: backwater/slough
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	Definition: a body of water (as an inlet or tributary) that is out of the main current of a larger body
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59	DD Name: IENC	Code: CATSEA_59	Date accepted: 2014-12-10
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	Name: bend
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	Definition: a curve or change in direction of a watercourse or river
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Attribute	Category of shoreline construction
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Acronym: CATSLC Code: 60

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATSLC_1 Date accepted: 2000-11-01 Name: breakwater Definition: a structure protecting a shore area, harbour, anchorage, or basin from waves. (IHO Dictionary, S-32, 5th Edition, 542)
2	DD Name: HYDRO Code: CATSLC_2 Date accepted: 2000-11-01 Name: groyne (groin) Definition: a low artificial wall-like structure of durable material extending from the land to seaward for a particular purpose, such as to prevent coast erosion (adapted from IHO Dictionary, S-32, 5th Edition, 2525 and IHO Chart Specifications, M-4)
4	DD Name: HYDRO Code: CATSLC_4 Date accepted: 2000-11-01 Name: pier (jetty) Definition: a long, narrow structure extending into the water to afford a berthing place for vessels, to serve as a promenade, etc. (IHO Dictionary, S-32, 5th Edition, 3833)
5	DD Name: HYDRO Code: CATSLC_5 Date accepted: 2000-11-01 Name: promenade pier Definition: a pier built only for recreational purposes. (IHO Chart Specifications, M-4)
6	DD Name: HYDRO Code: CATSLC_6 Date accepted: 2000-11-01 Name: wharf (quay) Definition: a structure serving as a berthing place for vessels. (IHO Dictionary, S-32, 5th Edition, 5985)

Value	Data Dictionary (DD) Reference		
7	DD Name: HYDRO	Code: CATSLC_7	Date accepted: 2000-11-01
	Name: training wall		
	Definition: a wall or bank, often submerged, built to direct or confine the flow of a river or tidal current, or to promote a scour action. (Adapted from IHO Dictionary, S-32, 5th Edition, 5586 and IHO Chart Specifications, M-4).		
8	DD Name: HYDRO	Code: CATSLC_8	Date accepted: 2000-11-01
	Name: rip rap		
	Definition: A layer of broken rock, cobbles, boulders, or fragments of sufficient size to resist the erosive forces of flowing water and wave action. (Adapted from Marine Chart Manual, US National Oceanic and Atmospheric Administration - NOAA, 1992)		
9	DD Name: HYDRO	Code: CATSLC_9	Date accepted: 2000-11-01
	Name: revetment		
	Definition: facing of stone or other material, either permanent or temporary, placed along the edge of a stream, river or canal to stabilize the bank and to protect it from the erosive action of the stream. (Adapted from IHO Dictionary, S-32, 5th Edition, 4379)		
10	DD Name: HYDRO	Code: CATSLC_10	Date accepted: 2000-11-01
	Name: sea wall		
	Definition: an embankment or wall for protection against waves or tidal action along a shore or water front. (IHO Dictionary, S-32, 5th Edition, 4584)		
11	DD Name: HYDRO	Code: CATSLC_11	Date accepted: 2000-11-01
	Name: landing steps		
	Definition: steps at the shoreline as the connection between land and water on different levels.		
12	DD Name: HYDRO	Code: CATSLC_12	Date accepted: 2000-11-01
	Name: ramp		
	Definition: a sloping structure that can either be used, as a landing place, at variable water levels, for small vessels, landing ships, or a ferry boat, or for hauling a cradle carrying a vessel, which may include rails. (Adapted from IHO Dictionary, S-32, 5th Edition, 4209)		
13	DD Name: HYDRO	Code: CATSLC_13	Date accepted: 2000-11-01
	Name: slipway		
	Definition: the prepared and usually reinforced inclined surface on which keel- and bilge-blocks are laid for supporting a vessel under construction. (IHO Dictionary, S-32, 5th Edition, 4775)		
14	DD Name: HYDRO	Code: CATSLC_14	Date accepted: 2000-11-01
	Name: fender		

Value	Data Dictionary (DD) Reference		
	Definition: a protective structure designed to cushion the impact of a vessel and prevent damage.		
15	DD Name: HYDRO	Code: CATSLC_15	Date accepted: 2000-11-01
	Name: solid face wharf		
	Definition: a wharf consisting of a solid wall of concrete, masonry, wood etc., such that the water cannot circulate freely under the wharf. The type of construction affects ship-handling; for example, a solid face wharf may give shelter from tidal streams, but under certain circumstances a cushion of water may build up between such a wharf and a ship attempting to berth at it, causing difficulties in ship handling. (Capt. A. Rae, pilot, Port of Halifax and Mr. R. Morash, wharf building engineer, Transport Canada)		
16	DD Name: HYDRO	Code: CATSLC_16	Date accepted: 2000-11-01
	Name: open face wharf		
	Definition: a wharf supported on piles or other structures which allow free circulation of water under the wharf. (Capt. A. Rae, pilot, Port of Halifax and Mr. R. Morash, wharf building engineer, Transport Canada)		

Attribute	Category of silo/tank
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Acronym: CATSIL Code: 63

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2008-01-31

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATSIL_1 Date accepted: 2000-11-01 Name: silo in general Definition: a generally cylindrical tower used for storing fodder or grain.
2	DD Name: HYDRO Code: CATSIL_2 Date accepted: 2000-11-01 Name: tank in general Definition: a fixed structure for storing liquids. (IHO Dictionary, S-32, 5th Edition, 5290)
3	DD Name: HYDRO Code: CATSIL_3 Date accepted: 2000-11-01 Name: grain elevator Definition: a storage building for grain. Usually a tall frame, metal or concrete structure with an especially compartmented interior. (The New Encyclopaedia Britannica Micropaedia, 15th Edition).
4	DD Name: HYDRO Code: CATSIL_4 Date accepted: 2000-11-01 Name: water tower Definition: a tower with an elevated container used to hold water.

Attribute	Category of slope
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Acronym: CATSLO Code: 64

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: CATSLO_2 Date accepted: 2000-11-01

Name: embankment

Definition: an artificial elevation constructed from earth, stone, etc. carrying a road, railway or similar or serving to dam water.

3 DD Name: HYDRO Code: CATSLO_3 Date accepted: 2000-11-01

Name: dune

Definition: a mound, ridge or hill of drifted material on the sea coast or in a desert. (adapted from IHO Dictionary, S-32, 5th Edition, 1496)

6 DD Name: HYDRO Code: CATSLO_6 Date accepted: 2000-11-01

Name: cliff

Definition: land rising abruptly for a considerable distance above the water or surrounding land. (IHO Dictionary, S-32, 5th Edition, 829)

1 DD Name: HYDRO Code: CATSLO_1 Date accepted: 2000-11-01

Name: cutting

Definition: an excavation through high ground for a road, canal, etc.

Attribute	Category of small craft facility
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Acronym: CATSCF Code: 65

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATSCF_1 Date accepted: 2000-11-01 Name: visitor's berth Definition: a berth set aside for the use of visiting vessels.
2	DD Name: HYDRO Code: CATSCF_2 Date accepted: 2000-11-01 Name: nautical club Definition: a club for mariners generally associated with other small craft facilities.
3	DD Name: HYDRO Code: CATSCF_3 Date accepted: 2000-11-01 Name: boat hoist Definition: a hoist for lifting boats out of the water.
4	DD Name: HYDRO Code: CATSCF_4 Date accepted: 2000-11-01 Name: sailmaker Definition: a place where sails are made or may be taken for repair.
5	DD Name: HYDRO Code: CATSCF_5 Date accepted: 2000-11-01 Name: boatyard Definition: a place on shore where boats may be built, stored and repaired.
6	DD Name: HYDRO Code: CATSCF_6 Date accepted: 2000-11-01 Name: public inn

Value	Data Dictionary (DD) Reference		
	Definition: a public house providing food, drink and accommodation. (The Collins Reference English Dictionary, 1992)		
7	DD Name: HYDRO	Code: CATSCF_7	Date accepted: 2000-11-01
	Name: restaurant		
	Definition: a commercial establishment serving food. (The Collins Reference Dictionary, 1992)		
8	DD Name: HYDRO	Code: CATSCF_8	Date accepted: 2000-11-01
	Name: chandler		
	Definition: a dealer in ships' supplies. (The Collins Reference Dictionary, 1992)		
9	DD Name: HYDRO	Code: CATSCF_9	Date accepted: 2000-11-01
	Name: provisions		
	Definition: a place where food and other such supplies are available.		
10	DD Name: HYDRO	Code: CATSCF_10	Date accepted: 2000-11-01
	Name: doctor		
	Definition: a place where a doctor is available to provide medical attention.		
11	DD Name: HYDRO	Code: CATSCF_11	Date accepted: 2000-11-01
	Name: pharmacy		
	Definition: a place where medical drugs are dispensed.		
12	DD Name: HYDRO	Code: CATSCF_12	Date accepted: 2000-11-01
	Name: water tap		
	Definition: a place where fresh water is available.		
13	DD Name: HYDRO	Code: CATSCF_13	Date accepted: 2000-11-01
	Name: fuel station		
	Definition: a place where fuel is available.		
14	DD Name: HYDRO	Code: CATSCF_14	Date accepted: 2000-11-01
	Name: electricity		
	Definition: a place where a connection to an electrical supply is available.		
15	DD Name: HYDRO	Code: CATSCF_15	Date accepted: 2000-11-01
	Name: bottle gas		
	Definition: a place where bottled gas is available.		

Value	Data Dictionary (DD) Reference		
16	DD Name: HYDRO Name: showers Definition: a place where showers are available.	Code: CATSCF_16	Date accepted: 2000-11-01
17	DD Name: HYDRO Name: launderette Definition: a place where there are facilities for washing clothes.	Code: CATSCF_17	Date accepted: 2000-11-01
18	DD Name: HYDRO Name: public toilets Definition: a place where toilets are available for public use.	Code: CATSCF_18	Date accepted: 2000-11-01
19	DD Name: HYDRO Name: post box Definition: a place where mail may be posted.	Code: CATSCF_19	Date accepted: 2000-11-01
20	DD Name: HYDRO Name: public telephone Definition: a place where a telephone is available for public use.	Code: CATSCF_20	Date accepted: 2000-11-01
21	DD Name: HYDRO Name: refuse bin Definition: a place where refuse may be dumped.	Code: CATSCF_21	Date accepted: 2000-11-01
22	DD Name: HYDRO Name: car park Definition: a place where cars may be parked.	Code: CATSCF_22	Date accepted: 2000-11-01
23	DD Name: HYDRO Name: parking for boats and trailers Definition: a place on shore where boats and/or trailers may be parked.	Code: CATSCF_23	Date accepted: 2000-11-01
24	DD Name: HYDRO Name: caravan site Definition: a place where caravans may be parked or where caravan accommodation is provided.	Code: CATSCF_24	Date accepted: 2000-11-01
25	DD Name: HYDRO	Code: CATSCF_25	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference			
	Name:	camping site		
	Definition:	a place where visitors may pitch tents and camp.		
26	DD Name:	HYDRO	Code: CATSCF_26	Date accepted: 2000-11-01
	Name:	sewerage pump-out station		
	Definition:	a place where sewerage may be pumped off a vessel.		
27	DD Name:	HYDRO	Code: CATSCF_27	Date accepted: 2000-11-01
	Name:	emergency telephone		
	Definition:	a place where a telephone is available for emergency use only.		
28	DD Name:	HYDRO	Code: CATSCF_28	Date accepted: 2000-11-01
	Name:	landing/launching place for boats		
	Definition:	a place where boats may be landed or launched.		
29	DD Name:	HYDRO	Code: CATSCF_29	Date accepted: 2000-11-01
	Name:	visitors mooring		
	Definition:	a mooring set aside for the use of visiting vessels.		
30	DD Name:	HYDRO	Code: CATSCF_30	Date accepted: 2000-11-01
	Name:	scrubbing berth		
	Definition:	a place where vessels may berth for the purpose of careening.		
31	DD Name:	HYDRO	Code: CATSCF_31	Date accepted: 2000-11-01
	Name:	picnic area		
	Definition:	a place where people may go to eat a picnic.		
32	DD Name:	HYDRO	Code: CATSCF_32	Date accepted: 2000-11-01
	Name:	mechanics workshop		
	Definition:	a place where mechanical repairs can be undertaken to engines or other vessel equipment.		
33	DD Name:	HYDRO	Code: CATSCF_33	Date accepted: 2000-11-01
	Name:	guard and/or security service		
	Definition:	a place where a vessel is patrolled by a security service or stored in a secure lockup.		

Attribute	Category of special purpose mark
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Acronym: CATSPM Code: 66

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

9 DD Name: HYDRO Code: CATSPM_9 Date accepted: 2000-11-01

Name: ODAS (Ocean-Data-Acquisition-System)

Definition: Ocean Data Acquisition System (IHO Dictionary, S-32, 5th Edition, 5953)

15 DD Name: HYDRO Code: CATSPM_15 Date accepted: 2000-11-01

Name: LANBY (Large Automatic Navigational Buoy)

Definition: a large buoy designed to take the place of a lightship where construction of an offshore light station is not feasible. (IHO Dictionary, S-32, 5th Edition, 2656)

53 DD Name: HYDRO Code: CATSPM_53 Date accepted: 2000-11-01

Name: wellhead mark

Definition: a mark indicating a borehole that produces or is capable of producing oil or natural gas. (adapted from IHO Dictionary, S-32, 5th Edition, 5971)

56 DD Name: HYDRO Code: CATSPM_56 Date accepted: 2000-11-01

Name: artificial reef mark

Definition: a mark indicating the existence or the extent of an artificial reef.

1 DD Name: HYDRO Code: CATSPM_1 Date accepted: 2000-11-01

Name: firing danger area mark

Definition: a mark used to indicate a firing danger area, usually at sea.

2 DD Name: HYDRO Code: CATSPM_2 Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	target mark	
	Definition:	any object toward which something is directed. The distinctive marking or instrumentation of a ground point to aid its identification on a photograph.	
3	DD Name:	HYDRO Code: CATSPM_3 Date accepted:	2000-11-01
	Name:	marker ship mark	
	Definition:	a mark marking the position of a ship which is used as a target during some military exercise.	
4	DD Name:	HYDRO Code: CATSPM_4 Date accepted:	2000-11-01
	Name:	degaussing range mark	
	Definition:	a mark used to indicate a degaussing range,	
5	DD Name:	HYDRO Code: CATSPM_5 Date accepted:	2000-11-01
	Name:	barge mark	
	Definition:	a mark of relevance to barges.	
6	DD Name:	HYDRO Code: CATSPM_6 Date accepted:	2000-11-01
	Name:	cable mark	
	Definition:	a mark used to indicate the position of submarine cables or the point at which they run on land.	
7	DD Name:	HYDRO Code: CATSPM_7 Date accepted:	2000-11-01
	Name:	spoil ground mark	
	Definition:	a mark used to indicate the limit of a spoil ground.	
8	DD Name:	HYDRO Code: CATSPM_8 Date accepted:	2000-11-01
	Name:	outfall mark	
	Definition:	a mark used to indicate the position of an outfall or the point at which it leaves the land.	
10	DD Name:	HYDRO Code: CATSPM_10 Date accepted:	2000-11-01
	Name:	recording mark	
	Definition:	a mark used to record data for scientific purposes.	
11	DD Name:	HYDRO Code: CATSPM_11 Date accepted:	2000-11-01
	Name:	seaplane anchorage mark	
	Definition:	a mark used to indicate a seaplane anchorage.	
12	DD Name:	HYDRO Code: CATSPM_12 Date accepted:	2000-11-01
	Name:	recreation zone mark	

Value	Data Dictionary (DD) Reference			
	Definition: a mark used to indicate a recreation zone.			
13	DD Name: HYDRO	Code: CATSPM_13	Date accepted: 2000-11-01	
	Name: private mark			
	Definition: a privately maintained mark			
14	DD Name: HYDRO	Code: CATSPM_14	Date accepted: 2000-11-01	
	Name: mooring mark			
	Definition: a mark indicating a mooring or moorings.			
16	DD Name: HYDRO	Code: CATSPM_16	Date accepted: 2000-11-01	
	Name: leading mark			
	Definition: aids to navigation or other indicators so located as to indicate the path to be followed. Leading marks identify a leading line when they are in transit			
17	DD Name: HYDRO	Code: CATSPM_17	Date accepted: 2000-11-01	
	Name: measured distance mark			
	Definition: a mark forming part of a transit indicating one end of a measured distance.			
18	DD Name: HYDRO	Code: CATSPM_18	Date accepted: 2000-11-01	
	Name: notice mark			
	Definition: a notice board or sign indicating information to the mariner.			
19	DD Name: HYDRO	Code: CATSPM_19	Date accepted: 2000-11-01	
	Name: TSS mark			
	Definition: a mark indicating a traffic separation scheme			
20	DD Name: HYDRO	Code: CATSPM_20	Date accepted: 2000-11-01	
	Name: anchoring prohibited mark			
	Definition: a mark indicating an anchoring prohibited area.			
21	DD Name: HYDRO	Code: CATSPM_21	Date accepted: 2000-11-01	
	Name: berthing prohibited mark			
	Definition: a mark indicating that berthing is prohibited.			
22	DD Name: HYDRO	Code: CATSPM_22	Date accepted: 2000-11-01	
	Name: overtaking prohibited mark			
	Definition: a mark indicating that overtaking is prohibited.			

Value	Data Dictionary (DD) Reference		
23	DD Name: HYDRO	Code: CATSPM_23	Date accepted: 2000-11-01
	Name: two-way traffic prohibited mark		
	Definition: a mark indicating a one-way route.		
24	DD Name: HYDRO	Code: CATSPM_24	Date accepted: 2000-11-01
	Name: 'reduced wake' mark		
	Definition: a mark indicating that vessels must not generate excessive wake.		
25	DD Name: HYDRO	Code: CATSPM_25	Date accepted: 2000-11-01
	Name: speed limit mark		
	Definition: a mark indicating that a speed limit applies.		
26	DD Name: HYDRO	Code: CATSPM_26	Date accepted: 2000-11-01
	Name: stop mark		
	Definition: a mark indicating the place where the bow of a ship must stop when traffic lights show red.		
27	DD Name: HYDRO	Code: CATSPM_27	Date accepted: 2000-11-01
	Name: general warning mark		
	Definition: a mark indicating that special caution must be exercised in the vicinity of the mark.		
28	DD Name: HYDRO	Code: CATSPM_28	Date accepted: 2000-11-01
	Name: 'sound ship's siren' mark		
	Definition: a mark indicating that a ship should sound its siren or horn.		
29	DD Name: HYDRO	Code: CATSPM_29	Date accepted: 2000-11-01
	Name: restricted vertical clearance mark		
	Definition: a mark indicating the minimum vertical space available for passage.		
30	DD Name: HYDRO	Code: CATSPM_30	Date accepted: 2000-11-01
	Name: maximum vessel's draught mark		
	Definition: a mark indicating the maximum draught of vessel permitted.		
31	DD Name: HYDRO	Code: CATSPM_31	Date accepted: 2000-11-01
	Name: restricted horizontal clearance mark		
	Definition: a mark indicating the minimum horizontal space available for passage.		
32	DD Name: HYDRO	Code: CATSPM_32	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	strong current warning mark	
	Definition:	a mark warning of strong currents.	
33	DD Name:	HYDRO Code: CATSPM_33	Date accepted: 2000-11-01
	Name:	berthing permitted mark	
	Definition:	a mark indicating that berthing is allowed.	
34	DD Name:	HYDRO Code: CATSPM_34	Date accepted: 2000-11-01
	Name:	overhead power cable mark	
	Definition:	a mark indicating an overhead power cable.	
35	DD Name:	HYDRO Code: CATSPM_35	Date accepted: 2000-11-01
	Name:	'channel edge gradient' mark	
	Definition:	a mark indicating the gradient of the slope of a dredge channel edge.	
36	DD Name:	HYDRO Code: CATSPM_36	Date accepted: 2000-11-01
	Name:	telephone mark	
	Definition:	a mark indicating the presence of a telephone.	
37	DD Name:	HYDRO Code: CATSPM_37	Date accepted: 2000-11-01
	Name:	ferry crossing mark	
	Definition:	a mark indicating that a ferry route crosses the ship route; often used with a 'sound ship's siren' mark.	
39	DD Name:	HYDRO Code: CATSPM_39	Date accepted: 2000-11-01
	Name:	pipeline mark	
	Definition:	a mark used to indicate the position of submarine pipelines or the point at which they run on to the land.	
40	DD Name:	HYDRO Code: CATSPM_40	Date accepted: 2000-11-01
	Name:	anchorage mark	
	Definition:	a mark indicating an anchorage area.	
41	DD Name:	HYDRO Code: CATSPM_41	Date accepted: 2000-11-01
	Name:	clearing mark	
	Definition:	a mark used to indicate a clearing line.	
42	DD Name:	HYDRO Code: CATSPM_42	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	control mark	
	Definition:	a mark indicating the location at which a restriction or requirement exists.	
43	DD Name:	HYDRO Code: CATSPM_43	Date accepted: 2000-11-01
	Name:	diving mark	
	Definition:	a mark indicating that diving may take place in the vicinity.	
44	DD Name:	HYDRO Code: CATSPM_44	Date accepted: 2000-11-01
	Name:	refuge beacon	
	Definition:	a mark providing or indicating a place of safety.	
45	DD Name:	HYDRO Code: CATSPM_45	Date accepted: 2000-11-01
	Name:	foul ground mark	
	Definition:	a mark indicating a foul ground.	
46	DD Name:	HYDRO Code: CATSPM_46	Date accepted: 2000-11-01
	Name:	yachting mark	
	Definition:	a mark installed for use by yachtsmen.	
47	DD Name:	HYDRO Code: CATSPM_47	Date accepted: 2000-11-01
	Name:	heliport mark	
	Definition:	a mark indicating an area where helicopters may land.	
48	DD Name:	HYDRO Code: CATSPM_48	Date accepted: 2000-11-01
	Name:	GPS mark	
	Definition:	a mark indicating a location at which a GPS position has been accurately determined.	
49	DD Name:	HYDRO Code: CATSPM_49	Date accepted: 2000-11-01
	Name:	seaplane landing mark	
	Definition:	a mark indicating an area where sea-planes land.	
50	DD Name:	HYDRO Code: CATSPM_50	Date accepted: 2000-11-01
	Name:	entry prohibited mark	
	Definition:	a mark indicating that entry is prohibited.	
51	DD Name:	HYDRO Code: CATSPM_51	Date accepted: 2000-11-01
	Name:	work in progress mark	
	Definition:	a mark indicating that work (generally construction) is in progress.	

Value	Data Dictionary (DD) Reference		
52	DD Name: HYDRO	Code: CATSPM_52	Date accepted: 2000-11-01
	Name:	mark with unknown purpose	
	Definition:	a mark whose detailed characteristics are unknown.	
54	DD Name: HYDRO	Code: CATSPM_54	Date accepted: 2000-11-01
	Name:	channel separation mark	
	Definition:	a mark indicating the point at which a channel divides separately into two channels.	
55	DD Name: HYDRO	Code: CATSPM_55	Date accepted: 2000-11-01
	Name:	marine farm mark	
	Definition:	a mark indicating the existence of a fish, mussel, oyster or pearl farm/culture.	

Attribute	Category of temporal variation
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Acronym: CATTEV

Code: 19000

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2019-05-13

Definition: An assessment of the likelihood of change over time.

Enumerations:

Value Data Dictionary (DD) Reference

4 DD Name: HYDRO Code: CATTEV_4 Date accepted: 2019-05-13

Name: likely to change

Definition:

5 DD Name: HYDRO Code: CATTEV_5 Date accepted: 2019-05-13

Name: unlikely to change

Definition:

6 DD Name: HYDRO Code: CATTEV_6 Date accepted: 2019-05-13

Name: unassessed

Definition:

Attribute	Category of Tidal stream
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Acronym: CAT_TS Code: 188
 Use Type: F
 Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
 Definition: Category of Tidal stream

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: HYDRO Code: CAT_TS_1 Date accepted: 2000-11-01
Name: flood stream
Definition: the horizontal movement of water associated with the rising tide. Flood streams generally set towards the shore, or in the direction of the tide progression. Also called flood, flood current or ingoing stream. (Adapted from IHO Dictionary, S-32, 5th Edition) |
| 2 | DD Name: HYDRO Code: CAT_TS_2 Date accepted: 2000-11-01
Name: ebb stream
Definition: the horizontal movement of water associated with falling tide. Ebb streams generally set seaward, or in the opposite direction to the tide progression. Also called ebb, ebb current or outgoing stream. (Adapted from IHO Dictionary, S-32, 5th Edition) |
| 3 | DD Name: HYDRO Code: CAT_TS_3 Date accepted: 2000-11-01
Name: other tidal flow
Definition: any other horizontal movement of water associated with tides, eg. rotary flow. |

Attribute	Category of traffic separation scheme
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Acronym: CATTSS Code: 67
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-08-11

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- 1 DD Name: HYDRO Code: CATTSS_1 Date accepted: 2010-08-11
Name: IMO - adopted
Definition: a defined Traffic Separation Scheme that has been adopted as an IMO routing measure.
- 2 DD Name: HYDRO Code: CATTSS_2 Date accepted: 2010-08-11
Name: not IMO - adopted
Definition: a defined Traffic Separation Scheme that has not been adopted as an IMO routing measure.

Attribute	Category of vegetation
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Acronym: CATVEG Code: 68

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
6	DD Name: HYDRO Code: CATVEG_6 Date accepted: 2000-11-01 Name: wood in general (inc mixed wood) Definition: growing trees densely occupying a tract of land. (The Concise Oxford Dictionary)
13	DD Name: HYDRO Code: CATVEG_13 Date accepted: 2000-11-01 Name: tree in general Definition: a woody perennial plant, having a self supporting main stem or trunk.
11	DD Name: HYDRO Code: CATVEG_11 Date accepted: 2000-11-01 Name: reed Definition: any of various water or marsh plants with a firm stem

Attribute	Category of water turbulence
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Acronym: CATWAT Code: 69
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
Definition: Category of water turbulence

Enumerations:

Value Data Dictionary (DD) Reference

6 DD Name: IENC Code: CATWAT_6 Date accepted: 2014-07-11
Name: under water turbulence
Definition:

Attribute	Category of weed/kelp
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Acronym: CATWED Code: 70

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Category of weed/kelp

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CATWED_1 Date accepted: 2000-11-01

Name: kelp

Definition: a giant plant sometimes 60 metres long with no roots, it is anchored by hold-fasts or tendrils up to 10 metres long, that cling to rock. Gas filled bubbles on fronds act as floats keeping the kelp just below the surface. (Earth Sciences References, Mary McNeil)

2 DD Name: HYDRO Code: CATWED_2 Date accepted: 2000-11-01

Name: sea weed

Definition: general name for marine plants of the algae class which grow in long narrow ribbons. Also called sea grass. (International Maritime Dictionary, 2nd Edition)

3 DD Name: HYDRO Code: CATWED_3 Date accepted: 2000-11-01

Name: sea grass

Definition: any grasslike marine alga. Eelgrass is one of the best known seagrasses. (IHO Dictionary, S-32, 5th Edition, 4565)

4 DD Name: HYDRO Code: CATWED_4 Date accepted: 2000-11-01

Name: sargasso

Definition: a certain type of sea weed, or more generally, a large floating mass of this sea weed. (IHO Dictionary, S-32, 5th Edition, 4501)

Attribute	Category of wreck
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Acronym: CATWRK Code: 71

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATWRK_1 Date accepted: 2000-11-01 Name: non-dangerous wreck Definition: a wreck which is not considered to be dangerous to surface navigation.
2	DD Name: HYDRO Code: CATWRK_2 Date accepted: 2000-11-01 Name: dangerous wreck Definition: a wreck which is considered to be dangerous to surface navigation.
3	DD Name: HYDRO Code: CATWRK_3 Date accepted: 2000-11-01 Name: distributed remains of wreck Definition: (foul ground) an area over which it is safe to navigate but which should be avoided for anchoring, taking the ground or ground fishing. (IHO Chart Specifications, M-4)
4	DD Name: HYDRO Code: CATWRK_4 Date accepted: 2000-11-01 Name: wreck showing mast/masts Definition: wreck of which only the mast(s) is visible at the sounding datum indicated.
5	DD Name: HYDRO Code: CATWRK_5 Date accepted: 2000-11-01 Name: wreck showing any portion of hull or superstructure Definition: wreck of which any portion of the hull or superstructure is visible at the sounding datum indicated.

Attribute	Category of zone of confidence in data
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Acronym: CATZOC Code: 72

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CATZOC_1 Date accepted: 2000-11-01 Name: zone of confidence A1 Definition:
2	DD Name: HYDRO Code: CATZOC_2 Date accepted: 2000-11-01 Name: zone of confidence A2 Definition:
3	DD Name: HYDRO Code: CATZOC_3 Date accepted: 2000-11-01 Name: zone of confidence B Definition:
4	DD Name: HYDRO Code: CATZOC_4 Date accepted: 2000-11-01 Name: zone of confidence C Definition:
5	DD Name: HYDRO Code: CATZOC_5 Date accepted: 2000-11-01 Name: zone of confidence D Definition:
6	DD Name: HYDRO Code: CATZOC_6 Date accepted: 2000-11-01 Name: zone of confidence U (data not assessed)

Value	Data Dictionary (DD) Reference
	Definition:

Attribute	Colour
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Acronym: COLOUR Code: 75

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: COLOUR_1 Date accepted: 2000-11-01 Name: white Definition:
2	DD Name: HYDRO Code: COLOUR_2 Date accepted: 2000-11-01 Name: black Definition:
3	DD Name: HYDRO Code: COLOUR_3 Date accepted: 2000-11-01 Name: red Definition:
4	DD Name: HYDRO Code: COLOUR_4 Date accepted: 2000-11-01 Name: green Definition:
5	DD Name: HYDRO Code: COLOUR_5 Date accepted: 2000-11-01 Name: blue Definition:
6	DD Name: HYDRO Code: COLOUR_6 Date accepted: 2000-11-01 Name: yellow

Value	Data Dictionary (DD) Reference			
	Definition:			
7	DD Name: HYDRO	Code: COLOUR_7	Date accepted:	2000-11-01
	Name: grey			
	Definition:			
8	DD Name: HYDRO	Code: COLOUR_8	Date accepted:	2000-11-01
	Name: brown			
	Definition:			
9	DD Name: HYDRO	Code: COLOUR_9	Date accepted:	2000-11-01
	Name: amber			
	Definition:			
10	DD Name: HYDRO	Code: COLOUR_10	Date accepted:	2000-11-01
	Name: violet			
	Definition:			
11	DD Name: HYDRO	Code: COLOUR_11	Date accepted:	2000-11-01
	Name: orange			
	Definition:			
12	DD Name: HYDRO	Code: COLOUR_12	Date accepted:	2000-11-01
	Name: magenta			
	Definition:			
13	DD Name: HYDRO	Code: COLOUR_13	Date accepted:	2000-11-01
	Name: pink			
	Definition:			

Attribute	Colour pattern
-----------	----------------

Acronym: COLPAT

Code: 76

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: COLPAT_1 Date accepted: 2000-11-01

Name: horizontal stripes

Definition: straight bands or stripes of differing colours painted horizontally.

2 DD Name: HYDRO Code: COLPAT_2 Date accepted: 2000-11-01

Name: vertical stripes

Definition: straight bands or stripes of differing colours painted vertically.

3 DD Name: HYDRO Code: COLPAT_3 Date accepted: 2000-11-01

Name: diagonal stripes

Definition: straight bands or stripes of differing colours painted diagonally (ie not horizontally or vertically).

4 DD Name: HYDRO Code: COLPAT_4 Date accepted: 2000-11-01

Name: squared

Definition: often referred to as checker plate, where alternate colours are used to create squares similar to a chess or draught board. The pattern may be straight or diagonal.

5 DD Name: HYDRO Code: COLPAT_5 Date accepted: 2000-11-01

Name: stripes (direction unknown)

Definition: straight bands or stripes of differing colours painted in an unknown direction.

6 DD Name: HYDRO Code: COLPAT_6 Date accepted: 2000-11-01

Name: border stripe

Value Data Dictionary (DD) Reference

Definition: a band or stripe of colour which is displayed around the outer edge of the object, which may also form a border to an inner pattern or plain colour.

Attribute	Communication channel
-----------	-----------------------

Acronym: COMCHA Code: 77
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A channel number assigned to a specific radio frequency, frequencies or frequency band.

Attribute	Condition
-----------	-----------

Acronym: CONDTN Code: 81

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-11-16

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CONDTN_1 Date accepted: 2010-11-16

Name: under construction

Definition: a structure that is in the process of being built.

2 DD Name: HYDRO Code: CONDTN_2 Date accepted: 2010-11-16

Name: ruined

Definition: a structure in a decayed or deteriorated condition resulting from neglect or disuse, or a damaged structure in need of repair.(IHO Dictionary, S-32, 5th Edition, 4456)

3 DD Name: HYDRO Code: CONDTN_3 Date accepted: 2010-11-16

Name: under reclamation

Definition: an area of the sea that is being reclaimed as land, usually by the dumping of earth and other material.

4 DD Name: HYDRO Code: CONDTN_4 Date accepted: 2010-11-16

Name: wingless

Definition: a windmill or windmotor from which the turbine blades are missing.

5 DD Name: HYDRO Code: CONDTN_5 Date accepted: 2010-11-16

Name: planned construction

Definition: an area where a future construction is planned

Attribute	Conspicuous, radar
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Acronym: CONRAD Code: 82
 Use Type: F
 Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: CONRAD_1 Date accepted: 2000-11-01 Name: radar conspicuous Definition: an object which returns a strong radar echo. (IHO Dictionary, S-32, 5th Edition, 4142)
2	DD Name: HYDRO Code: CONRAD_2 Date accepted: 2000-11-01 Name: not radar conspicuous Definition: an object which does not return a particularly strong radar echo.
3	DD Name: HYDRO Code: CONRAD_3 Date accepted: 2000-11-01 Name: radar conspicuous (has radar reflector) Definition: an object which returns a strong radar echo, having a radar reflector.

Attribute	Conspicuous, visually
-----------	-----------------------

Acronym: CONVIS Code: 83
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: CONVIS_1 Date accepted: 2000-11-01

Name: visually conspicuous

Definition: term applied to an object either natural or artificial which is distinctly and notably visible from seaward. (IHO Dictionary, S-32, 5th Edition, 984)

2 DD Name: HYDRO Code: CONVIS_2 Date accepted: 2000-11-01

Name: not visually conspicuous

Definition: an object which is visible from seaward, but is not conspicuous.

Attribute	Current velocity
-----------	------------------

Acronym: CURVEL Code: 84
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01
Definition: Current velocity

Attribute	Date end
-----------	----------

Acronym: DATEND Code: 85

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The latest date on which an object (e.g., a buoy) will be present.

Attribute	Date start
-----------	------------

Acronym: DATSTA Code: 86
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The earliest date on which an object (e.g., a buoy) will be present.

Attribute	Depth range value 1
-----------	---------------------

Acronym: DRVAL1 Code: 87

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The minimum (shoalest) value of a depth range.

Attribute	Depth range value 2
-----------	---------------------

Acronym: DRVAL2 Code: 88

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The maximum (deepest) value of a depth range.

Attribute	Elevation
-----------	-----------

Acronym: ELEVAT Code: 90
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The altitude of the ground level of an object, measured from a specified vertical datum.

Attribute	Exhibition condition of light
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Acronym: EXCLIT Code: 92

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | DD Name: HYDRO Code: EXCLIT_1 Date accepted: 2000-11-01
Name: light shown without change of character
Definition: a light shown throughout the 24 hours without change of character. IHO Chart Specifications, M-4 |
| 2 | DD Name: HYDRO Code: EXCLIT_2 Date accepted: 2000-11-01
Name: daytime light
Definition: a light which is only exhibited by day. |
| 3 | DD Name: HYDRO Code: EXCLIT_3 Date accepted: 2000-11-01
Name: fog light
Definition: a light which is exhibited in fog or conditions of reduced visibility. |
| 4 | DD Name: HYDRO Code: EXCLIT_4 Date accepted: 2000-11-01
Name: night light
Definition: a light which is only exhibited at night. |

Attribute	Exposition of sounding
-----------	------------------------

Acronym: EXPSOU Code: 93

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2010-08-12

Definition: Indicates objects with a 'value of sounding' not within the range of depth of the surrounding depth area.

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | <p>DD Name: HYDRO Code: EXPSOU_1 Date accepted: 2010-08-12</p> <p>Name: within the range of depth of the surrounding depth area</p> <p>Definition: the depth corresponds to the depth range of the surrounding depth area. i.e. the depth is not shoaler than the minimum depth of the surrounding depth area or deeper than the maximum depth of the surrounding depth area.</p> |
| 2 | <p>DD Name: HYDRO Code: EXPSOU_2 Date accepted: 2010-08-12</p> <p>Name: shoaler than the range of depth of the surrounding depth area</p> <p>Definition: the depth is shoaler than the minimum depth of the surrounding depth area.</p> |
| 3 | <p>DD Name: HYDRO Code: EXPSOU_3 Date accepted: 2010-08-12</p> <p>Name: deeper than the range of depth of the surrounding depth area</p> <p>Definition: the depth is deeper than the maximum depth of the surrounding depth area.</p> |

Attribute	Function
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Acronym: FUNCTN Code: 94

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: FUNCTN_2 Date accepted: 2000-11-01

Name: harbour-master's office

Definition: the office of the local official who has charge of mooring and berthing of vessels, collecting harbour fees, etc. (adapted from IHO Dictionary, S-32, 5th Edition, 2191)

3 DD Name: HYDRO Code: FUNCTN_3 Date accepted: 2000-11-01

Name: custom office

Definition: an office which is charged with enforcing customs regulations.

4 DD Name: HYDRO Code: FUNCTN_4 Date accepted: 2000-11-01

Name: health office

Definition: the office which is charged with the administration of health laws and sanitary inspections. (adapted from The New Shorter Oxford English Dictionary, 1993)

5 DD Name: HYDRO Code: FUNCTN_5 Date accepted: 2000-11-01

Name: hospital

Definition: an institution or establishment providing medical or surgical treatment for the ill or wounded. (The New Shorter Oxford English Dictionary, 1993)

6 DD Name: HYDRO Code: FUNCTN_6 Date accepted: 2000-11-01

Name: post office

Definition: the public department, agency or organisation responsible primarily for the collection, transmission and distribution of mail. (The New Shorter Oxford English Dictionary, 1993)

Value	Data Dictionary (DD) Reference		
7	DD Name: HYDRO	Code: FUNCTN_7	Date accepted: 2000-11-01
	Name: hotel		
	Definition: an establishment, especially of a comfortable or luxurious kind, where paying visitors are provided with accommodation, meals and other services. (The New Shorter Oxford English Dictionary, 1993)		
8	DD Name: HYDRO	Code: FUNCTN_8	Date accepted: 2000-11-01
	Name: railway station		
	Definition: a building with platforms where trains arrive, load, discharge and depart. (The New Shorter Oxford English Dictionary, 1993)		
9	DD Name: HYDRO	Code: FUNCTN_9	Date accepted: 2000-11-01
	Name: police station		
	Definition: the office of the local police force.		
10	DD Name: HYDRO	Code: FUNCTN_10	Date accepted: 2000-11-01
	Name: water-police station		
	Definition: the headquarters of a local water-police force.		
11	DD Name: HYDRO	Code: FUNCTN_11	Date accepted: 2000-11-01
	Name: pilot office		
	Definition: the office or headquarters of pilots; the place where the services of a pilot may be obtained. (IHO Dictionary, S-32, 5th Edition, 3845)		
12	DD Name: HYDRO	Code: FUNCTN_12	Date accepted: 2000-11-01
	Name: pilot lookout		
	Definition: a distinctive structure on shore from which personnel keep watch upon events at sea or along the coast. (IHO Dictionary, S-32, 5th Edition, 2917)		
13	DD Name: HYDRO	Code: FUNCTN_13	Date accepted: 2000-11-01
	Name: bank office		
	Definition: an office for custody, deposit, loan, exchange or issue of money. (adapted from The New Shorter Oxford English Dictionary, 1993)		
14	DD Name: HYDRO	Code: FUNCTN_14	Date accepted: 2000-11-01
	Name: headquarters for district control		
	Definition: the quarters of an executive officer (director, manager, etc.) with responsibility for an administrative area.		

Value	Data Dictionary (DD) Reference		
15	DD Name: HYDRO	Code: FUNCTN_15	Date accepted: 2000-11-01
	Name: transit shed/warehouse		
	Definition: a building or part of a building for storage of wares or goods. (adapted from The New Shorter Oxford English Dictionary, 1993)		
16	DD Name: HYDRO	Code: FUNCTN_16	Date accepted: 2000-11-01
	Name: factory		
	Definition: a building or buildings with equipment for manufacturing; a workshop. (The New Shorter Oxford English Dictionary, 1993)		
17	DD Name: HYDRO	Code: FUNCTN_17	Date accepted: 2000-11-01
	Name: power station		
	Definition: a stationary plant containing apparatus for large scale conversion of some form of energy (such as hydraulic, steam, chemical or nuclear energy) into electrical energy. (McGraw-Hill Dictionary of Scientific and Technical Terms, 3rd Edition, 1984)		
18	DD Name: HYDRO	Code: FUNCTN_18	Date accepted: 2000-11-01
	Name: administrative		
	Definition: a building for the management of affairs. (adapted from The New Shorter Oxford English Dictionary, 1993)		
19	DD Name: HYDRO	Code: FUNCTN_19	Date accepted: 2000-11-01
	Name: educational facility		
	Definition: a building concerned with education (eg. school, college, university, etc.)		
20	DD Name: HYDRO	Code: FUNCTN_20	Date accepted: 2000-11-01
	Name: church		
	Definition: a building for public Christian worship. (The New Shorter Oxford English Dictionary, 1993)		
21	DD Name: HYDRO	Code: FUNCTN_21	Date accepted: 2000-11-01
	Name: chapel		
	Definition: a place for Christian worship other than a parish, cathedral or church, especially one attached to a private house or institution. (The New Shorter Oxford English Dictionary, 1993)		
22	DD Name: HYDRO	Code: FUNCTN_22	Date accepted: 2000-11-01
	Name: temple		
	Definition: a building for public Jewish worship. (adapted from The New Shorter Oxford English Dictionary, 1993)		

Value	Data Dictionary (DD) Reference		
23	DD Name: HYDRO	Code: FUNCTN_23	Date accepted: 2000-11-01
	Name: pagoda		
	Definition: a Hindu or Buddhist temple or sacred building. (The New Shorter Oxford English Dictionary, 1993)		
24	DD Name: HYDRO	Code: FUNCTN_24	Date accepted: 2000-11-01
	Name: shinto shrine		
	Definition: a building for public Shinto worship. (adapted from The New Shorter Oxford English Dictionary, 1993)		
25	DD Name: HYDRO	Code: FUNCTN_25	Date accepted: 2000-11-01
	Name: buddhist temple		
	Definition: see pagoda.		
26	DD Name: HYDRO	Code: FUNCTN_26	Date accepted: 2000-11-01
	Name: mosque		
	Definition: a Muslim place of worship. (The New Shorter Oxford English Dictionary, 1993)		
27	DD Name: HYDRO	Code: FUNCTN_27	Date accepted: 2000-11-01
	Name: marabout		
	Definition: a shrine marking the burial place of a Muslim holy man. (The New Shorter Oxford English Dictionary, 1993)		
28	DD Name: HYDRO	Code: FUNCTN_28	Date accepted: 2000-11-01
	Name: lookout		
	Definition: keeping a watch upon events at sea or along the coast. (adapted from IHO Dictionary, S-32,5th Edition,2917)		
29	DD Name: HYDRO	Code: FUNCTN_29	Date accepted: 2000-11-01
	Name: communication		
	Definition: transmitting and/or receiving electronic communication signals. (adapted from Digital Geographic Information Standard - DIGEST)		
30	DD Name: HYDRO	Code: FUNCTN_30	Date accepted: 2000-11-01
	Name: television		
	Definition: broadcast of television signals.		
31	DD Name: HYDRO	Code: FUNCTN_31	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	radio	
	Definition:	broadcast of radio signals.	
32	DD Name:	HYDRO	Code: FUNCTN_32 Date accepted: 2000-11-01
	Name:	radar	
	Definition:	a method, system or technique of using beamed, reflected, and timed radio waves for detecting, locating, or tracking objects, and for measuring altitudes. (IHO Dictionary, S-32, 5th Edition,4158)	
33	DD Name:	HYDRO	Code: FUNCTN_33 Date accepted: 2000-11-01
	Name:	light support	
	Definition:	supporting a light	
34	DD Name:	HYDRO	Code: FUNCTN_34 Date accepted: 2000-11-01
	Name:	microwave	
	Definition:	broadcasting and receiving signals using microwaves.	
35	DD Name:	HYDRO	Code: FUNCTN_35 Date accepted: 2000-11-01
	Name:	cooling	
	Definition:	dissipating heat.	
36	DD Name:	HYDRO	Code: FUNCTN_36 Date accepted: 2000-11-01
	Name:	observation	
	Definition:	a place from which the surroundings can be observed but at which a watch is not habitually maintained. (adapted from IHO Dictionary, S-32, 5th Edition,2917)	
37	DD Name:	HYDRO	Code: FUNCTN_37 Date accepted: 2000-11-01
	Name:	time ball	
	Definition:	a visual time signal in form of a ball	
38	DD Name:	HYDRO	Code: FUNCTN_38 Date accepted: 2000-11-01
	Name:	clock	
	Definition:	visual time signal. (adapted from S-32, 5th Edition, 5536)	
39	DD Name:	HYDRO	Code: FUNCTN_39 Date accepted: 2000-11-01
	Name:	control	
	Definition:	used to control the flow of air, rail, or marine traffic. (Digital Geographic Information Standard - DIGEST)	
40	DD Name:	HYDRO	Code: FUNCTN_40 Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	airship mooring	
	Definition:	a facility to secure an airship. (adapted from Digital Geographic Information Standard - DIGEST)	
41	DD Name:	HYDRO	Code: FUNCTN_41 Date accepted: 2000-11-01
	Name:	stadium	
	Definition:	a large usually unroofed building with tiers of seats for spectators	
42	DD Name:	HYDRO	Code: FUNCTN_42 Date accepted: 2000-11-01
	Name:	bus station	
	Definition:	a location at which buses arrive and from which they depart.	

Attribute	Height
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Acronym: HEIGHT Code: 95

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The value of the vertical distance to the highest point of the object, measured from a specified vertical datum.

Attribute	Horizontal Accuracy
-----------	---------------------

Acronym: HORACC Code: 97
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The best estimate of the horizontal accuracy of horizontal clearance and distances.

Attribute	Horizontal clearance
-----------	----------------------

Acronym: HORCLR Code: 98

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The width of an object, such as a canal or a tunnel, which is available for safe navigation. This may, or may not, be the same as the total physical width of the object.

Attribute	Horizontal length
-----------	-------------------

Acronym: HORLEN Code: 99

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A measurement of the longer of two linear axis. (Digital Geographic Information Working Group -DGIWG, Oct.87)

Attribute	Horizontal width
-----------	------------------

Acronym: HORWID Code: 100
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: A measurement of the shorter of two linear axis. (Digital Geographic Information Working Group -DGIWG, Oct.87)

Attribute	Information
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Acronym: INFORM

Code: 102

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Textual information about the object.

Attribute	Information in national language
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Acronym: NINFOM

Code: 300

Use Type: N

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Textual information in national language characters

Attribute	Jurisdiction
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Acronym: JRSDTN Code: 103

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: JRSDTN_1 Date accepted: 2000-11-01 Name: international Definition: involving more than one country; covering more than one national area.
2	DD Name: HYDRO Code: JRSDTN_2 Date accepted: 2000-11-01 Name: national Definition: an area administered or controlled by a single nation.
3	DD Name: HYDRO Code: JRSDTN_3 Date accepted: 2000-11-01 Name: national sub-division Definition: an area smaller than the nation in which it lies.

Attribute	Light characteristic
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Acronym: LITCHR Code: 107

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: LITCHR_1 Date accepted: 2000-11-01 Name: fixed Definition: a signal light that shows continuously, in any given direction, with constant luminous intensity and colour. (IHO Dictionary, S-32, 5th Edition, 2780)
2	DD Name: HYDRO Code: LITCHR_2 Date accepted: 2000-11-01 Name: flashing Definition: a rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration. (IHO Dictionary, S-32, 5th Edition, 2783)
3	DD Name: HYDRO Code: LITCHR_3 Date accepted: 2000-11-01 Name: long-flashing Definition: a flashing light in which a single flash of not less than two seconds duration is regularly repeated. (IHO Dictionary, S-32, 5th Edition, 2796)
4	DD Name: HYDRO Code: LITCHR_4 Date accepted: 2000-11-01 Name: quick-flashing Definition: a light exhibiting without interruption very rapid regular alternations of light and darkness. (IHO Dictionary, S-32, 5th Edition, 2803)
7	DD Name: HYDRO Code: LITCHR_7 Date accepted: 2000-11-01 Name: isophased Definition: a light with all durations of light and darkness equal. (IHO Dictionary, S-32, 5th Edition, 2779)

Value	Data Dictionary (DD) Reference		
9	DD Name: HYDRO	Code: LITCHR_9	Date accepted: 2000-11-01
	Name: interrupted quick-flashing		
	Definition: a quick light in which the sequence of flashes is interrupted by regularly repeated eclipses of constant and long duration. (IHO Dictionary, S-32, 5th Edition, 2790)		
5	DD Name: HYDRO	Code: LITCHR_5	Date accepted: 2000-11-01
	Name: very quick-flashing		
	Definition: a flashing light in which flashes are repeated at a rate of not less than 80 flashes per minute but less than 160 flashes per minute		
6	DD Name: HYDRO	Code: LITCHR_6	Date accepted: 2000-11-01
	Name: ultra quick flashing		
	Definition: a flashing light in which flashes are repeated at a rate of not less than 160 flashes per minute		
8	DD Name: HYDRO	Code: LITCHR_8	Date accepted: 2000-11-01
	Name: occulting		
	Definition: a rhythmic light in which the total duration of light in a period is clearly longer than the total duration of darkness and all the eclipses are of equal duration		
10	DD Name: HYDRO	Code: LITCHR_10	Date accepted: 2000-11-01
	Name: interrupted very quick-flashing		
	Definition: a light in which the very rapid alterations of light and darkness are interrupted at regular intervals by eclipses of long duration		
11	DD Name: HYDRO	Code: LITCHR_11	Date accepted: 2000-11-01
	Name: interrupted ultra quick-flashing		
	Definition: a light in which the ultra quick flashes (160 or more per minute) are interrupted at regular intervals by eclipses of long duration		
12	DD Name: HYDRO	Code: LITCHR_12	Date accepted: 2000-11-01
	Name: morse		
	Definition: a rhythmic light in which appearances of light of two clearly different durations are grouped to represent a character or characters in the Morse code		
13	DD Name: HYDRO	Code: LITCHR_13	Date accepted: 2000-11-01
	Name: fixed/flash		
	Definition:		
28	DD Name: HYDRO	Code: LITCHR_28	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	alternating	
	Definition:	a signal light that shows, in any given direction, two or more colours in a regularly repeated sequence with a regular periodicity	
14	DD Name: HYDRO	Code: LITCHR_14	Date accepted: 2000-11-01
	Name:	flash/long-flash	
	Definition:		
15	DD Name: HYDRO	Code: LITCHR_15	Date accepted: 2000-11-01
	Name:	occulting/flash	
	Definition:		
16	DD Name: HYDRO	Code: LITCHR_16	Date accepted: 2000-11-01
	Name:	fixed/long-flash	
	Definition:		
17	DD Name: HYDRO	Code: LITCHR_17	Date accepted: 2000-11-01
	Name:	occulting alternating	
	Definition:		
18	DD Name: HYDRO	Code: LITCHR_18	Date accepted: 2000-11-01
	Name:	long-flash alternating	
	Definition:		
19	DD Name: HYDRO	Code: LITCHR_19	Date accepted: 2000-11-01
	Name:	flash alternating	
	Definition:		
20	DD Name: HYDRO	Code: LITCHR_20	Date accepted: 2000-11-01
	Name:	group alternating	
	Definition:		
25	DD Name: HYDRO	Code: LITCHR_25	Date accepted: 2000-11-01
	Name:	quick-flash plus long-flash	
	Definition:		
26	DD Name: HYDRO	Code: LITCHR_26	Date accepted: 2000-11-01
	Name:	very quick-flash plus long-flash	

Value	Data Dictionary (DD) Reference		
	Definition:		
27	DD Name: HYDRO	Code: LITCHR_27	Date accepted: 2000-11-01
	Name: ultra quick-flash plus long-flash		
	Definition:		
29	DD Name: HYDRO	Code: LITCHR_29	Date accepted: 2000-11-01
	Name: fixed and alternating flashing		
	Definition:		

Attribute	Light visibility
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Acronym: LITVIS Code: 108

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2008-01-31

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

4 DD Name: HYDRO Code: LITVIS_4 Date accepted: 2000-11-01

Name: intensified

Definition: a light in a sector is intensified (i.e. has longer range than other sectors). (Bundesamt für Seeschifffahrt und Hydrographie, Germany)

7 DD Name: HYDRO Code: LITVIS_7 Date accepted: 2000-11-01

Name: obscured

Definition: said of the arc of a light sector designated by its limiting bearings in which the light is not visible from seaward. (IHO Dictionary, S-32, 5th Edition, 3492)

8 DD Name: HYDRO Code: LITVIS_8 Date accepted: 2000-11-01

Name: partially obscured

Definition: this value specifies that parts of the sector are obscured.

3 DD Name: HYDRO Code: LITVIS_3 Date accepted: 2000-11-01

Name: faint

Definition: a decrease in the apparent intensity of a light which may occur in the case of partial obstructions.

Attribute	Marks navigational - System of
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Acronym: MARSYS Code: 109
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- 1 DD Name: HYDRO Code: MARSYS_1 Date accepted: 2000-11-01
Name: IALA A
Definition: navigational aids conform to the International Association of Lighthouse Authorities - IALA A system.
- 2 DD Name: HYDRO Code: MARSYS_2 Date accepted: 2000-11-01
Name: IALA B
Definition: navigational aids conform to the International Association of Lighthouse Authorities - IALA B system.

Attribute	Multiplicity of lights
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Acronym: MLTYLT

Code: 110

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The number of lights of identical character that exist as a co-located group.

Attribute	Nationality
-----------	-------------

Acronym: NATION Code: 111
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The nationality of the specific object.

Attribute	Natural surface
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Acronym: NATSUR Code: 113

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2008-01-31

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference		
1	DD Name: HYDRO	Code: NATSUR_1	Date accepted: 2000-11-01
	Name: mud		
	Definition: soft, wet earth		
2	DD Name: HYDRO	Code: NATSUR_2	Date accepted: 2000-11-01
	Name: clay		
	Definition: (particles of less than 0.002 mm); stiff, sticky earth that becomes hard when baked.		
3	DD Name: HYDRO	Code: NATSUR_3	Date accepted: 2000-11-01
	Name: silt		
	Definition: (particles of 0.002-0.0625 mm); when dried on hand will rub off easily.		
4	DD Name: HYDRO	Code: NATSUR_4	Date accepted: 2000-11-01
	Name: sand		
	Definition: (particles of 0.0625-2.0 mm); tiny grains of crushed or worn rock.		
5	DD Name: HYDRO	Code: NATSUR_5	Date accepted: 2000-11-01
	Name: stone		
	Definition: a general term for rock fragments ranging in size from pebbles and gravel to boulders or a large rock mass. (IHO Dictionary, S-32, 5th Edition, 5059)		
6	DD Name: HYDRO	Code: NATSUR_6	Date accepted: 2000-11-01
	Name: gravel		

Value	Data Dictionary (DD) Reference		
	Definition: (particles of 2.0-4.0 mm); small stones with coarse sand.		
7	DD Name: HYDRO	Code: NATSUR_7	Date accepted: 2000-11-01
	Name: pebbles		
	Definition: (particles of 4.0-64.0 mm); small stones made smooth and round by being rolled in water.		
8	DD Name: HYDRO	Code: NATSUR_8	Date accepted: 2000-11-01
	Name: cobbles		
	Definition: (particles of 64.0-256.0 mm); stones worn round and smooth by water and used for paving.		
9	DD Name: HYDRO	Code: NATSUR_9	Date accepted: 2000-11-01
	Name: rock		
	Definition: any formation of natural origin that constitutes an integral part of the lithosphere. The natural occurring material that forms firm, hard, and solid masses. (adapted from IHO Dictionary, S-32, 5th Edition, 4415)		
11	DD Name: HYDRO	Code: NATSUR_11	Date accepted: 2000-11-01
	Name: lava		
	Definition: the fluid or semi-fluid matter flowing from a volcano. The substance that results from the cooling of the molten rock. Part of the ocean bed is composed of lava. (IHO Dictionary, S-32, 5th Edition, 2680)		
14	DD Name: HYDRO	Code: NATSUR_14	Date accepted: 2000-11-01
	Name: coral		
	Definition: hard calcareous skeletons of many tribes of marine polyps. (IHO Dictionary, S-32, 5th Edition, 1061)		
17	DD Name: HYDRO	Code: NATSUR_17	Date accepted: 2000-11-01
	Name: shells		
	Definition: exoskeletons of various water dwelling animals. (adapted from IHO Dictionary, S-32, 5th Edition, 4680)		
18	DD Name: HYDRO	Code: NATSUR_18	Date accepted: 2000-11-01
	Name: boulder		
	Definition: a rounded rock with diameter of 256 mm or larger. (adapted from IHO Dictionary, S-32, 5th Edition, 527)		

Attribute	Nature of construction
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Acronym: NATCON Code: 112

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: HYDRO Code: NATCON_1 Date accepted: 2000-11-01 |
| | Name: masonry |
| | Definition: constructed of brick or stone. |
| 2 | DD Name: HYDRO Code: NATCON_2 Date accepted: 2000-11-01 |
| | Name: concreted |
| | Definition: constructed of concrete, a material made of sand and gravel that is united by cement into a hardened mass used for roads, foundations, etc. (adapted from the Illustrated Contemporary Dictionary, Encyclopaedic Edition, 1978) |
| 3 | DD Name: HYDRO Code: NATCON_3 Date accepted: 2000-11-01 |
| | Name: loose boulders |
| | Definition: constructed from large stones or blocks of concrete, often placed loosely for protection against waves or water turbulence. |
| 4 | DD Name: HYDRO Code: NATCON_4 Date accepted: 2000-11-01 |
| | Name: hard surfaced |
| | Definition: constructed with a surface of hard material, usually a term applied to roads surfaced with asphalt or concrete. |
| 5 | DD Name: HYDRO Code: NATCON_5 Date accepted: 2000-11-01 |
| | Name: unsurfaced |
| | Definition: constructed with no extra protection, usually a term applied to roads not surfaced with a hard material. |

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO Name: wooden Definition: constructed from wood.	Code: NATCON_6	Date accepted: 2000-11-01
7	DD Name: HYDRO Name: metal Definition: constructed from metal.	Code: NATCON_7	Date accepted: 2000-11-01
8	DD Name: HYDRO Name: glass reinforced plastic (GRP) Definition: constructed from a plastic material strengthened with fibres of glass.	Code: NATCON_8	Date accepted: 2000-11-01
9	DD Name: HYDRO Name: painted Definition: the application of paint to some other construction or natural feature.	Code: NATCON_9	Date accepted: 2000-11-01

Attribute	Nature of surface - qualifying terms
-----------	--------------------------------------

Acronym: NATQUA

Code: 114

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Nature of surface - qualifying terms

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: NATQUA_1 Date accepted: 2000-11-01

Name: Fine

Definition: falls within the smallest size continuum for a particular nature of surface term

2 DD Name: HYDRO Code: NATQUA_2 Date accepted: 2000-11-01

Name: Medium

Definition: falls within the moderate size continuum for a particular nature of surface term

3 DD Name: HYDRO Code: NATQUA_3 Date accepted: 2000-11-01

Name: Coarse

Definition: falls within the largest size continuum for a particular nature of surface term

4 DD Name: HYDRO Code: NATQUA_4 Date accepted: 2000-11-01

Name: Broken

Definition: fractured or in pieces

Attribute	Object class definition
-----------	-------------------------

Acronym: CLSDEF

Code: 18027

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2015-01-01

Definition: Specifies the defining characteristics of a 'new object'.

Attribute	Object class name
-----------	-------------------

Acronym: CLSNAM

Code: 18028

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2015-01-01

Definition: Specifies the descriptive name of a 'new object' feature object class.

Attribute	Object name
-----------	-------------

Acronym: OBJNAM Code: 116

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The individual name of an object.

Attribute	Object name in national language
-----------	----------------------------------

Acronym: NOBJNM

Code: 301

Use Type: N

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Name of object in national language characters

Attribute	Orientation
-----------	-------------

Acronym: ORIENT Code: 117

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The angular distance measured from true north to the major axis of the object. (Digital Geographic Information Working Group -DGIWG, Oct.87)

Attribute	Periodic date end
-----------	-------------------

Acronym: PEREND

Code: 118

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The end of the active period for a seasonal object (e.g. a buoy).

Attribute	Periodic date start
-----------	---------------------

Acronym: PERSTA

Code: 119

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The start of the active period for a seasonal object (e.g. a buoy).

Attribute	Pictorial representation
-----------	--------------------------

Acronym: PICREP Code: 120
Use Type: F
Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Indicates whether a pictorial representation of the object is available.

Attribute	Pilot district
-----------	----------------

Acronym: PILDST Code: 121

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The area within which a particular pilotage service operates.

Attribute	Pilot district in national language
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Acronym: NPLDST

Code: 302

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Pilot district (c...): string of national language characters

Attribute	Positional Accuracy
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Acronym: POSACC Code: 401
Use Type: F,S
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The best estimate of the accuracy of a position. The expected input is the maximum of the two-dimensional error. The error is assumed to be positive and negative. The plus/minus character shall not be encoded.

Acronym:	PRODC	Code:	123
Use Type:	F		
Value Type:	L		

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Value	Data Dictionary (DD) Reference
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1 DD Name: HYDRO Code: PRODC T 1 Date accepted: 2000-11-01

Name: oil

Definition: a thick, slippery liquid that will not dissolve in water, usually petroleum based in the context of storage tanks. (adapted from the Oxford Minidictionary, Third Edition)

2 DD Name: HYDRO Code: PRODUCT 2 Date accepted: 2000-11-01

Name: gas

Definition: a substance with particles that can move freely, usually a fuel substance in the context of storage tanks. (adapted from the Oxford Minidictionary, Third Edition)

3 DD Name: HYDRO Code: PRODC T 3 Date accepted: 2000-11-01

Name: water

Definition: a colourless, odourless, tasteless liquid that is a compound of hydrogen and oxygen. (adapted from the Oxford Minidictionary, Third Edition)

4 DD Name: HYDRO Code: PRODUCT 4 Date accepted: 2000-11-01

Name: stone

Definition: a general term for rock fragments. (IHO Dictionary, S-32, 5th Edition, 5059)

5 DD Name: HYDRO Code: PRODC T 5 Date accepted: 2000-11-01

Name: coal

Definition: a hard black mineral that is burned as fuel. (adapted from the Oxford Minidictionary, Third Edition)

Value	Data Dictionary (DD) Reference		
6	DD Name: HYDRO Name: ore Definition: a solid rock or mineral from which metal is obtained. (adapted form the Oxford Minidictionary, Third Edition)	Code: PRODUCT_6	Date accepted: 2000-11-01
7	DD Name: HYDRO Name: chemicals Definition: any substance obtained by or used in a chemical process. (adapted from the Oxford Minidictionary, Third Edition)	Code: PRODUCT_7	Date accepted: 2000-11-01
8	DD Name: HYDRO Name: drinking water Definition: water that is suitable for human consumption. (adapted from the Oxford Minidictionary, Third Edition)	Code: PRODUCT_8	Date accepted: 2000-11-01
14	DD Name: HYDRO Name: sand Definition: tiny grains of crushed or worn rock. (adapted from the Oxford Minidictionary, Third Edition)	Code: PRODUCT_14	Date accepted: 2000-11-01
15	DD Name: HYDRO Name: timber Definition: wood prepared for use in building or carpentry. (adapted from the Oxford Minidictionary, Third Edition)	Code: PRODUCT_15	Date accepted: 2000-11-01
17	DD Name: HYDRO Name: scrap metal Definition: discarded metal suitable for being reprocessed. (adapted from the Oxford Minidictionary, Third Edition)	Code: PRODUCT_17	Date accepted: 2000-11-01
21	DD Name: HYDRO Name: cement Definition: a substance made of powdered lime and clay, mixed with water. (adapted from the Websters New World Dictionary)	Code: PRODUCT_21	Date accepted: 2000-11-01
22	DD Name: HYDRO Name: grain Definition: a small hard seed, especially that of any cereal plant such as wheat, rice, corn, rye etc. (adapted from the Websters New World Dictionary)	Code: PRODUCT_22	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
18	DD Name: HYDRO	Code: PRODCT_18	Date accepted: 2000-11-01
	Name:	liquified natural gas (LNG)	
	Definition:	a compressed gas consisting of flammable light hydrocarbons and derived from natural gas.	
19	DD Name: HYDRO	Code: PRODCT_19	Date accepted: 2000-11-01
	Name:	liquified petroleum gas (LPG)	
	Definition:	a compressed gas consisting of flammable light hydrocarbons and derived from petroleum. (adapted from Websters Third New)	

Attribute	Quality of position
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Acronym: QUAPOS

Code: 402

Use Type: F,S

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

4 DD Name: HYDRO Code: QUAPOS_4 Date accepted: 2000-11-01

Name: approximate

Definition: a position that is considered to be less than third-order accuracy, but is generally considered to be within 30.5 metres of its correct geographic location. Also may apply to an object whose position does not remain fixed. (adapted from IHO Dictionary, S-32, 213, 3967, and IHO Specifications, M-4, 424.1)

10 DD Name: HYDRO Code: QUAPOS_10 Date accepted: 2000-11-01

Name: precisely known

Definition: a position that is of a known value, such as the position of an anchor berth or other defined object.

Attribute	Quality of sounding measurement
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Acronym: QUASOU

Code: 125

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: QUASOU_1 Date accepted: 2000-11-01

Name: depth known

Definition: the depth from chart datum to the bottom is a known value.

2 DD Name: HYDRO Code: QUASOU_2 Date accepted: 2000-11-01

Name: depth unknown

Definition: the depth from chart datum to the bottom is unknown.

3 DD Name: HYDRO Code: QUASOU_3 Date accepted: 2000-11-01

Name: doubtful sounding

Definition: a depth that may be less than indicated. (adapted from IHO Dictionary, S-32, 5th Edition, 4840)

4 DD Name: HYDRO Code: QUASOU_4 Date accepted: 2000-11-01

Name: unreliable sounding

Definition: a depth that is considered to be an unreliable value.

6 DD Name: HYDRO Code: QUASOU_6 Date accepted: 2000-11-01

Name: least depth known

Definition: the shoalest depth over a feature is of known value. (adapted from IHO Dictionary, S-32, 5th Edition, 2705)

7 DD Name: HYDRO Code: QUASOU_7 Date accepted: 2000-11-01

Name: least depth unknown, safe clearance at depth shown

Value	Data Dictionary (DD) Reference		
	Definition:	the least depth over a feature is unknown, but there is considered to be safe clearance at this depth.	
8	DD Name: HYDRO	Code: QUASOU_8	Date accepted: 2000-11-01
	Name:	value reported (not surveyed)	
	Definition:	depth value obtained from a report, but not fully surveyed.	
9	DD Name: HYDRO	Code: QUASOU_9	Date accepted: 2000-11-01
	Name:	value reported (not confirmed)	
	Definition:	depth value obtained from a report, which it has not been possible to confirm.	
10	DD Name: HYDRO	Code: QUASOU_10	Date accepted: 2000-11-01
	Name:	maintained depth	
	Definition:	the depth at which a channel is kept by human influence, usually by dredging. (IHO Dictionary, S-32, 5th Edition, 3057)	
11	DD Name: HYDRO	Code: QUASOU_11	Date accepted: 2000-11-01
	Name:	not regularly maintained	
	Definition:	depths may be altered by human influence, but will not be routinely maintained.	

Attribute	Radar wave length
-----------	-------------------

Acronym: RADWAL

Code: 126

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave in the radar band of the electromagnetic spectrum.

Attribute	Reference year for magnetic variation
-----------	---------------------------------------

Acronym: RYRMGV

Code: 130

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The reference calendar year for magnetic variation values.

Attribute	Restriction
-----------	-------------

Acronym:	RESTRN	Code:	131
Use Type:	F		
Value Type:	L		

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The official legal statute of each kind of restricted area.

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: RESTRN_1 Date accepted: 2000-11-01 Name: anchoring prohibited Definition: an area within which anchoring is not permitted.
2	DD Name: HYDRO Code: RESTRN_2 Date accepted: 2000-11-01 Name: anchoring restricted Definition: a specified area designated by appropriate authority, within which anchoring is restricted in accordance with certain specified conditions.
3	DD Name: HYDRO Code: RESTRN_3 Date accepted: 2000-11-01 Name: fishing prohibited Definition: an area within which fishing is not permitted.
4	DD Name: HYDRO Code: RESTRN_4 Date accepted: 2000-11-01 Name: fishing restricted Definition: a specified area designated by appropriate authority, within which fishing is restricted in accordance with certain specified conditions.
5	DD Name: HYDRO Code: RESTRN_5 Date accepted: 2000-11-01 Name: trawling prohibited Definition: an area within which trawling is not permitted.
6	DD Name: HYDRO Code: RESTRN 6 Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	trawling restricted	
	Definition:	a specified area designated by appropriate authority, within which trawling is restricted in accordance with certain specified conditions.	
7	DD Name:	HYDRO Code: RESTRN_7	Date accepted: 2000-11-01
	Name:	entry prohibited	
	Definition:	an area within which navigation and/or anchoring is prohibited.	
8	DD Name:	HYDRO Code: RESTRN_8	Date accepted: 2000-11-01
	Name:	entry restricted	
	Definition:	a specified area designated by appropriate authority, within which navigation is restricted in accordance with certain specified conditions.	
9	DD Name:	HYDRO Code: RESTRN_9	Date accepted: 2000-11-01
	Name:	dredging prohibited	
	Definition:	an area within which dredging is not permitted.	
10	DD Name:	HYDRO Code: RESTRN_10	Date accepted: 2000-11-01
	Name:	dredging restricted	
	Definition:	a specified area designated by appropriate authority, within which dredging is restricted in accordance with certain specified conditions.	
11	DD Name:	HYDRO Code: RESTRN_11	Date accepted: 2000-11-01
	Name:	diving prohibited	
	Definition:	an area within which diving is not permitted.	
12	DD Name:	HYDRO Code: RESTRN_12	Date accepted: 2000-11-01
	Name:	diving restricted	
	Definition:	a specified area designated by appropriate authority, within which diving is restricted in accordance with certain specified conditions.	
13	DD Name:	HYDRO Code: RESTRN_13	Date accepted: 2000-11-01
	Name:	no wake	
	Definition:	mariners must adjust the speed of their vessels to reduce the wave or wash which may cause erosion or disturb moored vessels.	
14	DD Name:	HYDRO Code: RESTRN_14	Date accepted: 2000-11-01
	Name:	area to be avoided	
	Definition:	an IMO designated area to be avoided, defined as a routeing measure.	

Value	Data Dictionary (DD) Reference		
15	DD Name: HYDRO	Code: RESTRN_15	Date accepted: 2000-11-01
	Name:	construction prohibited	
	Definition:	the erection of permanent or temporary fixed structures or artificial islands is prohibited.	
16	DD Name: HYDRO	Code: RESTRN_16	Date accepted: 2000-11-01
	Name:	discharging prohibited	
	Definition:	an area within which discharging or dumping is prohibited	
17	DD Name: HYDRO	Code: RESTRN_17	Date accepted: 2000-11-01
	Name:	discharging restricted	
	Definition:	a specified area designated by an appropriate authority, within which discharging or dumping is restricted in accordance with specified conditions.	
18	DD Name: HYDRO	Code: RESTRN_18	Date accepted: 2000-11-01
	Name:	industrial or mineral exploration/development prohibited	
	Definition:	an area within which industrial or mineral exploration and development are prohibited.	
19	DD Name: HYDRO	Code: RESTRN_19	Date accepted: 2000-11-01
	Name:	industrial or mineral exploration/development restricted	
	Definition:	a specified area designated by an appropriate authority, within which industrial or mineral exploration and development is restricted in accordance with certain specified conditions.	
20	DD Name: HYDRO	Code: RESTRN_20	Date accepted: 2000-11-01
	Name:	drilling prohibited	
	Definition:	an area within which excavating a hole on the sea-bottom with a drill is prohibited.	
21	DD Name: HYDRO	Code: RESTRN_21	Date accepted: 2000-11-01
	Name:	drilling restricted	
	Definition:	a specified area designated by an appropriate authority, within which excavating a hole on the sea-bottom with a drill is restricted in accordance with certain specified conditions.	
22	DD Name: HYDRO	Code: RESTRN_22	Date accepted: 2000-11-01
	Name:	removal of historical artifacts prohibited	
	Definition:	an area within which the removal of historical artifacts is prohibited.	
23	DD Name: HYDRO	Code: RESTRN_23	Date accepted: 2000-11-01
	Name:	cargo transshipment (lightering) prohibited	
	Definition:	an area in which cargo transshipment (lightering) is prohibited.	

Value	Data Dictionary (DD) Reference		
24	DD Name: HYDRO	Code: RESTRN_24	Date accepted: 2000-11-01
	Name:	dragging prohibited	
	Definition:	an area in which the dragging of anything along the bottom, e.g. bottom trawling, is prohibited.	
25	DD Name: HYDRO	Code: RESTRN_25	Date accepted: 2000-11-01
	Name:	stopping prohibited	
	Definition:	an area in which a vessel is prohibited from stopping.	
26	DD Name: HYDRO	Code: RESTRN_26	Date accepted: 2000-11-01
	Name:	landing prohibited	
	Definition:	an area in which landing is prohibited.	
27	DD Name: HYDRO	Code: RESTRN_27	Date accepted: 2000-11-01
	Name:	speed restricted	
	Definition:	an area within which speed is restricted.	
38	DD Name: IENC	Code: restrn_38	Date accepted: 2000-11-01
	Name:	use of spuds prohibited	
	Definition:	The use of anchoring spuds (telescopic piles) is prohibited	

Attribute	Scale minimum
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Acronym: SCAMIN

Code: 133

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The minimum scale at which the object may be used e.g. for ECDIS presentation.

Attribute	Sector limit one
-----------	------------------

Acronym: SECTR1 Code: 136

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2008-01-31

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.
(Advanced Learner=s Dictionary, 2nd Edition) Sector limit 1 specifies the first limit of the sector. The order of sector limit 1 and sector limit 2 is clockwise around the central object (e.g. a light).

Attribute	Sector limit two
-----------	------------------

Acronym: SECTR2 Code: 137

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2008-01-31

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.
(Advanced Learner=s Dictionary, 2nd Edition) The sector limit 2 specifies the second limit of the sector.
The order of sector limit 1 and sector limit 2 is clockwise around the central object (e.g. a light).

Attribute	Signal frequency
-----------	------------------

Acronym: SIGFRQ

Code: 139

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The frequency of a signal.

Attribute	Signal generation
-----------	-------------------

Acronym: SIGGEN Code: 140
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

- 1 DD Name: HYDRO Code: SIGGEN_1 Date accepted: 2000-11-01
Name: automatically
Definition: signal generation is initiated by a self regulating mechanism such as a timer or light sensor.
- 2 DD Name: HYDRO Code: SIGGEN_2 Date accepted: 2000-11-01
Name: by wave action
Definition: the signal is generated by the motion of the sea surface such as a bell in a buoy.

Attribute	Signal group
-----------	--------------

Acronym: SIGGRP Code: 141
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The number of signals, the combination of signals or the morse character(s) within one period of full sequence.

Attribute	Signal period
-----------	---------------

Acronym: SIGPER Code: 142
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The time occupied by an entire cycle of intervals of light and eclipse.

Attribute	Signal sequence
-----------	-----------------

Acronym: SIGSEQ Code: 143
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The sequence of times occupied by intervals of light and eclipse for all 'light characteristics' except for occulting where the sequence of times is occupied by intervals of eclipse and light.

Acronym:	SOUACC	Code:	144
Use Type:	F		
Value Type:	F		

DD Name: HYDRO Date accepted: 2000-11-01

IENC_FC_26_CORR3.DOCX

Attribute	Source date
-----------	-------------

Acronym: SORDAT Code: 147
Use Type: F
Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The production date of the source, e.g. the date of measurement.

Attribute	Source indication
-----------	-------------------

Acronym: SORINDCode: 148

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Information about the source of the object.

Attribute	Status
-----------	--------

Acronym: STATUS Code: 149

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The condition of an object at a given instant in time.

Enumerations:

Value	Data Dictionary (DD) Reference
2	DD Name: HYDRO Code: STATUS_2 Date accepted: 2000-11-01 Name: occasional Definition: acting on special occasions; happening irregularly. (The Concise Oxford Dictionary, 7th Edition)
3	DD Name: HYDRO Code: STATUS_3 Date accepted: 2000-11-01 Name: recommended Definition: presented as worthy of confidence, acceptance, use, etc. (The Macquarie Dictionary, 1988)
4	DD Name: HYDRO Code: STATUS_4 Date accepted: 2000-11-01 Name: not in use Definition: no longer used for the purpose intended; disused.
8	DD Name: HYDRO Code: STATUS_8 Date accepted: 2000-11-01 Name: private Definition: not in public ownership or operation.
9	DD Name: HYDRO Code: STATUS_9 Date accepted: 2000-11-01 Name: mandatory Definition: compulsory; enforced. (The Concise Oxford Dictionary, 7th Edition)
12	DD Name: HYDRO Code: STATUS_12 Date accepted: 2000-11-01 Name: illuminated

Value	Data Dictionary (DD) Reference		
	Definition:	lit by floodlights, strip lights, etc.	
14	DD Name: HYDRO	Code: STATUS_14	Date accepted: 2000-11-01
	Name:	public	
	Definition:	belonging to, available to, used or shared by, the community as a whole and not restricted to private use. (adapted from The New Shorter Oxford English Dictionary, 1993)	
16	DD Name: HYDRO	Code: STATUS_16	Date accepted: 2000-11-01
	Name:	watched	
	Definition:	looked at or observed over a period of time especially so as to be aware of any movement or change. (adapted from The New Shorter Oxford English Dictionary, 1993)	
17	DD Name: HYDRO	Code: STATUS_17	Date accepted: 2000-11-01
	Name:	un-watched	
	Definition:	usually automatic in operation, without any permanently-stationed personnel to superintend it. (adapted from IHO Dictionary, S-32, 5th Edition, 2814)	
18	DD Name: HYDRO	Code: STATUS_18	Date accepted: 2000-11-01
	Name:	existence doubtful	
	Definition:	an object that has been reported but has not been definitely determined to exist.	
7	DD Name: HYDRO	Code: STATUS_7	Date accepted: 2000-11-01
	Name:	temporary	
	Definition:	meant to last only for a time. (The Concise Oxford Dictionary)	
1	DD Name: HYDRO	Code: STATUS_1	Date accepted: 2000-11-01
	Name:	permanent	
	Definition:	intended to last or function indefinitely	
5	DD Name: HYDRO	Code: STATUS_5	Date accepted: 2000-11-01
	Name:	periodic/intermittent	
	Definition:	recurring at intervals	
6	DD Name: HYDRO	Code: STATUS_6	Date accepted: 2000-11-01
	Name:	reserved	
	Definition:	set apart for some specific use.	
11	DD Name: HYDRO	Code: STATUS_11	Date accepted: 2000-11-01
	Name:	extinguished	

Value	Data Dictionary (DD) Reference		
	Definition: no longer lit		
13	DD Name: HYDRO	Code: STATUS_13	Date accepted: 2000-11-01
	Name: historic		
	Definition: famous in history; of historical interest		
15	DD Name: HYDRO	Code: STATUS_15	Date accepted: 2000-11-01
	Name: synchronized		
	Definition: occur at a time, coincide in point of time, be contemporary or simultaneous		

Attribute	Survey authority
-----------	------------------

Acronym: SURATH

Code: 150

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The authority which was responsible for the survey.

Attribute	Survey date - end
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Acronym: SUREND

Code: 151

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The end date of the survey.

Attribute	Survey date - start
-----------	---------------------

Acronym: SURSTA

Code: 152

Use Type: F

Value Type: S

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The start date of the survey.

Attribute	Survey type
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Acronym: SURTYP Code: 153
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: HYDRO Code: SURTYP_2 Date accepted: 2000-11-01

Name: controlled survey

Definition: a thorough survey usually conducted with reference to guidelines.

Attribute	Symbol instruction
-----------	--------------------

Acronym: SYMINS

Code: 18029

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2015-01-01

Definition: This specifies the S-52 Presentation Library symbol instruction to be adopted in ECDIS for the new object class (as specified in the S-52 Symbol Library - Addendum to S-52 Presentation Library).

Attribute	Technique of sounding measurement
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Acronym: TECSOU Code: 156

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: TECSOU_1 Date accepted: 2000-11-01 Name: found by echo-sounder Definition: the depth was determined by using an instrument that determines depth of water by measuring the time interval between emission of a sonic or ultrasonic signal and return of its echo from the bottom. (adapted from IHO Dictionary, S-32, 1547)
2	DD Name: HYDRO Code: TECSOU_2 Date accepted: 2000-11-01 Name: found by side-scan-sonar Definition: the depth was computed from a record produced by active sonar in which fixed acoustic beams are directed into the water perpendicularly to the direction of travel to scan the bottom and generate a record of the bottom configuration. (adapted from IHO Dictionary, S-32, 4710)
3	DD Name: HYDRO Code: TECSOU_3 Date accepted: 2000-11-01 Name: found by multi-beam Definition: the depth was determined by using a wide swath echo sounder that uses multiple beams to measure depths directly below and transverse to the ship's track. (adapted from IHO Dictionary, S-32, 3339)
4	DD Name: HYDRO Code: TECSOU_4 Date accepted: 2000-11-01 Name: found by diver Definition: the depth was determined by a person skilled in the practice of diving. (adapted from IHO Dictionary, S-32, 1422)
5	DD Name: HYDRO Code: TECSOU_5 Date accepted: 2000-11-01 Name: found by lead-line

Value	Data Dictionary (DD) Reference		
	Definition: the depth was determined by using a line, graduated with attached marks and fastened to a sounding lead. (adapted from IHO Dictionary, S-32, 2698)		
6	DD Name: HYDRO	Code: TECSOU_6	Date accepted: 2000-11-01
	Name: swept by wire-drag		
	Definition: the given area was determined to be free from navigational dangers to a certain depth by towing a buoyed wire at the desired depth by two launches, or a least depth was identified using the same technique. (adapted from IHO Dictionary, S-32, 5248, 6013)		
8	DD Name: HYDRO	Code: TECSOU_8	Date accepted: 2000-11-01
	Name: swept by vertical acoustic system		
	Definition: the given area has been swept using a system comprised of multiple echo sounder transducers attached to booms deployed from the survey vessel.		
9	DD Name: HYDRO	Code: TECSOU_9	Date accepted: 2000-11-01
	Name: found by electromagnetic sensor		
	Definition: the depth was determined by using an instrument that compares electromagnetic signals. (adapted from IHO Dictionary, S-32, 1571)		
10	DD Name: HYDRO	Code: TECSOU_10	Date accepted: 2000-11-01
	Name: photogrammetry		
	Definition: the depth was determined by applying mathematical techniques to photographs. (adapted from IHO Dictionary, S-32, 3791)		
11	DD Name: HYDRO	Code: TECSOU_11	Date accepted: 2000-11-01
	Name: satellite imagery		
	Definition: the depth was determined by using instruments placed aboard an artificial satellite. (adapted from IHO Dictionary, S-32, 4509)		
12	DD Name: HYDRO	Code: TECSOU_12	Date accepted: 2000-11-01
	Name: found by levelling		
	Definition: the depth was determined by using levelling techniques to find the elevation of the point relative to a datum. (adapted from IHO Dictionary, S-32, 2741)		
13	DD Name: HYDRO	Code: TECSOU_13	Date accepted: 2000-11-01
	Name: swept by side-scan-sonar		
	Definition: the given area was determined to be free from navigational dangers to a certain depth by towing a side-scan-sonar. (adapted from IHO Dictionary, S-32, 5248, 4710) [415.2]		
15	DD Name: HYDRO	Code: TECSOU_15	Date accepted: 2020-04-21

Value	Data Dictionary (DD) Reference		
	Name:	found by LIDAR	
	Definition:	the depth was measured by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.	
16	DD Name: HYDRO	Code: TECSOU_16	Date accepted: 2020-04-21
	Name:	synthetic aperture radar	
	Definition:	the depth was determined using radar with a synthetic aperture antenna which is composed of a large number of elementary transducing elements. The signals are electronically combined into a resulting signal equivalent to that of a single antenna of a given aperture in a given direction.	
17	DD Name: HYDRO	Code: TECSOU_17	Date accepted: 2020-04-21
	Name:	hyperspectral imagery	
	Definition:	the depth was determined using imagery derived from subdividing the electromagnetic spectrum into very narrow bandwidths. These bandwidths may be combined with or subtracted from each other in various ways to form images useful in precise terrain or target analysis.	

Attribute	Textual description
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Acronym: TXTDSC Code: 158
Use Type: F
Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The string encodes the file name of an external text file that contains the text in English

Attribute	Textual description in national language
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Acronym: NTXTDS Code: 304
Use Type: N
Value Type: T

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The file name of an external text file that contains the text in a national language.

Attribute	Topmark/daymark shape
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Acronym: TOPSHP Code: 171

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: HYDRO Code: TOPSHP_1 Date accepted: 2000-11-01 Name: cone, point up Definition: is where the vertex points up.
2	DD Name: HYDRO Code: TOPSHP_2 Date accepted: 2000-11-01 Name: cone, point down Definition: is where the vertex points down.
3	DD Name: HYDRO Code: TOPSHP_3 Date accepted: 2000-11-01 Name: sphere Definition: a body the surface of which is at all points equidistant from the centre. (The New Shorter Oxford English Dictionary. 1993. vol 2). Spheres are commonly used as International Association of Lighthouse Authorities - IALA topmarks (safe water).
4	DD Name: HYDRO Code: TOPSHP_4 Date accepted: 2000-11-01 Name: 2 spheres Definition: two black spheres are commonly used as an International Association of Lighthouse Authorities - IALA topmark (isolated danger).
5	DD Name: HYDRO Code: TOPSHP_5 Date accepted: 2000-11-01 Name: cylinder (can) Definition: a solid geometrical figure generated by straight lines fixed in direction and describing with one of point a closed curve, especially a circle (in which case the figure is circular cylinder, it's ends being parallel circles). (The New Shorter Oxford English Dictionary. 1993. vol 2). Cylinders are commonly used as

Value	Data Dictionary (DD) Reference		
	International Association of Lighthouse Authorities - IALA topmarks (lateral).		
6	DD Name: HYDRO	Code: TOPSHP_6	Date accepted: 2000-11-01
	Name: board		
	Definition: usually of rectangular shape, made from timber or metal and used to provide a contrast with the natural background of a daymark. The actual daymark is often painted on to this board.		
7	DD Name: HYDRO	Code: TOPSHP_7	Date accepted: 2000-11-01
	Name: x-shape (St. Andrew's cross)		
	Definition: having a shape or a cross-section like the capital letter X. (The New Shorter Oxford English Dictionary. 1993. vol 2). An x-shape as an International Association of Lighthouse Authorities - IALA topmark should be 3 dimensional in shape. It is made of at least three crossed bars.		
8	DD Name: HYDRO	Code: TOPSHP_8	Date accepted: 2000-11-01
	Name: upright cross (St George's cross)		
	Definition: a cross with one vertical member and one horizontal member, i.e. similar in shape to the character '+'.		
9	DD Name: HYDRO	Code: TOPSHP_9	Date accepted: 2000-11-01
	Name: cube, point up		
	Definition: a cube standing on one of its vertexes.		
10	DD Name: HYDRO	Code: TOPSHP_10	Date accepted: 2000-11-01
	Name: 2 cones, point to point		
	Definition: 2 cones, one above the other, with their vertices together in the centre.		
11	DD Name: HYDRO	Code: TOPSHP_11	Date accepted: 2000-11-01
	Name: 2 cones, base to base		
	Definition: 2 cones, one above the other, with their bases together in the centre and their vertices pointing up and down.		
12	DD Name: HYDRO	Code: TOPSHP_12	Date accepted: 2000-11-01
	Name: rhombus (diamond)		
	Definition: a plane figure having four equal sides and equal opposite angles (two acute and two obtuse); an oblique equilateral parallelogram. (The New Shorter Oxford English Dictionary. 1993. vol 2)		
13	DD Name: HYDRO	Code: TOPSHP_13	Date accepted: 2000-11-01
	Name: 2 cones (points upward)		
	Definition: 2 cones, one above the other, with their their vertices pointing up		

Value	Data Dictionary (DD) Reference		
14	DD Name: HYDRO	Code: TOPSHP_14	Date accepted: 2000-11-01
	Name: 2 cones (points downward)		
	Definition: 2 cones, one above the other, with their their vertices pointing down		
15	DD Name: HYDRO	Code: TOPSHP_15	Date accepted: 2000-11-01
	Name: besom, point up (broom or perch)		
	Definition: a bundle of rods or twigs. (The New Shorter Oxford English Dictionary. 1993. vol 1)		
16	DD Name: HYDRO	Code: TOPSHP_16	Date accepted: 2000-11-01
	Name: besom, point down (broom or perch)		
	Definition: a bundle of rods or twigs. (The New Shorter Oxford English Dictionary. 1993. vol 1)		
17	DD Name: HYDRO	Code: TOPSHP_17	Date accepted: 2000-11-01
	Name: flag		
	Definition: a flag mounted on a short pole.		
18	DD Name: HYDRO	Code: TOPSHP_18	Date accepted: 2000-11-01
	Name: sphere over rhombus		
	Definition: A sphere located above a rhombus.		
19	DD Name: HYDRO	Code: TOPSHP_19	Date accepted: 2000-11-01
	Name: square		
	Definition: a plane figure with four right angles and four equal straight sides (The New Shorter Oxford English Dictionary. 1993. vol 2)		
20	DD Name: HYDRO	Code: TOPSHP_20	Date accepted: 2000-11-01
	Name: rectangle, horizontal		
	Definition: a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length where the two longer opposite sides are standing horizontally (The New Shorter Oxford English Dictionary. 1993. vol 2).		
21	DD Name: HYDRO	Code: TOPSHP_21	Date accepted: 2000-11-01
	Name: rectangle, vertical		
	Definition: a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length where the two longer opposite sides are standing vertically (The New Shorter Oxford English Dictionary. 1993. vol 2).		
22	DD Name: HYDRO	Code: TOPSHP_22	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	trapezium, up	
	Definition:	a quadrilateral having one pair of opposite sides parallel which stands on its longer parallel side. (The New Shorter Oxford English Dictionary. 1993. vol 2).	
23	DD Name: HYDRO	Code: TOPSHP_23	Date accepted: 2000-11-01
	Name:	trapezium, down	
	Definition:	a quadrilateral having one pair of opposite sides parallel which stands on its shorter parallel side. (The New Shorter Oxford English Dictionary. 1993. vol 2).	
24	DD Name: HYDRO	Code: TOPSHP_24	Date accepted: 2000-11-01
	Name:	triangle, point up	
	Definition:	a figure having three angles and three sides with point up. (New Shorter Oxford English Dictionary. 1993. vol 2)	
25	DD Name: HYDRO	Code: TOPSHP_25	Date accepted: 2000-11-01
	Name:	triangle, point down	
	Definition:	a figure having three angles and three sides with point down. (New Shorter Oxford English Dictionary. 1993. vol 2)	
26	DD Name: HYDRO	Code: TOPSHP_26	Date accepted: 2000-11-01
	Name:	circle	
	Definition:	a perfectly round plane figure whose circumference is everywhere equidistant from its centre. (The New Shorter Oxford English Dictionary. 1993. vol 1)	
27	DD Name: HYDRO	Code: TOPSHP_27	Date accepted: 2000-11-01
	Name:	two upright crosses (one over the other)	
	Definition:	two upright crosses, generally vertically disposed one above the other.	
28	DD Name: HYDRO	Code: TOPSHP_28	Date accepted: 2000-11-01
	Name:	T-shape	
	Definition:	having a shape like the capital letter T.	
29	DD Name: HYDRO	Code: TOPSHP_29	Date accepted: 2000-11-01
	Name:	triangle pointing up over a circle	
	Definition:	a triangle, vertex uppermost, located above a circle.	
30	DD Name: HYDRO	Code: TOPSHP_30	Date accepted: 2000-11-01
	Name:	upright cross over a circle	
	Definition:	an upright cross located above a circle.	

Value	Data Dictionary (DD) Reference		
31	DD Name: HYDRO	Code: TOPSHP_31	Date accepted: 2000-11-01
	Name:	rhombus over a circle	
	Definition:	a rhombus located above a circle.	
32	DD Name: HYDRO	Code: TOPSHP_32	Date accepted: 2000-11-01
	Name:	circle over a triangle pointing up	
	Definition:	a circle located over a triangle, vertex uppermost.	
33	DD Name: HYDRO	Code: TOPSHP_33	Date accepted: 2000-11-01
	Name:	other shape (see INFORM)	
	Definition:		

Attribute	Traffic flow
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Acronym: TRAFIC Code: 172
Use Type: F
Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: TRAFIC_1 Date accepted: 2000-11-01

Name: inbound

Definition: traffic flow in a general direction toward a port or similar destination.

2 DD Name: HYDRO Code: TRAFIC_2 Date accepted: 2000-11-01

Name: outbound

Definition: traffic flow in a general direction away from a port or similar point of origin.

3 DD Name: HYDRO Code: TRAFIC_3 Date accepted: 2000-11-01

Name: one-way

Definition: traffic flow in one general direction only.

4 DD Name: HYDRO Code: TRAFIC_4 Date accepted: 2000-11-01

Name: two-way

Definition: traffic flow in two generally opposite directions.

Attribute	Value of annual change in magnetic variation
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Acronym: VALACM

Code: 173

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The annual change in magnetic variation values.

Attribute	Value of depth contour
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Acronym: VALDCO

Code: 174

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The depth of a sea bottom contour.

Attribute	Value of magnetic variation
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Acronym: VALMAG

Code: 176

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The magnetic variation value.

Attribute	Value of maximum range
-----------	------------------------

Acronym: VALMXR

Code: 177

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The extreme distance at which an object can be seen or a signal detected.

Attribute	Value of nominal range
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Acronym: VALNMR

Code: 178

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The nominal range at which an object can be seen or a signal detected

Attribute	Value of sounding
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Acronym: VALSOU

Code: 179

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The value of the measurement of a sounding relative to the chart datum.

Attribute	Vertical Accuracy
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Acronym: VERACC Code: 180
Use Type: F
Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The one-dimensional error. The error is assumed to be positive and negative. The plus/minus character shall not be encoded.

Attribute	Vertical clearance
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Acronym: VERCLR

Code: 181

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The vertical clearance measured from the plane towards the object overhead.

Attribute	Vertical clearance, closed
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Acronym: VERCCL Code: 182

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The vertical clearance of an object in closed condition (e.g. a closed lifting bridge) measured from the plane towards the object overhead.

Attribute	Vertical clearance, open
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Acronym: VERCOP Code: 183

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: The vertical clearance of an object in opened condition (e.g. an opened lifting bridge) measured from the plane towards the object overhead.

Attribute	Vertical datum
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Acronym: VERDAT Code: 185

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2001-05-31

Definition: Vertical datum

Enumerations:

Value	Data Dictionary (DD) Reference
4	DD Name: HYDRO Code: VERDAT_4 Date accepted: 2001-05-31 Name: Lowest low water Definition: an arbitrary level conforming to the lowest tide observed at a place, or some what lower.
24	DD Name: HYDRO Code: VERDAT_24 Date accepted: 2000-11-01 Name: Local datum Definition: an arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
23	DD Name: HYDRO Code: VERDAT_23 Date accepted: 2000-11-01 Name: Lowest astronomical tide Definition: (LAT) - the lowest tide level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions. (IHO Dictionary, S-32, 5th Edition, 2936)
30	DD Name: HYDRO Code: VERDAT_30 Date accepted: 2000-11-01 Name: Highest astronomical tide Definition: (HAT) - the highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions. (IHO Dictionary, S-32, 5th Edition, 2244).

Attribute	Vertical length
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Acronym: VERLEN

Code: 186

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition: Vertical length

Attribute	Water level effect
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Acronym: WATLEV

Code: 187

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: HYDRO Date accepted: 2000-11-01

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: HYDRO Code: WATLEV_1 Date accepted: 2000-11-01

Name: partly submerged at high water

Definition: partially covered and partially dry at high water.

2 DD Name: HYDRO Code: WATLEV_2 Date accepted: 2000-11-01

Name: always dry

Definition: not covered at high water under average meteorological conditions.

3 DD Name: HYDRO Code: WATLEV_3 Date accepted: 2000-11-01

Name: always under water/submerged

Definition: remains covered by water at all times under average meteorological conditions.

4 DD Name: HYDRO Code: WATLEV_4 Date accepted: 2000-11-01

Name: covers and uncovers

Definition: expression intended to indicate an area of a reef or other projection from the bottom of a body of water which periodically extends above and is submerged below the surface. Also referred to as dries or uncovers. (IHO Dictionary, S-32, 5th Edition, 1111)

5 DD Name: HYDRO Code: WATLEV_5 Date accepted: 2000-11-01

Name: awash

Definition: flush with, or washed by the waves at low water under average meteorological conditions. (adapted from IHO Dictionary, S-32, 5th Edition, 308)

7 DD Name: HYDRO Code: WATLEV_7 Date accepted: 2000-11-01

Value Data Dictionary (DD) Reference

Name: floating

Definition: resting or moving on the surface of a liquid without sinking (Concise Oxford Dictionary)

Attribute	Additional mark
-----------	-----------------

Acronym: addmrk

Code: 17050

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Shape and position of an additional board on a notice mark

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: addmrk_1 Date accepted: 2001-05-31

Name: top (board)

Definition: a rectangular board at the top of the main sign

2 DD Name: IENC Code: addmrk_2 Date accepted: 2001-05-31

Name: bottom (board)

Definition: a rectangular board at the bottom of the main sign

3 DD Name: IENC Code: addmrk_3 Date accepted: 2001-05-31

Name: right (triangle to the right)

Definition: a triangular board at the right side of the main sign

4 DD Name: IENC Code: addmrk_4 Date accepted: 2001-05-31

Name: left (triangle to the left)

Definition: a triangular board at the left side of the main sign

5 DD Name: IENC Code: addmrk_5 Date accepted: 2001-05-31

Name: bottom (triangle to the bottom)

Definition: a triangular board at the bottom of the main sign

Attribute	Allowed consumption
-----------	---------------------

Acronym: allcon

Code: 18033

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26

Definition: The maximum allowed power that may be used by the vessel

Attribute	Amount of amperage
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Acronym: amoamp

Code: 18032

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26

Definition: The maximum electric amperage possible

Attribute	Assemblies of ship (excluding)
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Acronym: lc_ase

Code: 18015

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Excluding list of assemblies of ships for the applicability of a feature

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lc_ase_1 Date accepted: 2001-05-31

Name: all types

Definition:

2 DD Name: IENC Code: lc_ase_2 Date accepted: 2001-05-31

Name: other

Definition:

3 DD Name: IENC Code: lc_ase_3 Date accepted: 2001-05-31

Name: single vessel

Definition:

5 DD Name: IENC Code: lc_ase_5 Date accepted: 2001-05-31

Name: convoy

Definition: a rigid or towed convoy of craft

6 DD Name: IENC Code: lc_ase_6 Date accepted: 2001-05-31

Name: formation

Definition: the manner in which a convoy is assembled

7 DD Name: IENC Code: lc_ase_7 Date accepted: 2001-05-31

Name: rigid convoy

Value	Data Dictionary (DD) Reference		
	Definition: a pushed convoy or breasted up formation		
8	DD Name: IENC	Code: lc_ase_8	Date accepted: 2001-05-31
	Name: pushed convoy		
	Definition: a rigid assembly of craft of which at least one is positioned in front of the craft providing the power for propelling the convoy, known as the "pusher(s)"; a convoy composed of a pusher craft and a pushed craft coupled so as to permit guided articulation is also considered as rigid		
9	DD Name: IENC	Code: lc_ase_9	Date accepted: 2001-05-31
	Name: breasted up formation		
	Definition: an assembly of craft coupled rigidly side by side, none of which is positioned in front of the craft propelling the assembly		
10	DD Name: IENC	Code: lc_ase_10	Date accepted: 2001-05-31
	Name: towed convoy		
	Definition: an assembly of one or more craft, floating establishments or floating installations towed by one or more self-propelled craft forming part of the convoy		

Attribute	Assemblies of ship (including)
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Acronym: lc_asl

Code: 18014

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Including list of assemblies of ships for the applicability of a feature

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lc_asl_1 Date accepted: 2001-05-31

Name: all types

Definition:

2 DD Name: IENC Code: lc_asl_2 Date accepted: 2001-05-31

Name: other

Definition:

3 DD Name: IENC Code: lc_asl_3 Date accepted: 2001-05-31

Name: single vessel

Definition:

5 DD Name: IENC Code: lc_asl_5 Date accepted: 2001-05-31

Name: convoy

Definition: a rigid or towed convoy of craft

6 DD Name: IENC Code: lc_asl_6 Date accepted: 2001-05-31

Name: formation

Definition: the manner in which a convoy is assembled

7 DD Name: IENC Code: lc_asl_7 Date accepted: 2001-05-31

Name: rigid convoy

Value	Data Dictionary (DD) Reference		
	Definition: a pushed convoy or breasted up formation		
8	DD Name: IENC	Code: lc_asl_8	Date accepted: 2001-05-31
	Name: pushed convoy		
	Definition: a rigid assembly of craft of which at least one is positioned in front of the craft providing the power for propelling the convoy, known as the "pusher(s)"; a convoy composed of a pusher craft and a pushed craft coupled so as to permit guided articulation is also considered as rigid		
9	DD Name: IENC	Code: lc_asl_9	Date accepted: 2001-05-31
	Name: breasted up formation		
	Definition: an assembly of craft coupled rigidly side by side, none of which is positioned in front of the craft propelling the assembly		
10	DD Name: IENC	Code: lc_asl_10	Date accepted: 2001-05-31
	Name: towed convoy		
	Definition: an assembly of one or more craft, floating establishments or floating installations towed by one or more self-propelled craft forming part of the convoy		

Attribute	Average Passing Time Reference
-----------	--------------------------------

Acronym: aptref

Code: 17099

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The string encodes the file name of an external file

Attribute	Bank of the waterway
-----------	----------------------

Acronym: bnkwtw

Code: 17105

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2009-09-11

Definition: Bank of the river (waterway)

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: bnkwtw_1 Date accepted: 2009-09-11

Name: left

Definition: left bank of the river

2 DD Name: IENC Code: bnkwtw_2 Date accepted: 2009-09-11

Name: right

Definition: right bank of the river

Attribute	Bunker vessel, availability
-----------	-----------------------------

Acronym: bunves

Code: 17065

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Indication of the availability of a bunker vessel

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: bunves_1 Date accepted: 2001-05-31

Name: bunker vessel available

Definition: a bunker vessel is available

2 DD Name: IENC Code: bunves_2 Date accepted: 2001-05-31

Name: no bunker vessel available

Definition: a bunker vessel is not available

Attribute	Category of anchorage
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Acronym: catach

Code: 17000

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of anchorage

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: IENC Code: catach_1 Date accepted: 2001-05-31
Name: unrestricted anchorage
Definition: an area in which vessels anchor or may anchor. (IHO Dictionary, S-32, 5th Edition, 130) |
| 2 | DD Name: IENC Code: catach_2 Date accepted: 2001-05-31
Name: deep water anchorage
Definition: an area in which vessels of deep draught anchor or may anchor. |
| 3 | DD Name: IENC Code: catach_3 Date accepted: 2001-05-31
Name: tanker anchorage
Definition: an area in which tankers anchor or may anchor. |
| 4 | DD Name: IENC Code: catach_4 Date accepted: 2001-05-31
Name: explosives anchorage
Definition: an area set apart for anchored ships discharging or receiving explosives. (IHO Dictionary, S-32, 5th Edition, 1732) |
| 5 | DD Name: IENC Code: catach_5 Date accepted: 2001-05-31
Name: quarantine anchorage
Definition: an area where a vessel anchors when satisfying quarantine regulations. (IHO Dictionary, S-32, 5th Edition, 4117) |
| 6 | DD Name: IENC Code: catach_6 Date accepted: 2001-05-31 |

Value	Data Dictionary (DD) Reference		
	Name:	sea-plane anchorage	
	Definition:	an area in which sea-planes anchor or may anchor.	
7	DD Name: IENC	Code: catach_7	Date accepted: 2001-05-31
	Name:	small craft anchorage	
	Definition:	an area in which yachts and small boats anchor or may anchor.	
9	DD Name: IENC	Code: catach_9	Date accepted: 2001-05-31
	Name:	anchorage for periods up to 24 hours	
	Definition:	an area in which vessels anchor or may anchor for periods of up to 24 hours.	
10	DD Name: IENC	Code: catach_10	Date accepted: 2001-05-31
	Name:	anchorage for pushing-navigation vessels	
	Definition:	an area where pushing-navigation vessels may anchor	
11	DD Name: IENC	Code: catach_11	Date accepted: 2001-05-31
	Name:	anchorage for other vessels than pushing-navigation vessels	
	Definition:	an area where other vessels than pushing-navigation vessels may anchor	
12	DD Name: IENC	Code: catach_12	Date accepted: 2009-12-09
	Name:	anchorage for dry cargo vessels	
	Definition:	an area where dry cargo vessels may anchor	
13	DD Name: IENC	Code: catach_13	Date accepted: 2009-12-09
	Name:	anchorage for rafts	
	Definition:	an area where rafts may anchor	

Attribute	Category of berth
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Acronym: catbrt

Code: 17066

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of berth

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catbrt_1 Date accepted: 2001-05-31

Name: loading

Definition: A place where vessels may berth for loading cargo.

2 DD Name: IENC Code: catbrt_2 Date accepted: 2001-05-31

Name: unloading

Definition: A place where vessels may berth for unloading cargo.

3 DD Name: IENC Code: catbrt_3 Date accepted: 2001-05-31

Name: overnight accommodation

Definition: Berths that are suitable/ meant for berthing overnight.

4 DD Name: IENC Code: catbrt_4 Date accepted: 2001-05-31

Name: berth for pushing-navigation vessels

Definition: an place where pushing-navigation vessels may berth.

5 DD Name: IENC Code: catbrt_5 Date accepted: 2001-05-31

Name: berth for other vessels than pushing-navigation vessels

Definition: an place where other vessels than pushing-navigation vessels may berth.

6 DD Name: IENC Code: catbrt_6 Date accepted: 2001-05-31

Name: fleeting area

Value	Data Dictionary (DD) Reference		
	Definition: A legally permitted area in or near the waterway designated for temporary barge mooring.		
7	DD Name: IENC	Code: catbrt_7	Date accepted: 2001-05-31
	Name: first class landing		
	Definition: A federally designated area that provides tie-ups and at least 9 feet (2.7m) of water depth during low water level.		
8	DD Name: IENC	Code: catbrt_8	Date accepted: 2001-05-31
	Name: second class landing		
	Definition: A federally designated area that provides tie-ups and at least 9 feet (2.7m) of water depth normal pool level.		
9	DD Name: IENC	Code: catbrt_9	Date accepted: 2012-12-19
	Name: berth for passenger vessels		
	Definition:		
10	DD Name: IENC	Code: catbrt_10	Date accepted: 2023-11-22
	Name: waiting berth		
	Definition: a berth that is dedicated for vessels waiting for the opening of a bridge, lockage or the use of other infrastructure. When a berth is dedicated to waiting, but can be used for resting with permission of the operator of the infrastructure, it is also a waiting berth.		

Attribute	Category of bunker station
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Acronym: catbun

Code: 17067

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of bunker station

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catbun_1 Date accepted: 2001-05-31

Name: diesel oil

Definition: diesel oil available

2 DD Name: IENC Code: catbun_2 Date accepted: 2001-05-31

Name: water

Definition: water available

3 DD Name: IENC Code: catbun_3 Date accepted: 2001-05-31

Name: ballast

Definition: ballast available

4 DD Name: IENC Code: catbun_4 Date accepted: 2001-05-31

Name: power

Definition: power supply available

5 DD Name: IENC Code: catbun_5 Date accepted: 2021-04-01

Name: compressed hydrogen bunkering

Definition: compressed hydrogen available for bunkering

6 DD Name: IENC Code: catbun_6 Date accepted: 2024-04-01

Name: liquefied hydrogen bunkering

Value	Data Dictionary (DD) Reference		
	Definition: liquefied hydrogen available for bunkering		
7	DD Name: IENC	Code: catbun_7	Date accepted: 2024-04-01
	Name: methanol bunkering		
	Definition: methanol available for bunkering		
8	DD Name: IENC	Code: catbun_8	Date accepted: 2024-04-01
	Name: ammonia bunkering		
	Definition: ammonia available for bunkering		

Attribute	Category of cable
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Acronym: catchbl

Code: 17101

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of cable

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catchbl_1 Date accepted: 2001-05-31

Name: power line

Definition: a cable used for the supply of electricity.

3 DD Name: IENC Code: catchbl_3 Date accepted: 2001-05-31

Name: transmission line

Definition: multiple un-insulated cables usually supported by steel lattice towers. Such features are generally more prominent than normal power lines.

4 DD Name: IENC Code: catchbl_4 Date accepted: 2001-05-31

Name: telephone

Definition: a cable used for the transmission of telephone signals.

5 DD Name: IENC Code: catchbl_5 Date accepted: 2001-05-31

Name: telegraph

Definition: a cable used for the transmission of telegraph signals.

6 DD Name: IENC Code: catchbl_6 Date accepted: 2001-05-31

Name: mooring cable/chain

Definition: a cable or chain used to secure a mooring buoy or other floating structure.

7 DD Name: IENC Code: catchbl_7 Date accepted: 2001-05-31

Name: ferry cable

Value Data Dictionary (DD) Reference

Definition: a cable where a cable ferry is connected to

Attribute	Category of cargo (excluding)
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Acronym: lc_cce

Code: 18017

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Excluding list of categories of cargo for the applicability of a feature

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lc_cce_1 Date accepted: 2001-05-31

Name: all types

Definition:

2 DD Name: IENC Code: lc_cce_2 Date accepted: 2001-05-31

Name: other

Definition:

4 DD Name: IENC Code: lc_cce_4 Date accepted: 2001-05-31

Name: bulk

Definition: unpacked homogenous cargo poured loose in a certain space of a vessel e.g. oil or grain

5 DD Name: IENC Code: lc_cce_5 Date accepted: 2001-05-31

Name: dry cargo

Definition:

6 DD Name: IENC Code: lc_cce_6 Date accepted: 2001-05-31

Name: liquid cargo

Definition:

7 DD Name: IENC Code: lc_cce_7 Date accepted: 2001-05-31

Name: liquid cargo (type N)

Value	Data Dictionary (DD) Reference
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	Definition:
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8	DD Name: IENC	Code: lc_cce_8	Date accepted: 2001-05-31
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	Name: liquid cargo (type C)
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	Definition:
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9	DD Name: IENC	Code: lc_cce_9	Date accepted: 2001-05-31
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	Name: gas
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	Definition:
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Attribute	Category of cargo (including)
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Acronym: lc_cci Code: 18016

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Including list of categories of cargo for the applicability of a feature

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: IENC Code: lc_cci_1 Date accepted: 2001-05-31 Name: all types Definition:
2	DD Name: IENC Code: lc_cci_2 Date accepted: 2001-05-31 Name: other Definition:
4	DD Name: IENC Code: lc_cci_4 Date accepted: 2001-05-31 Name: bulk Definition: unpacked homogenous cargo poured loose in a certain space of a vessel e.g. oil or grain
5	DD Name: IENC Code: lc_cci_5 Date accepted: 2001-05-31 Name: dry cargo Definition:
6	DD Name: IENC Code: lc_cci_6 Date accepted: 2001-05-31 Name: liquid cargo Definition:
7	DD Name: IENC Code: lc_cci_7 Date accepted: 2001-05-31 Name: liquid cargo (type N)

Value	Data Dictionary (DD) Reference
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	Definition:
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8	DD Name: IENC	Code: lc_cci_8	Date accepted: 2001-05-31
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	Name: liquid cargo (type C)
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	Definition:
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9	DD Name: IENC	Code: lc_cci_9	Date accepted: 2001-05-31
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	Name: gas
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	Definition:
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Attribute	Category of CEMT class
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Acronym: catccl

Code: 17068

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of CEMT class

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catccl_1 Date accepted: 2001-05-31

Name: 0 small vessels and pleasure craft

Definition: designated for small vessels and pleasure crafts only

2 DD Name: IENC Code: catccl_2 Date accepted: 2001-05-31

Name: I peniche

Definition: designated for barges of type "Péniche" (west of river Elbe) or of type "Gross Finow" (east of river Elbe)

3 DD Name: IENC Code: catccl_3 Date accepted: 2001-05-31

Name: II campine barge

Definition: designated for barges of type "Kempenaar" (west of river Elbe) or of type "BM-500" (east of river Elbe)

4 DD Name: IENC Code: catccl_4 Date accepted: 2001-05-31

Name: III Dortmund-Ems barge

Definition: designated for barges of type "Gustav Koenigs" (west of river Elbe) or of a similar type concerning the dimensions (east of river Elbe)

5 DD Name: IENC Code: catccl_5 Date accepted: 2001-05-31

Name: IV Rhine-Herne barge

Definition: designated for barges of type "Johann Welker"

6 DD Name: IENC Code: catccl_6 Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	Va Large Rhine barge; 1-barge push-tow unit	
	Definition:	designated for barges of type "Large Rhine barge" or pushed convoys with one barge	
7	DD Name: IENC	Code: catccl_7	Date accepted: 2001-05-31
	Name:	Vb 2-barge push-tow unit; long formation	
	Definition:	designated for pushed convoys with two barges, long formation	
8	DD Name: IENC	Code: catccl_8	Date accepted: 2001-05-31
	Name:	Vla 2-barge push-tow unit; wide formation	
	Definition:	designated for pushed convoys with two barges, wide formation	
9	DD Name: IENC	Code: catccl_9	Date accepted: 2001-05-31
	Name:	Vlb 4-barge push-tow unit	
	Definition:	designated for pushed convoys with four barges	
10	DD Name: IENC	Code: catccl_10	Date accepted: 2001-05-31
	Name:	Vlc 6-barge push-tow unit	
	Definition:	designated for pushed convoys with six barges	
11	DD Name: IENC	Code: catccl_11	Date accepted: 2001-05-31
	Name:	No CEMT class	
	Definition:		
12	DD Name: IENC	Code: catccl_12	Date accepted: 2011-10-04
	Name:	VII 9-barge push-town unit	
	Definition:		

Attribute	Category of checkpoint
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Acronym: catchp Code: 17010
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31
Definition: Category of checkpoint

Enumerations:

Value Data Dictionary (DD) Reference

- 1 DD Name: IENC Code: catchp_1 Date accepted: 2001-05-31
Name: custom
Definition: an office, especially in ports, at which customs dues are collected or administrated. (adapted from The New Shorter Oxford English Dictionary, 1993)
- 2 DD Name: IENC Code: catchp_2 Date accepted: 2001-05-31
Name: border
Definition: an office, at which immigration control takes place

Attribute	Category of communication
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Acronym: catcom

Code: 17069

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of communication

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catcom_1 Date accepted: 2001-05-31

Name: VTS centre

Definition: The centre from which Vessel Traffic Services are operated. A VTS is a service implemented by a competent authority, designed to improve the safety and efficiency of vessel traffic and to protect the environment. The services should have the capability to interact with the traffic and to respond to traffic situations developing in the area.

2 DD Name: IENC Code: catcom_2 Date accepted: 2001-05-31

Name: VTS sector

Definition: The service area of a VTS centre.

3 DD Name: IENC Code: catcom_3 Date accepted: 2001-05-31

Name: IVS point

Definition: A reporting point of the "Informatie en Volgsysteem voor de Scheepvaart" in the Netherlands.

4 DD Name: IENC Code: catcom_4 Date accepted: 2001-05-31

Name: MIB

Definition: A reporting point of the "Melde- und Informationssystem Binnenschifffahrt" in Germany.

5 DD Name: IENC Code: catcom_5 Date accepted: 2001-05-31

Name: lock

Definition: A reporting point for vessels at a lock.

6 DD Name: IENC Code: catcom_6 Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	bridge	
	Definition:	A reporting point for vessels at a movable bridge.	
7	DD Name: IENC	Code: catcom_7	Date accepted: 2001-05-31
	Name:	custom	
	Definition:	A reporting point of the customs services for vessels.	
8	DD Name: IENC	Code: catcom_8	Date accepted: 2001-05-31
	Name:	harbour	
	Definition:	A reporting point of a harbour.	
9	DD Name: IENC	Code: catcom_9	Date accepted: 2011-12-09
	Name:	WLAN area	
	Definition:	An area where free wireless network is available	

Attribute	Category of exceptional structure
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Acronym: catexs

Code: 17100

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of exceptional navigational structure

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: IENC Code: catexs_1 Date accepted: 2001-05-31
Name: Lift-Lock
Definition: A lock of which the lock chamber itself is lifted vertically to level with the next waterway section |
| 2 | DD Name: IENC Code: catexs_2 Date accepted: 2001-05-31
Name: Aqueduct
Definition: A structure (similar to the ancient aqueducts), for conveying a canal over a river or hollow; more properly called an aqueduct bridge. (From Webster's Revised Unabridged Dictionary, 1913) |
| 3 | DD Name: IENC Code: catexs_3 Date accepted: 2001-05-31
Name: Sloping plane lock
Definition: A lock of which the lock chamber itself travels over a sloping plane to level with the next waterway section |
| 4 | DD Name: IENC Code: catexs_4 Date accepted: 2001-05-31
Name: Water slope lock (Pente d'Eau)
Definition: In French "Pente d'Eau". A lock of which the lock chamber is formed by a sloping plane and moving gate, which is pushing a triangular section of water up along the slope to level with the next waterway section |
| 5 | DD Name: IENC Code: catexs_5 Date accepted: 2001-05-31
Name: Other
Definition: other categories of an exceptional structure |

Attribute	Category of ferry
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Acronym: catfry Code: 17007

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of ferry

Enumerations:

Value Data Dictionary (DD) Reference

4 DD Name: IENC Code: catfry_4 Date accepted: 2001-05-31

Name: swinging wire ferry

Definition: ferry connected to a fixed point (e.g., an anchor in the middle of the waterway) and swings around this point from shore to shore via a cable to an anchor. The cable runs more or less parallel to the current

Attribute	Category of frequency
-----------	-----------------------

Acronym: catfrq Code: 18030
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26
Definition: The electrical frequency provided by the power supply station

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: IENC Code: catfrq_1 Date accepted: 2014-11-26 Name: 50Hz Definition: 50 Hertz
2	DD Name: IENC Code: catfrq_2 Date accepted: 2014-11-26 Name: 60Hz Definition: 60 Hertz

Attribute	Category of harbour area
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Acronym: cathbr

Code: 17070

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of harbour

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: cathbr_1 Date accepted: 2001-05-31

Name: custom harbour

Definition: A harbour that is administered by the customs. It may be a free harbour.

2 DD Name: IENC Code: cathbr_2 Date accepted: 2001-05-31

Name: port of refuge

Definition: A harbour that can be used to find shelter for bad environmental conditions or where efforts to mitigate larger damage or threat(s) of damage to either the vessel, her crew or the environment can be rendered.

3 DD Name: IENC Code: cathbr_3 Date accepted: 2001-05-31

Name: yacht harbour/marina

Definition: a harbour with facilities for small boats and yachts (IHO Dictionary, S-32, 5th Edition, 3095).

4 DD Name: IENC Code: cathbr_4 Date accepted: 2001-05-31

Name: fishing harbour

Definition: a harbour with facilities for fishing boats.

5 DD Name: IENC Code: cathbr_5 Date accepted: 2001-05-31

Name: private harbour

Definition: a harbour operated by a private body.

Attribute	Category of harbour facility
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Acronym: cathaf

Code: 17008

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of harbour facility

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: cathaf_1 Date accepted: 2001-05-31

Name: RoRo-terminal

Definition: a terminal for roll-on roll-off ferries.

3 DD Name: IENC Code: cathaf_3 Date accepted: 2001-05-31

Name: ferry terminal

Definition: a terminal for passenger and vehicle ferries.

4 DD Name: IENC Code: cathaf_4 Date accepted: 2001-05-31

Name: fishing harbour

Definition: a harbour with facilities for fishing boats.

6 DD Name: IENC Code: cathaf_6 Date accepted: 2001-05-31

Name: naval base

Definition: a centre of operations for naval vessels (adapted from The Collins Dictionary).

7 DD Name: IENC Code: cathaf_7 Date accepted: 2001-05-31

Name: tanker terminal

Definition: a terminal for the bulk handling of liquid cargoes.

8 DD Name: IENC Code: cathaf_8 Date accepted: 2001-05-31

Name: passenger terminal

Value	Data Dictionary (DD) Reference		
	Definition:	a terminal for the loading and unloading of passengers.	
9	DD Name: IENC	Code: cathaf_9	Date accepted: 2001-05-31
	Name:	shipyard	
	Definition:	a place where ships are built or repaired (IHO Dictionary, S-32, 5th Edition, 4686).	
10	DD Name: IENC	Code: cathaf_10	Date accepted: 2001-05-31
	Name:	container terminal	
	Definition:	a terminal for container ships.	
11	DD Name: IENC	Code: cathaf_11	Date accepted: 2001-05-31
	Name:	bulk terminal	
	Definition:	a terminal for the handling of bulk materials such as iron ore, coal, etc.	
12	DD Name: IENC	Code: cathaf_12	Date accepted: 2001-05-31
	Name:	syncrolift	
	Definition:	a platform powered by synchronous electric motors used to lift vessels (larger than boats) in and out of the water.	
13	DD Name: IENC	Code: cathaf_13	Date accepted: 2001-05-31
	Name:	straddle carrier	
	Definition:	a wheeled vehicle designed to lift and carry containers or vessels within its own framework. It is used for moving, and sometimes stacking, shipping containers and vessels.	
16	DD Name: IENC	Code: cathaf_16	Date accepted: 2001-05-31
	Name:	service and repair	
	Definition:	a place where mechanical services or repairs can be undertaken to engines or other vessel equipment.	
17	DD Name: IENC	Code: cathaf_17	Date accepted: 2001-05-31
	Name:	quarantine station	
	Definition:	A medical control center located in an isolated spot ashore where patients with contagious diseases from vessel in quarantine are taken.	

Attribute	Category of hulk
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Acronym: cathlk

Code: 17102

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of hulk

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: IENC Code: cathlk_1 Date accepted: 2001-05-31
Name: floating restaurant
Definition: a permanently moored floating structure, such as an old ship, used as a restaurant. |
| 2 | DD Name: IENC Code: cathlk_2 Date accepted: 2001-05-31
Name: historic ship
Definition: a ship of historical interest permanently moored as a tourist attraction. |
| 3 | DD Name: IENC Code: cathlk_3 Date accepted: 2001-05-31
Name: museum
Definition: a permanently moored floating structure, such as an old ship, used as a museum. |
| 4 | DD Name: IENC Code: cathlk_4 Date accepted: 2001-05-31
Name: accommodation
Definition: a permanently moored floating structure, such as an old ship, used for accommodation. |
| 5 | DD Name: IENC Code: cathlk_5 Date accepted: 2001-05-31
Name: floating breakwater
Definition: a permanently moored floating structure, often constructed from old ships, used as a breakwater. |
| 6 | DD Name: IENC Code: cathlk_6 Date accepted: 2001-05-31
Name: casino boat |

Value Data Dictionary (DD) Reference

Definition: a permanently moored floating structure, such as an old ship, used as a casino boat

Attribute	Category of lateral mark
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Acronym: catlam

Code: 17011

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of lateral mark

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | <p>DD Name: IENC Code: catlam_1 Date accepted: 2001-05-31</p> <p>Name: port-hand lateral mark</p> <p>Definition: indicates the port boundary of a navigational channel or suggested route when proceeding in the 'conventional direction of buoyage'.</p> |
| 2 | <p>DD Name: IENC Code: catlam_2 Date accepted: 2001-05-31</p> <p>Name: starboard-hand lateral mark</p> <p>Definition: indicates the starboard boundary of a navigational channel or suggested route when proceeding in the 'conventional direction of buoyage'.</p> |
| 3 | <p>DD Name: IENC Code: catlam_3 Date accepted: 2001-05-31</p> <p>Name: preferred channel to starboard lateral mark</p> <p>Definition: at a point where a channel divides, when proceeding in the 'conventional direction of buoyage', the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.</p> |
| 4 | <p>DD Name: IENC Code: catlam_4 Date accepted: 2001-05-31</p> <p>Name: preferred channel to port lateral mark</p> <p>Definition: at a point where a channel divides, when proceeding in the 'conventional direction of buoyage', the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.</p> |
| 5 | <p>DD Name: IENC Code: catlam_5 Date accepted: 2001-05-31</p> <p>Name: right-hand side of the waterway</p> <p>Definition: indicates the right-hand side of the inland waterway</p> |

Value	Data Dictionary (DD) Reference		
6	DD Name: IENC	Code: catlam_6	Date accepted: 2001-05-31
	Name: left-hand side of the waterway		
	Definition: indicates the left-hand side of the inland waterway		
7	DD Name: IENC	Code: catlam_7	Date accepted: 2001-05-31
	Name: right-hand side of the channel		
	Definition: indicates the right-hand side of a channel of an inland waterway		
8	DD Name: IENC	Code: catlam_8	Date accepted: 2001-05-31
	Name: left-hand side of the channel		
	Definition: indicates the left-hand side of a channel of an inland waterway		
9	DD Name: IENC	Code: catlam_9	Date accepted: 2001-05-31
	Name: bifurcation of the waterway		
	Definition: indicates a bifurcation of the inland waterway		
10	DD Name: IENC	Code: catlam_10	Date accepted: 2001-05-31
	Name: bifurcation of the channel		
	Definition: indicates a bifurcation of a channel of an inland waterway		
11	DD Name: IENC	Code: catlam_11	Date accepted: 2001-05-31
	Name: channel near the right bank		
	Definition: indicates that the channel is near the right bank		
12	DD Name: IENC	Code: catlam_12	Date accepted: 2001-05-31
	Name: channel near the left bank		
	Definition: indicates that the channel is near the left bank		
13	DD Name: IENC	Code: catlam_13	Date accepted: 2001-05-31
	Name: channel cross-over to the right bank		
	Definition: indicates that the channel crosses from the left to the right bank		
14	DD Name: IENC	Code: catlam_14	Date accepted: 2001-05-31
	Name: channel cross-over to the left bank		
	Definition: indicates that the channel crosses from the right to the left bank		
15	DD Name: IENC	Code: catlam_15	Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	danger point or obstacles at the right-hand side	
	Definition:	indicates a danger point or obstacles at the right-hand side	
16	DD Name: IENC	Code: catlam_16	Date accepted: 2001-05-31
	Name:	danger point or obstacles at the left-hand side	
	Definition:	indicates a danger point or obstacles at the left-hand side	
17	DD Name: IENC	Code: catlam_17	Date accepted: 2001-05-31
	Name:	turn off at the right-hand side	
	Definition:	indicates a turn off at the right-hand side	
18	DD Name: IENC	Code: catlam_18	Date accepted: 2001-05-31
	Name:	turn off at the left-hand side	
	Definition:	indicates a turn off at the left-hand side	
19	DD Name: IENC	Code: catlam_19	Date accepted: 2001-05-31
	Name:	junction at the right-hand side	
	Definition:	indicates a junction at the right-hand side	
20	DD Name: IENC	Code: catlam_20	Date accepted: 2001-05-31
	Name:	junction at the left-hand side	
	Definition:	indicates a junction at the left-hand side	
21	DD Name: IENC	Code: catlam_21	Date accepted: 2001-05-31
	Name:	harbour entry at the right-hand side	
	Definition:	indicates a harbour entry at the right-hand side	
22	DD Name: IENC	Code: catlam_22	Date accepted: 2001-05-31
	Name:	harbour entry at the left-hand side	
	Definition:	indicates a harbour entry at the left-hand side	
23	DD Name: IENC	Code: catlam_23	Date accepted: 2001-05-31
	Name:	bridge pier mark	
	Definition:	indicates a bridge pier in a inland waterway	
24	DD Name: IENC	Code: catlam_24	Date accepted: 2013-01-01
	Name:	entry from a lake to a narrower waterway, right bank	
	Definition:	indicates the right bank of the entry from a lake or a lake-like expansion to a section of the	

Value	Data Dictionary (DD) Reference		
	waterway which is narrower		
25	DD Name: IENC	Code: catlam_25	Date accepted: 2013-01-01
	Name: entry from a lake to a narrower waterway, left bank		
	Definition: indicates the left bank of the entry from a lake or a lakelike expansion to a section of the waterway which is narrower		
26	DD Name: IENC	Code: catlam_26	Date accepted: 2013-12-30
	Name: change bank		
	Definition:		
27	DD Name: IENC	Code: catlam_27	Date accepted: 2013-12-30
	Name: continue along bank		
	Definition:		

Attribute	Category of notice mark
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Acronym: catnmk

Code: 17052

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of notice mark

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catnmk_1 Date accepted: 2001-05-31

Name: (A.1) no entry (general sign)

Definition: Prohibition mark A.1: no entry (general sign) (Source: Annex AA of the Encoding Guide for Inland ENCs)

2 DD Name: IENC Code: catnmk_2 Date accepted: 2001-05-31

Name: (A.1.1) sections closed to use, no entry except for non-motorized small craft

Definition: Prohibition mark A.1.1: sections closed to use, no entry except for non-motorized small craft (Source: Annex AA of the Encoding Guide for Inland ENCs)

3 DD Name: IENC Code: catnmk_3 Date accepted: 2001-05-31

Name: (A.2) no overtaking

Definition: Prohibition mark A.2: no overtaking (Source: Annex AA of the Encoding Guide for Inland ENCs)

4 DD Name: IENC Code: catnmk_4 Date accepted: 2001-05-31

Name: (A.3) no overtaking of convoys by convoys

Definition: Prohibition mark A.3: no overtaking of convoys by convoys (Source: Annex AA of the Encoding Guide for Inland ENCs)

5 DD Name: IENC Code: catnmk_5 Date accepted: 2001-05-31

Name: (A.4) no passing or overtaking

Definition: Prohibition mark A.4: no passing or overtaking (Source: Annex AA of the Encoding Guide for Inland ENCs)

Value	Data Dictionary (DD) Reference		
6	DD Name: IENC	Code: catnmk_6	Date accepted: 2001-05-31
	Name: (A.5) no berthing on the side of the waterway on which the sign is placed		
	Definition: Prohibition mark A.5: no berthing (i.e. no anchoring or making fast to the bank) on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
7	DD Name: IENC	Code: catnmk_7	Date accepted: 2001-05-31
	Name: (A.5.1) no berthing on the stretch of water whose breadth, measured from the sign, is shown in metres on the sign		
	Definition: Prohibition mark A.5.1: no berthing on the stretch of water whose breadth, measured from the sign, is shown in metres on the sign (Source: Annex AA of the Encoding Guide for Inland ENCs)		
8	DD Name: IENC	Code: catnmk_8	Date accepted: 2001-05-31
	Name: (A.6) no anchoring or trailing of anchors, cables or chains		
	Definition: Prohibition mark A.6: no anchoring or trailing of anchors, cables or chains on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
9	DD Name: IENC	Code: catnmk_9	Date accepted: 2001-05-31
	Name: (A.7) no making fast to the bank		
	Definition: Prohibition mark A.7: no making fast to the bank on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
10	DD Name: IENC	Code: catnmk_10	Date accepted: 2001-05-31
	Name: (A.8) no turning		
	Definition: Prohibition mark A.8: no turning (Source: Annex AA of the Encoding Guide for Inland ENCs)		
11	DD Name: IENC	Code: catnmk_11	Date accepted: 2001-05-31
	Name: (A.9) Do not create wash		
	Definition: Prohibition mark A.9: do not create wash likely to cause damage (Source: Annex AA of the Encoding Guide for Inland ENCs)		
12	DD Name: IENC	Code: catnmk_12	Date accepted: 2001-05-31
	Name: (A.10) no passing on left side (in openings of bridges or weirs)		
	Definition: Prohibition mark A.10: no passing on left side (in openings of bridges or weirs) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
13	DD Name: IENC	Code: catnmk_13	Date accepted: 2001-05-31
	Name: (A.10) no passing on right side (in openings of bridges or weirs)		
	Definition: Prohibition mark A.10: no passing on right side (in openings of bridges or weirs) (Source: Annex AA of the Encoding Guide for Inland ENCs)		

Value	Data Dictionary (DD) Reference		
14	DD Name: IENC	Code: catnmk_14	Date accepted: 2001-05-31
	Name: (A.12) motorized craft prohibited		
	Definition: Prohibition mark A.12: motorized craft prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
15	DD Name: IENC	Code: catnmk_15	Date accepted: 2001-05-31
	Name: (A.13) sports and pleasure craft prohibited		
	Definition: Prohibition mark A.13: sports and pleasure craft prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
16	DD Name: IENC	Code: catnmk_16	Date accepted: 2001-05-31
	Name: (A.14) water skiing prohibited		
	Definition: Prohibition mark A.14: water skiing prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
17	DD Name: IENC	Code: catnmk_17	Date accepted: 2001-05-31
	Name: (A.15) sailing vessels prohibited		
	Definition: Prohibition mark A.15: sailing vessels prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
18	DD Name: IENC	Code: catnmk_18	Date accepted: 2001-05-31
	Name: (A.16) all craft other than motorized vessels or sailing craft prohibited		
	Definition: Prohibition mark A.16: all craft other than motorized vessels or sailing craft prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
19	DD Name: IENC	Code: catnmk_19	Date accepted: 2001-05-31
	Name: (A.17) use of sailboards prohibited		
	Definition: Prohibition mark A.17: use of sailboards prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
20	DD Name: IENC	Code: catnmk_20	Date accepted: 2001-05-31
	Name: (A.20) water bikes prohibited		
	Definition: Prohibition mark A.20: water bikes prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
21	DD Name: IENC	Code: catnmk_21	Date accepted: 2001-05-31
	Name: (A.18) end of zone authorized for high speed navigation of small sport and pleasure craft prohibited		
	Definition: Prohibition mark A.18: end of zone authorized for high speed navigation of small sport and		

Value	Data Dictionary (DD) Reference		
	pleasure craft prohibited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
22	DD Name: IENC	Code: catnmk_22	Date accepted: 2001-05-31
	Name: (A.19) no launching or beaching of vessels		
	Definition: Prohibition mark A.19: no launching or beaching of vessels (Source: Annex AA of the Encoding Guide for Inland ENCs)		
23	DD Name: IENC	Code: catnmk_23	Date accepted: 2001-05-31
	Name: (B.1) proceed in left direction		
	Definition: Regulation mark B.1: proceed in left direction (Source: Annex AA of the Encoding Guide for Inland ENCs)		
24	DD Name: IENC	Code: catnmk_24	Date accepted: 2001-05-31
	Name: (B.1) proceed in right direction		
	Definition: Regulation mark B.1: proceed in right direction (Source: Annex AA of the Encoding Guide for Inland ENCs)		
25	DD Name: IENC	Code: catnmk_25	Date accepted: 2001-05-31
	Name: (B.2a) move to the side of the fairway on your port side		
	Definition: Regulation mark B.2a: move to the side of the fairway on your port side (Source: Annex AA of the Encoding Guide for Inland ENCs)		
26	DD Name: IENC	Code: catnmk_26	Date accepted: 2001-05-31
	Name: (B.2b) move to the side of the fairway on your starboard side		
	Definition: Regulation mark B.2b: move to the side of the fairway on your starboard side (Source: Annex AA of the Encoding Guide for Inland ENCs)		
27	DD Name: IENC	Code: catnmk_27	Date accepted: 2001-05-31
	Name: (B.3a) keep on the side of the fairway on your port side		
	Definition: Regulation mark B.3a: keep on the side of the fairway on your port side (Source: Annex AA of the Encoding Guide for Inland ENCs)		
28	DD Name: IENC	Code: catnmk_28	Date accepted: 2001-05-31
	Name: (B.3b) keep on the side of the fairway on your starboard side		
	Definition: Regulation mark B.3b: keep on the side of the fairway on your starboard side (Source: Annex AA of the Encoding Guide for Inland ENCs)		
29	DD Name: IENC	Code: catnmk_29	Date accepted: 2001-05-31
	Name: (B.4a) cross fairway to port		
	Definition: Regulation mark B.4a: cross fairway to port (Source: Annex AA of the Encoding Guide for		

Value	Data Dictionary (DD) Reference		
	Inland ENCs)		
30	DD Name: IENC	Code: catnmk_30	Date accepted: 2001-05-31
	Name: (B.4b) cross fairway to starboard		
	Definition: Regulation mark B.4b: cross fairway to starboard (Source: Annex AA of the Encoding Guide for Inland ENCs)		
31	DD Name: IENC	Code: catnmk_31	Date accepted: 2001-05-31
	Name: (B.5) stop as prescribed in the regulations		
	Definition: Regulation mark B.5: stop as prescribed in the regulations (Source: Annex AA of the Encoding Guide for Inland ENCs)		
32	DD Name: IENC	Code: catnmk_32	Date accepted: 2001-05-31
	Name: (B.6) do not exceed the speed indicated (in km/h)		
	Definition: Regulation mark B.6: do not exceed the speed indicated (in km/h) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
33	DD Name: IENC	Code: catnmk_33	Date accepted: 2001-05-31
	Name: (B.7) give a sound signal		
	Definition: Regulation mark B.7: give a sound signal (Source: Annex AA of the Encoding Guide for Inland ENCs)		
34	DD Name: IENC	Code: catnmk_34	Date accepted: 2001-05-31
	Name: (B.8) keep a particularly sharp lookout		
	Definition: Regulation mark B.8: keep a particularly sharp lookout (Source: Annex AA of the Encoding Guide for Inland ENCs)		
35	DD Name: IENC	Code: catnmk_35	Date accepted: 2001-05-31
	Name: (B.9a) do not enter the main waterway until certain that this will not oblige vessels proceeding on it to change their course or speed		
	Definition: Regulation mark B.9a: do not enter the main waterway until certain that this will not oblige vessels proceeding on it to change their course or speed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
36	DD Name: IENC	Code: catnmk_36	Date accepted: 2001-05-31
	Name: (B.9b) do not cross the main waterway until certain that this will not oblige vessels proceeding on it to change their course or speed		
	Definition: Regulation mark B.9b: do not cross the main waterway until certain that this will not oblige vessels proceeding on it to change their course or speed (Source: Annex AA of the Encoding Guide for Inland ENCs)		

Value	Data Dictionary (DD) Reference		
37	DD Name: IENC	Code: catnmk_37	Date accepted: 2001-05-31
	Name: (B.11) obligation to enter into a radiotelephone link on the channel as indicated on the board		
	Definition: Regulation mark B.11: obligation to enter into a radiotelephone link on the channel as indicated on the board (Source: Annex AA of the Encoding Guide for Inland ENCs)		
38	DD Name: IENC	Code: catnmk_38	Date accepted: 2001-05-31
	Name: (C.1) depth of water limited		
	Definition: Restriction mark C.1: depth of water limited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
39	DD Name: IENC	Code: catnmk_39	Date accepted: 2001-05-31
	Name: (C.2) headroom limited		
	Definition: Restriction mark C.2: headroom limited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
40	DD Name: IENC	Code: catnmk_40	Date accepted: 2001-05-31
	Name: (C.3) width of passage or channel limited		
	Definition: Restriction mark C.3: width of passage or channel limited (Source: Annex AA of the Encoding Guide for Inland ENCs)		
41	DD Name: IENC	Code: catnmk_41	Date accepted: 2001-05-31
	Name: (C.4) there are restrictions on navigation		
	Definition: Restriction mark C.4: there are restrictions on navigation: see the information plate below the sign (Source: Annex AA of the Encoding Guide for Inland ENCs)		
42	DD Name: IENC	Code: catnmk_42	Date accepted: 2001-05-31
	Name: (C.5) the channel lies at a distance from the left bank		
	Definition: Restriction mark C.5: the channel lies at a distance from the left bank; the figure shown on the sign indicates the distance in metres, measured from the sign, to which vessels should keep. (Source: Annex AA of the Encoding Guide for Inland ENCs)		
43	DD Name: IENC	Code: catnmk_43	Date accepted: 2001-05-31
	Name: (C.5) the channel lies at a distance from the right bank		
	Definition: Restriction mark C.5: the channel lies at a distance from the right bank; ; the figure shown on the sign indicates the distance in metres, measured from the sign, to which vessels should keep. (Source: Annex AA of the Encoding Guide for Inland ENCs)		
44	DD Name: IENC	Code: catnmk_44	Date accepted: 2001-05-31
	Name: (D.1a) recommended channel in both directions		

Value	Data Dictionary (DD) Reference		
	Definition: Recommendation mark D.1a: recommended channel in both directions (Source: Annex AA of the Encoding Guide for Inland ENCs)		
45	DD Name: IENC	Code: catnmk_45	Date accepted: 2001-05-31
	Name: (D.1b) recommended channel only in the direction indicated, passage in the opposite direction prohibited (at bridges)		
	Definition: Recommendation mark D.1b: recommended channel only in the direction indicated, passage in the opposite direction prohibited (at bridges) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
46	DD Name: IENC	Code: catnmk_46	Date accepted: 2001-05-31
	Name: (D.2) you are recommended to keep on right side (in openings of bridges and weirs)		
	Definition: Recommendation mark D.2: you are recommended to keep on right side (in openings of bridges and weirs) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
47	DD Name: IENC	Code: catnmk_47	Date accepted: 2001-05-31
	Name: (D.2) you are recommended to keep on left side (in openings of bridges and weirs)		
	Definition: Recommendation mark D.2: you are recommended to keep on left side (in openings of bridges and weirs) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
48	DD Name: IENC	Code: catnmk_48	Date accepted: 2001-05-31
	Name: (D.3) you are recommended to proceed in the left direction		
	Definition: Recommendation mark D.3: you are recommended to proceed in the left direction (Source: Annex AA of the Encoding Guide for Inland ENCs)		
49	DD Name: IENC	Code: catnmk_49	Date accepted: 2001-05-31
	Name: (D.3) you are recommended to proceed in the right direction		
	Definition: Recommendation mark D.3: you are recommended to proceed in the right direction (Source: Annex AA of the Encoding Guide for Inland ENCs)		
50	DD Name: IENC	Code: catnmk_50	Date accepted: 2001-05-31
	Name: (E.1) entry permitted (general sign)		
	Definition: Information mark E.1: entry permitted (general sign) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
51	DD Name: IENC	Code: catnmk_51	Date accepted: 2001-05-31
	Name: (E.2) overhead cable crossing		
	Definition: Information mark E.2: overhead cable crossing (Source: Annex AA of the Encoding Guide for Inland ENCs)		
52	DD Name: IENC	Code: catnmk_52	Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	(E.3) weir	
	Definition:	Information mark E.3: weir (Source: Annex AA of the Encoding Guide for Inland ENCs)	
53	DD Name:	IENC	Code: catnmk_53 Date accepted: 2001-05-31
	Name:	(E.4a) ferry-boat not moving independently	
	Definition:	Information mark E.4a: ferry-boat not moving independently (Source: Annex AA of the Encoding Guide for Inland ENCs)	
54	DD Name:	IENC	Code: catnmk_54 Date accepted: 2001-05-31
	Name:	(E.4b) ferry-boat moving independently	
	Definition:	Information mark E.4b: ferry-boat moving independently (Source: Annex AA of the Encoding Guide for Inland ENCs)	
55	DD Name:	IENC	Code: catnmk_55 Date accepted: 2001-05-31
	Name:	(E.5) berthing (i.e. no anchoring or making fast to the bank) permitted	
	Definition:	Information mark E.5: berthing (i.e. no anchoring or making fast to the bank) permitted on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)	
56	DD Name:	IENC	Code: catnmk_56 Date accepted: 2001-05-31
	Name:	(E.5.1) berthing permitted on the stretch of water of the breadth measured from, and shown on the board in meters	
	Definition:	Information mark E.5.1: berthing permitted on the stretch of water of the breadth measured from, and shown on the board in meters (Source: Annex AA of the Encoding Guide for Inland ENCs)	
57	DD Name:	IENC	Code: catnmk_57 Date accepted: 2001-05-31
	Name:	(E.5.2) berthing permitted on the stretch of water bounded by the distances measured from, and shown on the board in meters	
	Definition:	Information mark E.5.2: berthing permitted on the stretch of water bounded by the distances measured from, and shown on the board in meters (Source: Annex AA of the Encoding Guide for Inland ENCs)	
58	DD Name:	IENC	Code: catnmk_58 Date accepted: 2001-05-31
	Name:	(E.5.3) maximum number of vessels permitted to berth abreast	
	Definition:	Information mark E.5.3: maximum number of vessels permitted to berth abreast on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)	
59	DD Name:	IENC	Code: catnmk_59 Date accepted: 2001-05-31
	Name:	(E.5.4) berthing area reserved for pushing-navigation vessels that are not required to carry blue lights or blue cones	

Value	Data Dictionary (DD) Reference
	Definition: Information mark E.5.4: berthing area reserved for pushing-navigation vessels that are not required to carry blue lights or blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
60	DD Name: IENC Code: catnmk_60 Date accepted: 2001-05-31 Name: (E.5.5) berthing area reserved for pushing-navigation vessels that are required to carry one blue light or one blue cone Definition: Information mark E.5.5: berthing area reserved for pushing-navigation vessels that are required to carry one blue light or one blue cone on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
61	DD Name: IENC Code: catnmk_61 Date accepted: 2001-05-31 Name: (E.5.6) berthing area reserved for pushing-navigation vessels that are required to carry two blue lights or two blue cones Definition: Information mark E.5.6: berthing area reserved for pushing-navigation vessels that are required to carry two blue lights or two blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
62	DD Name: IENC Code: catnmk_62 Date accepted: 2001-05-31 Name: (E.5.7) berthing area reserved for pushing-navigation vessels that are required to carry three blue lights or three blue cones Definition: Information mark E.5.7: berthing area reserved for pushing-navigation vessels that are required to carry three blue lights or three blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
63	DD Name: IENC Code: catnmk_63 Date accepted: 2001-05-31 Name: (E.5.8) berthing area reserved for vessels other than pushing-navigation vessels that are not required to carry blue lights or blue cones Definition: Information mark E.5.8: berthing area reserved for vessels other than pushing-navigation vessels that are not required to carry blue lights or blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
64	DD Name: IENC Code: catnmk_64 Date accepted: 2001-05-31 Name: (E.5.9) berthing area reserved for vessels other than for pushing-navigation vessels that are required to carry one blue light or one blue cone Definition: Information mark E.5.9: berthing area reserved for vessels other than pushing-navigation vessels that are required to carry one blue light or one blue cone on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
65	DD Name: IENC Code: catnmk_65 Date accepted: 2001-05-31 Name: (E.5.10) berthing area reserved for vessels other than for pushing-navigation vessels that are required to carry two blue lights or two blue cones

Value	Data Dictionary (DD) Reference
	Definition: Information mark E.5.10: berthing area reserved for vessels other than pushing-navigation vessels that are required to carry two blue lights or two blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
66	DD Name: IENC Code: catnmk_66 Date accepted: 2001-05-31 Name: (E.5.11) berthing area reserved for vessels other than for pushing-navigation vessels that are required to carry three blue lights or three blue cones Definition: Information mark E.5.11: berthing area reserved for vessels other than pushing-navigation vessels that are required to carry three blue lights or three blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
67	DD Name: IENC Code: catnmk_67 Date accepted: 2001-05-31 Name: (E.5.12) berthing area reserved for all vessels that are not required to carry blue lights or blue cones Definition: Information mark E.5.12: berthing area reserved for all vessels that are not required to carry blue lights or blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
68	DD Name: IENC Code: catnmk_68 Date accepted: 2001-05-31 Name: (E.5.13) berthing area reserved for all vessels that are required to carry one blue light or one blue cone Definition: Information mark E.5.13: berthing area reserved for all vessels that are required to carry one blue light or one blue cone on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
69	DD Name: IENC Code: catnmk_69 Date accepted: 2001-05-31 Name: (E.5.14) berthing area reserved for all vessels that are required to carry two blue lights or two blue cones Definition: Information mark E.5.14: berthing area reserved for all vessels that are required to carry two blue lights or two blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
70	DD Name: IENC Code: catnmk_70 Date accepted: 2001-05-31 Name: (E.5.15) berthing area reserved for all vessels that are required to carry three blue lights or three blue cones Definition: Information mark E.5.15: berthing area reserved for all vessels that are required to carry three blue lights or three blue cones on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)
71	DD Name: IENC Code: catnmk_71 Date accepted: 2001-05-31 Name: (E.6) anchoring or trailing of anchors, cables or chains permitted Definition: Information mark E.6: anchoring or trailing of anchors, cables or chains permitted on the side of

Value	Data Dictionary (DD) Reference		
	the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
72	DD Name: IENC	Code: catnmk_72	Date accepted: 2001-05-31
	Name: (E.7) making fast to the bank permitted		
	Definition: Information mark E.7: making fast to the bank permitted on the side of the waterway on which the sign is placed (Source: Annex AA of the Encoding Guide for Inland ENCs)		
73	DD Name: IENC	Code: catnmk_73	Date accepted: 2001-05-31
	Name: (E.7.1) berthing area reserved for loading and unloading of vehicles		
	Definition: Information mark E.7.1: berthing area reserved for loading and unloading of vehicles (Source: Annex AA of the Encoding Guide for Inland ENCs)		
74	DD Name: IENC	Code: catnmk_74	Date accepted: 2001-05-31
	Name: (E.8) turning area		
	Definition: Information mark E.8: turning area (Source: Annex AA of the Encoding Guide for Inland ENCs)		
75	DD Name: IENC	Code: catnmk_75	Date accepted: 2001-05-31
	Name: (E.9a) crossing with secondary waterway ahead		
	Definition: Information mark E.9a: crossing with secondary waterway ahead (Source: Annex AA of the Encoding Guide for Inland ENCs)		
76	DD Name: IENC	Code: catnmk_76	Date accepted: 2001-05-31
	Name: (E.9b) secondary waterway ahead on the right		
	Definition: Information mark E.9b: secondary waterway ahead on the right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
77	DD Name: IENC	Code: catnmk_77	Date accepted: 2001-05-31
	Name: (E.9c) secondary waterway ahead on the left		
	Definition: Information mark E.9c: secondary waterway ahead on the left (Source: Annex AA of the Encoding Guide for Inland ENCs)		
78	DD Name: IENC	Code: catnmk_78	Date accepted: 2001-05-31
	Name: (E.9d) secondary waterway ahead, main waterway on the right		
	Definition: Information mark E.9d: secondary waterway ahead, main waterway on the right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
79	DD Name: IENC	Code: catnmk_79	Date accepted: 2001-05-31
	Name: (E.9e) secondary waterway ahead, main waterway on the left		
	Definition: Information mark E.9e: secondary waterway ahead, main waterway on the left (Source: Annex		

Value	Data Dictionary (DD) Reference		
	AA of the Encoding Guide for Inland ENCs)		
80	DD Name: IENC	Code: catnmk_80	Date accepted: 2001-05-31
	Name: (E.9f) secondary waterway on the left, main waterway on the right		
	Definition: Information mark E.9f: secondary waterway on the left, main waterway on the right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
81	DD Name: IENC	Code: catnmk_81	Date accepted: 2001-05-31
	Name: (E.9g) secondary waterway on the right, main waterway on the left		
	Definition: Information mark E.9g: secondary waterway on the right, main waterway on the left (Source: Annex AA of the Encoding Guide for Inland ENCs)		
82	DD Name: IENC	Code: catnmk_82	Date accepted: 2001-05-31
	Name: (E.9h) secondary waterway ahead and on the left, main waterway on the right		
	Definition: Information mark E.9h: secondary waterway ahead and on the left, main waterway on the right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
83	DD Name: IENC	Code: catnmk_83	Date accepted: 2001-05-31
	Name: (E.9i) secondary waterway ahead and on the right, main waterway on the left		
	Definition: Information mark E.9i: secondary waterway ahead and on the right, main waterway on the left (Source: Annex AA of the Encoding Guide for Inland ENCs)		
84	DD Name: IENC	Code: catnmk_84	Date accepted: 2001-05-31
	Name: (E.10a) crossing with main waterway ahead		
	Definition: Information mark E.10a: crossing with main waterway ahead (Source: Annex AA of the Encoding Guide for Inland ENCs)		
85	DD Name: IENC	Code: catnmk_85	Date accepted: 2001-05-31
	Name: (E.10b) main waterway ahead		
	Definition: Information mark E.10b: main waterway ahead (Source: Annex AA of the Encoding Guide for Inland ENCs)		
86	DD Name: IENC	Code: catnmk_86	Date accepted: 2001-05-31
	Name: (E.10c) junction with main waterway ahead and right		
	Definition: Information mark E.10c: junction with main waterway ahead and right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
87	DD Name: IENC	Code: catnmk_87	Date accepted: 2001-05-31
	Name: (E.10d) junction with main waterway ahead and left		
	Definition: Information mark E.10d: junction with main waterway ahead and left (Source: Annex AA of the		

Value	Data Dictionary (DD) Reference		
	Encoding Guide for Inland ENCs)		
88	DD Name: IENC	Code: catnmk_88	Date accepted: 2001-05-31
	Name: (E.10e) junction with main waterway ahead and right, secondary waterway on the left		
	Definition: Information mark E.10e: junction with main waterway ahead and right, secondary waterway on the left (Source: Annex AA of the Encoding Guide for Inland ENCs)		
89	DD Name: IENC	Code: catnmk_89	Date accepted: 2001-05-31
	Name: (E.10f) junction with main waterway ahead and left, secondary waterway on the right		
	Definition: Information mark E.10f: junction with main waterway ahead and left, secondary waterway on the right (Source: Annex AA of the Encoding Guide for Inland ENCs)		
90	DD Name: IENC	Code: catnmk_90	Date accepted: 2001-05-31
	Name: (E.11) end of prohibition or obligation applying to traffic in one direction only, or end of a restriction		
	Definition: Information mark E.11: end of prohibition or obligation applying to traffic in one direction only, or end of a restriction (Source: Annex AA of the Encoding Guide for Inland ENCs)		
91	DD Name: IENC	Code: catnmk_91	Date accepted: 2001-05-31
	Name: (E.13) drinking water supply		
	Definition: Information mark E.13: drinking water supply (Source: Annex AA of the Encoding Guide for Inland ENCs)		
92	DD Name: IENC	Code: catnmk_92	Date accepted: 2001-05-31
	Name: (E.14) telephone		
	Definition: Information mark E.14: telephone (Source: Annex AA of the Encoding Guide for Inland ENCs)		
93	DD Name: IENC	Code: catnmk_93	Date accepted: 2001-05-31
	Name: (E.15) motorized vessels permitted		
	Definition: Information mark E.15: motorized vessels permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
94	DD Name: IENC	Code: catnmk_94	Date accepted: 2001-05-31
	Name: (E.16) sport and pleasure craft permitted		
	Definition: Information mark E.16: sport and pleasure craft permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
95	DD Name: IENC	Code: catnmk_95	Date accepted: 2001-05-31
	Name: (E.17) water skiing permitted		

Value	Data Dictionary (DD) Reference		
	Definition: Information mark E.17: water skiing permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
96	DD Name: IENC	Code: catnmk_96	Date accepted: 2001-05-31
	Name: (E.18) sailing vessels permitted		
	Definition: Information mark E.18: sailing vessels permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
97	DD Name: IENC	Code: catnmk_97	Date accepted: 2001-05-31
	Name: (E.19) craft other than motorized vessels or sailing craft permitted		
	Definition: Information mark E.19: craft other than motorized vessels or sailing craft permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
98	DD Name: IENC	Code: catnmk_98	Date accepted: 2001-05-31
	Name: (E.20) use of sailboards permitted		
	Definition: Information mark E.20: use of sailboards permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
99	DD Name: IENC	Code: catnmk_99	Date accepted: 2001-05-31
	Name: (E.23) possibility of obtaining nautical information by radiotelephone on the channel indicated		
	Definition: Information mark E.23: possibility of obtaining nautical information by radiotelephone on the channel indicated (Source: Annex AA of the Encoding Guide for Inland ENCs)		
100	DD Name: IENC	Code: catnmk_100	Date accepted: 2001-05-31
	Name: (E.24) water bikes permitted		
	Definition: Information mark E.24: water bikes permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
101	DD Name: IENC	Code: catnmk_101	Date accepted: 2001-05-31
	Name: (E.21) zone authorized for high speed navigation of small sport and pleasure craft		
	Definition: Information mark E.21: zone authorized for high speed navigation of small sport and pleasure craft (Source: Annex AA of the Encoding Guide for Inland ENCs)		
102	DD Name: IENC	Code: catnmk_102	Date accepted: 2001-05-31
	Name: (E.22) launching or beaching of small craft permitted		
	Definition: Information mark E.22: launching and beaching of small craft permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
103	DD Name: IENC	Code: catnmk_103	Date accepted: 2009-09-11
	Name: (BR) proceed close to the margin on your port side		

Value	Data Dictionary (DD) Reference		
	Definition:	regulation mark (BR): proceed close to the margin on your port side	
104	DD Name: IENC	Code: catnmk_104	Date accepted: 2009-09-11
	Name:	(BR) proceed close to the margin on your starboard side	
	Definition:	regulation mark (BR): proceed close to the margin on your starboard side	
105	DD Name: IENC	Code: catnmk_105	Date accepted: 2009-09-11
	Name:	(BR) proceed in the middle of the river	
	Definition:	regulation mark (BR): proceed in the middle of the river	
106	DD Name: IENC	Code: catnmk_106	Date accepted: 2009-09-11
	Name:	(BR) cross river to port	
	Definition:	regulation mark (BR): cross river to port	
107	DD Name: IENC	Code: catnmk_107	Date accepted: 2009-09-11
	Name:	(BR) cross river to starboard	
	Definition:	regulation mark (BR): cross river to starboard	
108	DD Name: IENC	Code: catnmk_108	Date accepted: 2009-09-11
	Name:	(BR) traffic between margins	
	Definition:	information mark (BR): traffic between margins	
109	DD Name: IENC	Code: catnmk_109	Date accepted: 2009-09-11
	Name:	(BR) reduce speed	
	Definition:	regulation mark (BR): reduce speed	
110	DD Name: IENC	Code: catnmk_110	Date accepted: 2009-09-11
	Name:	wreck pontoon, passage allowed on side showing red-white sign	
	Definition:	a red-white sign shown on a wreck pontoon to indicate the side on which passage is permitted (without wash of waves) and a red sign on the side on which passage is not permitted	
111	DD Name: IENC	Code: catnmk_111	Date accepted: 2009-09-11
	Name:	wreck pontoon, passage allowed on both sides	
	Definition:	red-white signs shown on a wreck pontoon to indicate that passage is permitted on both sides (without wash of waves)	
112	DD Name: IENC	Code: catnmk_112	Date accepted: 2009-12-09
	Name:	no passing or overtaking of convoys	

Value	Data Dictionary (DD) Reference			
	Definition:	Russian notice mark: 1.2, no passing or overtaking of convoys		
113	DD Name:	IENC	Code: catnmk_113	Date accepted: 2009-12-09
	Name:	small crafts prohibited		
	Definition:	Russian notice mark: 1.5, small crafts prohibited		
114	DD Name:	IENC	Code: catnmk_114	Date accepted: 2009-12-09
	Name:	Attention! (Keep caution)		
	Definition:	Russian notice mark: 2.1, Attention! (keep caution)		
115	DD Name:	IENC	Code: catnmk_115	Date accepted: 2009-12-09
	Name:	fairway crossing		
	Definition:	Russian notice mark: 2.2, fairway crossing		
116	DD Name:	IENC	Code: catnmk_116	Date accepted: 2009-12-09
	Name:	shipping inspection point		
	Definition:	Russian notice mark: 3.3, shipping inspection point		
124	DD Name:	IENC	Code: catnmk_124	Date accepted: 2023-11-10
	Name:	right pillar in passage for Tiete-Parana Waterway		
	Definition:	right pillar in passage for Tiete-Parana Waterway (at bridges).		
125	DD Name:	IENC	Code: catnmk_125	Date accepted: 2023-11-10
	Name:	left pillar in passage for Tiete-Parana Waterway		
	Definition:	left pillar in passage for Tiete-Parana Waterway (at bridges).		
126	DD Name:	IENC	Code: catnmk_126	Date accepted: 2023-11-10
	Name:	best transit point		
	Definition:	best transit point (at bridges).		
127	DD Name:	IENC	Code: catnmk_127	Date accepted: 2023-11-10
	Name:	mandatory stopping point for Tiete-Parana Waterway		
	Definition:	mandatory stopping point for Tiete-Parana Waterway		
117	DD Name:	IENC	Code: catnmk_117	Date accepted: 2014-10-20
	Name:	(E.25) electrical power supply point		
	Definition:	Information mark E.25: electrical power supply point (Source: Annex AA of the Encoding Guide for Inland ENCs)		

Value	Data Dictionary (DD) Reference		
118	DD Name: IENC	Code: catnmk_118	Date accepted: 2014-10-20
	Name: (E.26) winter harbour		
	Definition: Information mark E.26: winter harbour (Source: Annex AA of the Encoding Guide for Inland ENCs)		
119	DD Name: IENC	Code: catnmk_119	Date accepted: 2014-10-20
	Name: (E.26.1) maximum number of vessels permitted to berth in winter harbour		
	Definition: Information mark E.26.1: maximum number of vessels permitted to berth in winter harbour (Source: Annex AA of the Encoding Guide for Inland ENCs)		
120	DD Name: IENC	Code: catnmk_120	Date accepted: 2014-10-20
	Name: (E.27) winter shelter		
	Definition: Information mark E.27: winter shelter (Source: Annex AA of the Encoding Guide for Inland ENCs)		
121	DD Name: IENC	Code: catnmk_121	Date accepted: 2014-10-20
	Name: (E.27.1) maximum number of vessels permitted to berth in winter shelter; maximum number of vessels permitted to berth abreast; maximum number of rows of vessels which are berthed abreast		
	Definition: Information mark E.27.1: maximum number of vessels permitted to berth in winter shelter; maximum number of vessels permitted to berth abreast; maximum number of rows of vessels which are berthed abreast (Source: Annex AA of the Encoding Guide for Inland ENCs)		
122	DD Name: IENC	Code: catnmk_122	Date accepted: 2014-10-20
	Name: (E.6.1) use of spuds permitted		
	Definition: Information mark E.6.1: use of spuds permitted (Source: Annex AA of the Encoding Guide for Inland ENCs)		
123	DD Name: IENC	Code: catnmk_123	Date accepted: 2019-03-12
	Name: (B.12) obligation to use onshore power supply point		
	Definition: Regulation mark B.12: obligation to use onshore power supply point (all vessels are obligated to use the power supply for their electrical power needs) (Source: Annex AA of the Encoding Guide for Inland ENCs)		
124	DD Name: IENC	Code: catnmk_124	Date accepted: 2023-11-10
	Name: right pillar in passage for Tiete-Parana Waterway		
	Definition: right pillar in passage for Tiete-Parana Waterway (at bridges).		
125	DD Name: IENC	Code: catnmk_125	Date accepted: 2023-11-10
	Name: left pillar in passage for Tiete-Parana Waterway		

Value	Data Dictionary (DD) Reference		
	Definition:	left pillar in passage for Tiete-Parana Waterway (at bridges).	
126	DD Name: IENC	Code: catnmk_126	Date accepted: 2023-11-10
	Name:	best transit point	
	Definition:	best transit point (at bridges).	
127	DD Name: IENC	Code: catnmk_127	Date accepted: 2023-11-10
	Name:	mandatory stopping point for Tiete-Parana Waterway	
	Definition:	mandatory stopping point for Tiete-Parana Waterway	

Attribute	Category of plug
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Acronym: catplg

Code: 18034

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26

Definition: The type of plug(s) available at the power supply station.

Attribute	Category of refuse dump
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Acronym: catrfd

Code: 17071

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of refuse dump

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catrfd_1 Date accepted: 2001-05-31

Name: cargo residue/slop

Definition: A facility where vessels can dispose of cargo residues and/or slops.

2 DD Name: IENC Code: catrfd_2 Date accepted: 2001-05-31

Name: waste oil

Definition: A facility where vessels can dispose of waste oil.

3 DD Name: IENC Code: catrfd_3 Date accepted: 2001-05-31

Name: grey/black water

Definition: A facility where vessels can dispose of grey and/or black waste water.

4 DD Name: IENC Code: catrfd_4 Date accepted: 2001-05-31

Name: domestic refuse

Definition: A facility where vessels can dispose of domestic refuse.

Attribute	Category of rescue station
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Acronym: catrsc

Code: 17106

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2010-09-17

Definition: Category of rescue station

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catrsc_1 Date accepted: 2010-09-17

Name: rescue station with life boat

Definition: a place where equipment for saving life at sea is maintained; the type of lifeboat may vary from fast, long distance boats to inflatable inshore boats. (IHO Chart Specifications, M-4).

2 DD Name: IENC Code: catrsc_2 Date accepted: 2010-09-17

Name: rescue station with rocket

Definition: rocket - a pyrotechnic projectile used for signalling or for life-saving purposes. (IHO Dictionary, S-32, 5th Edition, 4418).

3 DD Name: IENC Code: catrsc_3 Date accepted: 2010-09-17

Name: not in use

Definition:

4 DD Name: IENC Code: catrsc_4 Date accepted: 2010-09-17

Name: refuge for ship-wrecked mariners

Definition: shelter or protection from danger or distress at sea.

5 DD Name: IENC Code: catrsc_5 Date accepted: 2010-09-17

Name: refuge for intertidal area walkers

Definition: shelter or protection from danger in areas exposed to extreme and sudden tides or tidal streams.

6 DD Name: IENC Code: catrsc_6 Date accepted: 2010-09-17

Value	Data Dictionary (DD) Reference		
	Name:	lifeboat lying at a mooring	
	Definition:	a place where a lifeboat is moored ready for use.	
7	DD Name: IENC	Code: catrsc_7	Date accepted: 2010-09-17
	Name:	aid radio station	
	Definition:	a radio station reserved for emergency situations, might also be a public telephone.	
8	DD Name: IENC	Code: catrsc_8	Date accepted: 2010-09-17
	Name:	first aid equipment	
	Definition:	a place where first aid equipment is available.	
9	DD Name: IENC	Code: catrsc_9	Date accepted: 2010-09-17
	Name:	lifebuoy, ring buoy, life ring, life saver	
	Definition:	a "kisby ring" or "perry buoy" designed to be thrown to a person in the water, to provide buoyancy and to prevent drowning.	

Attribute	Category of sensor
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Acronym: catsenCode: 18019

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENCDate accepted: 2011-12-19

Definition: Category of sensor

Enumerations:

Value	Data Dictionary (DD) Reference		
3	DD Name: IENCCode: catsen_3Date accepted: 2024-04-05		
	Name: radio activated		
	Definition: A sensor which is activated by radio		
1	DD Name: IENCCode: catsen_1Date accepted: 2011-12-19		
	Name: light activated		
	Definition: A sensor which is activated by a spotlight		
2	DD Name: IENCCode: catsen_2Date accepted: 2011-12-19		
	Name: telephone activated		
	Definition: A sensor which is activated by telephone		

Attribute	Category of ship (excluding)
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Acronym: lc_cse

Code: 18013

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lc_cse_1 Date accepted: 2001-05-31

Name: all types

Definition:

2 DD Name: IENC Code: lc_cse_2 Date accepted: 2001-05-31

Name: other

Definition:

3 DD Name: IENC Code: lc_cse_3 Date accepted: 2001-05-31

Name: non-motorized vessel

Definition:

5 DD Name: IENC Code: lc_cse_5 Date accepted: 2001-05-31

Name: craft

Definition: a vessel or item of floating equipment

6 DD Name: IENC Code: lc_cse_6 Date accepted: 2001-05-31

Name: vessel

Definition: an inland waterway vessel or sea going ship

7 DD Name: IENC Code: lc_cse_7 Date accepted: 2001-05-31

Name: inland waterway vessel

Value	Data Dictionary (DD) Reference		
	Definition:	a vessel intended solely or mainly for navigation on inland waterways	
8	DD Name: IENC	Code: lc_cse_8	Date accepted: 2001-05-31
	Name:	sea going ship	
	Definition:	a vessel certificated for sea-going service	
9	DD Name: IENC	Code: lc_cse_9	Date accepted: 2001-05-31
	Name:	motor vessel	
	Definition:	a motor cargo vessel or a motor tanker	
10	DD Name: IENC	Code: lc_cse_10	Date accepted: 2001-05-31
	Name:	motor tanker	
	Definition:	a vessel intended for the carriage of goods in fixed tanks and built to navigate independently under its own motive power	
11	DD Name: IENC	Code: lc_cse_11	Date accepted: 2001-05-31
	Name:	motor cargo vessel	
	Definition:	a vessel, other than a motor tanker, intended for the carriage of goods and built to navigate independently under its own motive power	
12	DD Name: IENC	Code: lc_cse_12	Date accepted: 2001-05-31
	Name:	canal barge	
	Definition:	an inland waterway vessel not exceeding 38.5 m in length and 5.05 m in breadth and usually operating on the Rhine-Rhône-Canal	
13	DD Name: IENC	Code: lc_cse_13	Date accepted: 2001-05-31
	Name:	tug	
	Definition:	a vessel specially built to perform towing operations	
14	DD Name: IENC	Code: lc_cse_14	Date accepted: 2001-05-31
	Name:	pusher	
	Definition:	a vessel specially built to propel a pushed convoy	
15	DD Name: IENC	Code: lc_cse_15	Date accepted: 2001-05-31
	Name:	barge	
	Definition:	a dumb barge or tank barge	
16	DD Name: IENC	Code: lc_cse_16	Date accepted: 2001-05-31
	Name:	tank barge	

Value	Data Dictionary (DD) Reference		
	Definition: a vessel intended for the carriage of goods in fixed tanks and built to be towed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres		
17	DD Name: IENC	Code: lc_cse_17	Date accepted: 2001-05-31
	Name: dumb barge		
	Definition: a vessel, other than a tank barge, intended for the carriage of goods and built to be towed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres		
18	DD Name: IENC	Code: lc_cse_18	Date accepted: 2001-05-31
	Name: lighter		
	Definition: a tank lighter, cargo lighter or ship borne lighter		
19	DD Name: IENC	Code: lc_cse_19	Date accepted: 2001-05-31
	Name: tank lighter		
	Definition: a vessel intended for the carriage of goods in fixed tanks, built or specially modified to be pushed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres when not part of a pushed convoy		
20	DD Name: IENC	Code: lc_cse_20	Date accepted: 2001-05-31
	Name: cargo lighter		
	Definition: a vessel, other than a tank lighter, intended for the carriage of goods and built or specially modified to be pushed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres when not part of a pushed convoy		
21	DD Name: IENC	Code: lc_cse_21	Date accepted: 2001-05-31
	Name: ship borne lighter		
	Definition: a lighter built to be carried aboard sea going ships and to navigate on inland waterways		
22	DD Name: IENC	Code: lc_cse_22	Date accepted: 2001-05-31
	Name: passenger vessel		
	Definition: a day trip or cabin vessel constructed and equipped to carry more than 12 passengers		
23	DD Name: IENC	Code: lc_cse_23	Date accepted: 2001-05-31
	Name: passenger sailing vessel		
	Definition: a passenger vessel fitted out mainly with a view to propulsion under sail		
24	DD Name: IENC	Code: lc_cse_24	Date accepted: 2001-05-31
	Name: day trip vessel		

Value	Data Dictionary (DD) Reference			
	Definition:	a passenger vessel without overnight passenger cabins		
25	DD Name: IENC	Code: lc_cse_25	Date accepted:	2001-05-31
	Name:	cabin vessel		
	Definition:	a passenger vessel with overnight passenger cabins		
26	DD Name: IENC	Code: lc_cse_26	Date accepted:	2001-05-31
	Name:	High-speed vessel		
	Definition:	a motorised vessel capable of reaching speeds over 40km/h with respect to water		
27	DD Name: IENC	Code: lc_cse_27	Date accepted:	2001-05-31
	Name:	floating equipment		
	Definition:	a floating installation carrying working gear such as cranes, dredging equipment, pile drivers or elevators		
28	DD Name: IENC	Code: lc_cse_28	Date accepted:	2001-05-31
	Name:	worksite craft		
	Definition:	a vessel, appropriately built and equipped for use at worksites, such as a reclamation barge, hopper or pontoon barge, pontoon or stone-dumping vessel		
29	DD Name: IENC	Code: lc_cse_29	Date accepted:	2001-05-31
	Name:	recreational craft		
	Definition:	a vessel other than a passenger vessel, intended for sport or pleasure		
30	DD Name: IENC	Code: lc_cse_30	Date accepted:	2001-05-31
	Name:	Dinghy		
	Definition:	a boat for use in transport, rescue, salvage and work duties		
31	DD Name: IENC	Code: lc_cse_31	Date accepted:	2001-05-31
	Name:	floating establishment		
	Definition:	any floating installation not normally intended to be moved, such as a swimming bath, dock, jetty or boathouse		
32	DD Name: IENC	Code: lc_cse_32	Date accepted:	2001-05-31
	Name:	floating object		
	Definition:	a raft or other structure, object or assembly capable of navigation, not being a vessel or floating equipment or establishment		

Attribute	Category of ship (including)
-----------	------------------------------

Acronym: lc_csi

Code: 18012

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition:

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lc_csi_1 Date accepted: 2001-05-31

Name: all types

Definition:

2 DD Name: IENC Code: lc_csi_2 Date accepted: 2001-05-31

Name: other

Definition:

3 DD Name: IENC Code: lc_csi_3 Date accepted: 2001-05-31

Name: non-motorized vessel

Definition: non-motorized vessel

5 DD Name: IENC Code: lc_csi_5 Date accepted: 2001-05-31

Name: craft

Definition: a vessel or item of floating equipment

6 DD Name: IENC Code: lc_csi_6 Date accepted: 2001-05-31

Name: vessel

Definition: an inland waterway vessel or sea going ship

7 DD Name: IENC Code: lc_csi_7 Date accepted: 2001-05-31

Name: inland waterway vessel

Value	Data Dictionary (DD) Reference		
	Definition:	a vessel intended solely or mainly for navigation on inland waterways	
8	DD Name: IENC	Code: lc_csi_8	Date accepted: 2001-05-31
	Name:	sea going ship	
	Definition:	a vessel certificated for sea-going service	
9	DD Name: IENC	Code: lc_csi_9	Date accepted: 2001-05-31
	Name:	motor vessel	
	Definition:	a motor cargo vessel or a motor tanker	
10	DD Name: IENC	Code: lc_csi_10	Date accepted: 2001-05-31
	Name:	motor tanker	
	Definition:	a vessel intended for the carriage of goods in fixed tanks and built to navigate independently under its own motive power	
11	DD Name: IENC	Code: lc_csi_11	Date accepted: 2001-05-31
	Name:	motor cargo vessel	
	Definition:	a vessel, other than a motor tanker, intended for the carriage of goods and built to navigate independently under its own motive power	
12	DD Name: IENC	Code: lc_csi_12	Date accepted: 2001-05-31
	Name:	canal barge	
	Definition:	an inland waterway vessel not exceeding 38.5 m in length and 5.05 m in breadth and usually operating on the Rhine-Rhône-Canal	
13	DD Name: IENC	Code: lc_csi_13	Date accepted: 2001-05-31
	Name:	tug	
	Definition:	a vessel specially built to perform towing operations	
14	DD Name: IENC	Code: lc_csi_14	Date accepted: 2001-05-31
	Name:	pusher	
	Definition:	a vessel specially built to propel a pushed convoy	
15	DD Name: IENC	Code: lc_csi_15	Date accepted: 2001-05-31
	Name:	barge	
	Definition:	a dumb barge or tank barge	
16	DD Name: IENC	Code: lc_csi_16	Date accepted: 2001-05-31
	Name:	tank barge	

Value	Data Dictionary (DD) Reference		
	Definition: a vessel intended for the carriage of goods in fixed tanks and built to be towed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres		
17	DD Name: IENC	Code: lc_csi_17	Date accepted: 2001-05-31
	Name: dumb barge		
	Definition: a vessel, other than a tank barge, intended for the carriage of goods and built to be towed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres		
18	DD Name: IENC	Code: lc_csi_18	Date accepted: 2001-05-31
	Name: lighter		
	Definition: a tank lighter, cargo lighter or ship borne lighter		
19	DD Name: IENC	Code: lc_csi_19	Date accepted: 2001-05-31
	Name: tank lighter		
	Definition: a vessel intended for the carriage of goods in fixed tanks, built or specially modified to be pushed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres when not part of a pushed convoy		
20	DD Name: IENC	Code: lc_csi_20	Date accepted: 2001-05-31
	Name: cargo lighter		
	Definition: a vessel, other than a tank lighter, intended for the carriage of goods and built or specially modified to be pushed, either having no motive power of its own or having only sufficient motive power to perform restricted manoeuvres when not part of a pushed convoy		
21	DD Name: IENC	Code: lc_csi_21	Date accepted: 2001-05-31
	Name: ship borne lighter		
	Definition: a lighter built to be carried aboard sea going ships and to navigate on inland waterways		
22	DD Name: IENC	Code: lc_csi_22	Date accepted: 2001-05-31
	Name: passenger vessel		
	Definition: a day trip or cabin vessel constructed and equipped to carry more than 12 passengers		
23	DD Name: IENC	Code: lc_csi_23	Date accepted: 2001-05-31
	Name: passenger sailing vessel		
	Definition: a passenger vessel fitted out mainly with a view to propulsion under sail		
24	DD Name: IENC	Code: lc_csi_24	Date accepted: 2001-05-31
	Name: day trip vessel		

Value	Data Dictionary (DD) Reference		
	Definition:	a passenger vessel without overnight passenger cabins	
25	DD Name: IENC	Code: lc_csi_25	Date accepted: 2001-05-31
	Name:	cabin vessel	
	Definition:	a passenger vessel with overnight passenger cabins	
26	DD Name: IENC	Code: lc_csi_26	Date accepted: 2001-05-31
	Name:	High-speed vessel	
	Definition:	a motorised vessel capable of reaching speeds over 40km/h with respect to water	
27	DD Name: IENC	Code: lc_csi_27	Date accepted: 2001-05-31
	Name:	floating equipment	
	Definition:	a floating installation carrying working gear such as cranes, dredging equipment, pile drivers or elevators	
28	DD Name: IENC	Code: lc_csi_28	Date accepted: 2001-05-31
	Name:	worksite craft	
	Definition:	a vessel, appropriately built and equipped for use at worksites, such as a reclamation barge, hopper or pontoon barge, pontoon or stone-dumping vessel	
29	DD Name: IENC	Code: lc_csi_29	Date accepted: 2001-05-31
	Name:	recreational craft	
	Definition:	a vessel other than a passenger vessel, intended for sport or pleasure	
30	DD Name: IENC	Code: lc_csi_30	Date accepted: 2001-05-31
	Name:	Dinghy	
	Definition:	a boat for use in transport, rescue, salvage and work duties	
31	DD Name: IENC	Code: lc_csi_31	Date accepted: 2001-05-31
	Name:	floating establishment	
	Definition:	any floating installation not normally intended to be moved, such as a swimming bath, dock, jetty or boathouse	
32	DD Name: IENC	Code: lc_csi_32	Date accepted: 2001-05-31
	Name:	floating object	
	Definition:	a raft or other structure, object or assembly capable of navigation, not being a vessel or floating equipment or establishment	

Attribute	Category of shoreline construction
-----------	------------------------------------

Acronym: catslc

Code: 17012

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2007-10-12

Definition: Category of shoreline construction

Enumerations:

Value Data Dictionary (DD) Reference

7 DD Name: IENC Code: catslc_7 Date accepted: 2007-10-12

Name: training wall

Definition: a wall or bank, often submerged, built to direct or confine the flow of a river or tidal current, or to promote a scour action. (Adapted from IHO Dictionary, S-32, 5th Edition, 5586 and IHO Chart Specifications, M-4).

18 DD Name: IENC Code: catslc_18 Date accepted: 2007-10-12

Name: lock/guide wall

Definition: permanent structure bounding a lock and including guide walls (USACE)

19 DD Name: IENC Code: catslc_19 Date accepted: 2009-09-11

Name: ice breaker

Definition: an often wedge-like structure used for protecting a bridge pier, dock, facility, etc. from floating ice or other debris.

2 DD Name: IENC Code: catslc_2 Date accepted: 2011-05-31

Name: groyne (groin)

Definition: a low artificial wall-like structure of durable material extending from the land to seaward for a particular purpose, such as to prevent coast erosion (adapted from IHO Dictionary, S-32, 5th Edition, 2525 and IHO Chart Specifications, M-4)

8 DD Name: IENC Code: catslc_8 Date accepted: 2013-01-23

Name: rip rap

Definition: A layer of broken rock, cobbles, boulders, or fragments of sufficient size to resist the erosive

Value	Data Dictionary (DD) Reference		
	forces of flowing water and wave action. (Adapted from Marine Chart Manual, US National Oceanic and Atmospheric Administration - NOAA, 1992)		
9	DD Name: IENC	Code: catslc_9	Date accepted: 2013-01-23
	Name: revetment		
	Definition: facing of stone or other material, either permanent or temporary, placed along the edge of a stream, river or canal to stabilize the bank and to protect it from the erosive action of the stream. (Adapted from IHO Dictionary, S-32, 5th Edition, 4379)		
20	DD Name: IENC	Code: catslc_20	Date accepted: 2013-09-23
	Name: water intake structure		
	Definition: a structure designed to divert water from a river or channel for the purpose of water supply, hydroelectric power or irrigation.		

Attribute	Category of signal station, traffic
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Acronym: catsit

Code: 17002

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of signal station, traffic

Enumerations:

Value Data Dictionary (DD) Reference

2 DD Name: IENC Code: catsit_2 Date accepted: 2001-05-31

Name: port entry and departure

Definition: a signal station for the control of vessels entering or leaving a port.

6 DD Name: IENC Code: catsit_6 Date accepted: 2001-05-31

Name: lock

Definition: a signal station for the control of vessels entering or leaving a lock.

8 DD Name: IENC Code: catsit_8 Date accepted: 2001-05-31

Name: bridge passage

Definition: a signal station for the control of vessels wishing to pass under a bridge.

10 DD Name: IENC Code: catsit_10 Date accepted: 2001-05-31

Name: oncoming traffic indication

Definition: indicates the oncoming traffic on an inland waterway

Attribute	Category of signal station, warning
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Acronym: catsiw

Code: 17003

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of signal station, warning

Enumerations:

Value Data Dictionary (DD) Reference

15 DD Name: IENC Code: catsiw_15 Date accepted: 2001-05-31

Name: high water mark

Definition: An indication of the official high water level.

16 DD Name: IENC Code: catsiw_16 Date accepted: 2001-05-31

Name: vertical clearance indication

Definition: An indication of the vertical clearance of a bridge, overhead cable, etc.

18 DD Name: IENC Code: catsiw_18 Date accepted: 2001-05-31

Name: depth indication

Definition: An indication of the local depth.

1 DD Name: HYDRO Code: catsiw_1 Date accepted: 2001-11-01

Name: danger

Definition: A signal or message warning of the presence of a danger to navigation.

Attribute	Category of time and behaviour
-----------	--------------------------------

Acronym: cattab

Code: 17092

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of time and behaviour

Enumerations:

Value Data Dictionary (DD) Reference

- | | | | |
|---|--|----------------|---------------------------|
| 1 | DD Name: IENC | Code: cattab_1 | Date accepted: 2001-05-31 |
| | Name: operational period | | |
| | Definition: being in a position or adjustment to permit passage or to perform an operation | | |
| 2 | DD Name: IENC | Code: cattab_2 | Date accepted: 2001-05-31 |
| | Name: non-operational period | | |
| | Definition: being in a position or adjustment to prevent passage | | |

Attribute	Category of vehicle transfer
-----------	------------------------------

Acronym: catvtr

Code: 17091

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of vehicle transfer

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catvtr_1 Date accepted: 2001-05-31

Name: official

Definition: Of or relating to an office or a post of authority

2 DD Name: IENC Code: catvtr_2 Date accepted: 2001-05-31

Name: private

Definition: Belonging to a particular person or persons, as opposed to the public or the government

3 DD Name: IENC Code: catvtr_3 Date accepted: 2001-05-31

Name: suitable for car cranes

Definition: Vehicle transfer location is suitable for car cranes

4 DD Name: IENC Code: catvtr_4 Date accepted: 2001-05-31

Name: suitable for car planks

Definition: Vehicle transfer location is suitable for car planks

5 DD Name: IENC Code: catvtr_5 Date accepted: 2001-05-31

Name: permission required

Definition: The transfer of a vehicle requires permission.

6 DD Name: IENC Code: catvtr_6 Date accepted: 2001-05-31

Name: locked gate

Value	Data Dictionary (DD) Reference
	Definition: The access to the public road is locked.

Attribute	Category of voltage
-----------	---------------------

Acronym: catvol

Code: 18031

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26

Definition: The electrical voltage provided by the power supply station

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: catvol_1 Date accepted: 2014-11-26

Name: 230V

Definition: 230 Volts

2 DD Name: IENC Code: catvol_2 Date accepted: 2014-11-26

Name: 400V

Definition: 400 Volts

Attribute	Category of waterway gauge
-----------	----------------------------

Acronym: catgag

Code: 17078

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Category of waterway gauge

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|--|
| 1 | <p>DD Name: IENC Code: catgag_1 Date accepted: 2001-05-31</p> <p>Name: water level staff / pole</p> <p>Definition: level indicator consisting of a calibrated staff/pole and the associated bench mark. (DIN 4049 Section 3, Oct. 1994)</p> |
| 2 | <p>DD Name: IENC Code: catgag_2 Date accepted: 2001-05-31</p> <p>Name: recording water level gauge</p> <p>Definition: analog or digital water level measuring and recording device. (adopted from DIN 4049 – Section 3, Oct. 1994)</p> |
| 3 | <p>DD Name: IENC Code: catgag_3 Date accepted: 2001-05-31</p> <p>Name: recording water level gauge with remote access</p> <p>Definition: recording water level gauge providing information remotely by any method (adopted from DIN 4049 – Section 3, Oct. 1994)</p> |
| 4 | <p>DD Name: IENC Code: catgag_4 Date accepted: 2001-05-31</p> <p>Name: recording water level gauge with external indicator</p> <p>Definition: recording gauge providing information of the water level via a large external indicator (adopted from DIN 4049 – Section 3, Oct. 1994)</p> |
| 5 | <p>DD Name: IENC Code: catgag_5 Date accepted: 2001-05-31</p> <p>Name: recording water level gauge with remote access and remote indicator</p> <p>Definition: recording gauge providing information remotely by any method and providing information of the water level via a large external indicator. (adopted from DIN 4049 – Section 3, Oct. 1994)</p> |

Attribute	Class of dangerous cargo
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Acronym: clsdng

Code: 17055

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Class of dangerous cargo

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|---|---|
| 1 | DD Name: IENC Code: clsdng_1 Date accepted: 2001-05-31
Name: one blue light / cone
Definition: Vessels carrying out transport operations involving certain flammable substances |
| 2 | DD Name: IENC Code: clsdng_2 Date accepted: 2001-05-31
Name: two blue lights / cones
Definition: Vessels carrying out transport operations involving certain substances constituting health hazards |
| 3 | DD Name: IENC Code: clsdng_3 Date accepted: 2001-05-31
Name: three blue lights / cones
Definition: Vessels carrying out transport operations involving certain explosives |
| 4 | DD Name: IENC Code: clsdng_4 Date accepted: 2001-05-31
Name: no blue light / cone
Definition: Vessels carrying out transport operations for which no blue light or blue cone is required |
| 5 | DD Name: IENC Code: clsdng_5 Date accepted: 2009-12-09
Name: one red light / red cone top down
Definition: Russian inland waterway regulations: vessels with one red light / red cone top down |

Attribute	Current velocity at high water level
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Acronym: curvhw

Code: 17095

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The rate of travel of a current at a high water level

Attribute	Current velocity at low water level
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Acronym: curvlw

Code: 17096

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The rate of travel of a current at a low water level

Attribute	Current velocity at mean water level
-----------	--------------------------------------

Acronym: curvmw

Code: 17097

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The rate of travel of a current at a mean water level

Attribute	Current velocity at other water level
-----------	---------------------------------------

Acronym: curvow

Code: 17098

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The rate of travel of a current at an other water level

Attribute	Description of legal conditions
-----------	---------------------------------

Acronym: lg_des Code: 18010

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Additional textual information which is related to the numerical description of the particular article/clause of the applicable law/regulation

Attribute	Direction of impact
-----------	---------------------

Acronym: dirimp

Code: 17056

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Direction of impact

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: dirimp_1 Date accepted: 2001-05-31

Name: upstream

Definition: toward the source of a stream

2 DD Name: IENC Code: dirimp_2 Date accepted: 2001-05-31

Name: downstream

Definition: in the direction of flow of a current or stream

3 DD Name: IENC Code: dirimp_3 Date accepted: 2001-05-31

Name: to the left bank

Definition: toward the left side of the bank

4 DD Name: IENC Code: dirimp_4 Date accepted: 2001-05-31

Name: to the right bank

Definition: toward the right side of the bank

5 DD Name: IENC Code: dirimp_5 Date accepted: 2010-08-12

Name: to harbor

Definition: to an harbor

Attribute	Distance from notice mark, first
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Acronym: disbk1

Code: 17057

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Minimum distance of the impact of an area, which is signed by notice marks. The distance is measured from the notice mark rectangular to the bank

Attribute	Distance from notice mark, second
-----------	-----------------------------------

Acronym: disbk2

Code: 17058

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Maximum distance of the impact of an area, which is signed by notice marks. The distance is measured from the notice mark rectangular to the bank

Attribute	Distance of impact, downstream
-----------	--------------------------------

Acronym: disipd

Code: 17060

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Downstream distance of the impact of an area, which is signed by notice marks. The distance is normally given on an additional mark left and/or right of the notice mark

Attribute	Distance of impact, upstream
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Acronym: disipu

Code: 17059

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Upstream distance of the impact of an area, which is signed by notice marks. The distance is normally given on an additional mark left and/or right of the notice mark

Attribute	Elevation 1 of surface (m)
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Acronym: eleva1

Code: 17061

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximum elevation of the bottom of a river within a depth contour and referred to a gravitational reference level.

Attribute	Elevation 2 of surface (m)
-----------	----------------------------

Acronym: eleva2

Code: 17062

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The minimum elevation of the bottom of a river within a depth contour and referred to a gravitational reference level.

Attribute	Elevation of water level
-----------	--------------------------

Acronym: elevwl

Code: 18050

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2020-01-24

Definition: Elevation of the water level of a specified object point measured from the reference gravitational level defined in reflev

Attribute	Function of notice mark
-----------	-------------------------

Acronym: fnctnm

Code: 17063

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Function of a notice mark

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: fnctnm_1 Date accepted: 2001-05-31

Name: prohibition mark

Definition: marks which indicate a prohibition

2 DD Name: IENC Code: fnctnm_2 Date accepted: 2001-05-31

Name: regulation mark

Definition: marks which indicate a regulation

3 DD Name: IENC Code: fnctnm_3 Date accepted: 2001-05-31

Name: restriction mark

Definition: marks which indicate a restriction

4 DD Name: IENC Code: fnctnm_4 Date accepted: 2001-05-31

Name: recommendation mark

Definition: marks which indicate a recommendation

5 DD Name: IENC Code: fnctnm_5 Date accepted: 2001-05-31

Name: information mark

Definition: marks with general information

Attribute	Function of sensor
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Acronym: fnctsnCode: 18020

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2011-12-19

Definition: Function of sensor

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: fnctsn_1 Date accepted: 2011-12-19

 Name: reduce bridge lighting

 Definition:

Attribute	Height/length units
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Acronym: hunits

Code: 17103

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Units of measure of waterway distances

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: hunits_1 Date accepted: 2001-05-31

Name: metres

Definition: heights/lengths are specified in metres (SI units of length)

2 DD Name: IENC Code: hunits_2 Date accepted: 2001-05-31

Name: feet

Definition: heights/lengths are specified in feet (imperial units of length)

3 DD Name: IENC Code: hunits_3 Date accepted: 2001-05-31

Name: kilometres

Definition: heights/lengths are specified in kilometres (1000 metres)

4 DD Name: IENC Code: hunits_4 Date accepted: 2001-05-31

Name: hectometres

Definition: heights/lengths are specified in hectometres (100 metres)

5 DD Name: IENC Code: hunits_5 Date accepted: 2001-05-31

Name: statute miles

Definition: heights/lengths are specified in statute (land) miles

6 DD Name: IENC Code: hunits_6 Date accepted: 2001-05-31

Name: nautical miles

Value	Data Dictionary (DD) Reference
	Definition: heights/lengths are specified in nautical (sea) miles

Attribute	Horizontal clearance length
-----------	-----------------------------

Acronym: horcll

Code: 17074

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The length of an object, such as a lock or basin, which is available for safe navigation. This may, or may not, be the same as the total physical length (HORLEN) of the object

Attribute	Horizontal clearance width
-----------	----------------------------

Acronym: horclw

Code: 17075

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The width of an object, such as a lock or basin, which is available for safe navigation. This may, or may not, be the same as the total physical width (HORWID) of the object

Attribute	Location Maritime Resource Name
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Acronym: locmrn

Code: 18051

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2023-11-22

Definition: The attribute locMRN should be used to encode the Maritime Resource Name (MRN)

Attribute	Marks navigational - System of
-----------	--------------------------------

Acronym: marsys

Code: 17009

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: System of navigational marks

Enumerations:

Value Data Dictionary (DD) Reference

- | | |
|----|--|
| 1 | DD Name: IENC Code: marsys_1 Date accepted: 2001-05-31
Name: IALA A
Definition: navigational aids conform to the International Association of Lighthouse Authorities - IALA A system. |
| 2 | DD Name: IENC Code: marsys_2 Date accepted: 2001-05-31
Name: IALA B
Definition: navigational aids conform to the International Association of Lighthouse Authorities - IALA B system. |
| 9 | DD Name: IENC Code: marsys_9 Date accepted: 2001-05-31
Name: no system
Definition: navigational aids do not conform to any defined system. |
| 10 | DD Name: IENC Code: marsys_10 Date accepted: 2001-05-31
Name: other system
Definition: navigational aids conform to a defined system other than International Association of Lighthouse Authorities -IALA. |
| 11 | DD Name: IENC Code: marsys_11 Date accepted: 2001-05-31
Name: main European inland waterway marking system
Definition: navigational aids as required in international, national or regional regulations that contain the same navigational aids as the European Code for Inland Waterways of UN/ECE, or if there is no regulation for a waterway, navigational aids as recommended in the European Code for Inland Waterways of UN/ECE |

Value	Data Dictionary (DD) Reference		
	(for notice marks see Annex AA of the Encoding Guide for Inland ENCs)		
12	DD Name: IENC	Code: marsys_12	Date accepted: 2001-05-31
	Name:	Russian inland waterway regulations	
	Definition:	navigational aids conform to the Russian inland waterway regulations.	
13	DD Name: IENC	Code: marsys_13	Date accepted: 2009-09-11
	Name:	Brazilian national inland waterway regulation	
	Definition:	navigational aids conform to the Brazilian national inland waterway regulation.	
15	DD Name: IENC	Code: marsys_15	Date accepted: 2009-09-11
	Name:	Paraguay-Parana waterway - Brazilian complementary aids	
	Definition:	Brazilian complementary navigational aids on the Paraguay-Parana waterway	

Attribute	Maximal permitted beam
-----------	------------------------

Acronym: lg_bme

Code: 18003

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximal permitted beam (width of a ship's hull) of a vessel or convoy according to the particular article/clause of the applicable law/regulation

Attribute	Maximal permitted draught
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Acronym: lg_drt

Code: 18005

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximal permitted draught of a vessel or convoy according to the particular article/clause of the applicable law/regulation

Attribute	Maximal permitted length
-----------	--------------------------

Acronym: lg_lgs

Code: 18004

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximal permitted length of a vessel or convoy according to the particular article/clause of the applicable law/regulation

Attribute	Maximal permitted speed
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Acronym: lg_spd

Code: 18001

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximal permitted vessel speed according to the particular article/clause of the applicable law/regulation

Attribute	Maximal permitted water displacement
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Acronym: lg_wdp

Code: 18006

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The maximal permitted water displacement of a vessel or convoy according to the particular article/clause of the applicable law/regulation

Attribute	MMSI Code
-----------	-----------

Acronym: mmsico

Code: 18046

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2017-05-30

Definition: Maritime Mobile Service Identity code

Attribute	Name of other locally relevant water level
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Acronym: othnam

Code: 17087

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level, which is used for the attribute othwat (value at other locally relevant water level) including version identification, for example year of issue or period

Attribute	Name of relevant high water level
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Acronym: highnam

Code: 17081

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level, which is used for the attribute higwat (value at relevant high water level) including version identification, for example year of issue or period

Attribute	Name of relevant low water level
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Acronym: lownam

Code: 17083

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level, which is used for the attribute lowwat (value at relevant low water level) including version identification, for example year of issue or period

Attribute	Name of relevant mean water level
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Acronym: meanam

Code: 17085

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level, which is used for the attribute meawat (value at relevant mean water level) including version identification, for example year of issue or period

Attribute	Name of Sounding datum reference level
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Acronym: sdrlev

Code: 17089

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level depth values are referred to

Attribute	Name of vertical river datum reference level
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Acronym: vcrlev

Code: 17090

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Name of the water level vertical clearance values are referred to

Attribute	Number of shore connectors
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Acronym: shrnum

Code: 18035

Use Type: F

Value Type: I

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2014-11-26

Definition: The number of shore connectors available at the power supply station.

Attribute	Publication reference
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Acronym: lg_pbr Code: 18011
Use Type: F
Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Waterway or waterway section for which a juridical regulation with respect to the maximum permitted vessel dimensions exists

Attribute	Reference Gauge
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Acronym: refgag

Code: 18018

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2013-01-11

Definition: The ISRS code of the gauge, or the acronym of the gauge, which can be used to calculate the vertical clearance.

Attribute	Reference gravitational level
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Acronym: reflev

Code: 17088

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Gravitational reference level

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: reflev_1 Date accepted: 2001-05-31

Name: Baltic datum

Definition: Baltic Datum (Baltic Heights System) - The unified State system for absolute heights reckoning from Kronshtadt Tide-gauge Datum that is accepted in Russian Federation. (Hydrographic Terminology Dictionary, HDNO, 1984)-

2 DD Name: IENC Code: reflev_2 Date accepted: 2001-05-31

Name: Adriatic level

Definition: The average height of the surface of the Adriatic sea at the tide station of Trieste in Italy.

3 DD Name: IENC Code: reflev_3 Date accepted: 2001-05-31

Name: Amsterdam Ordnance Datum (NAP)

Definition: Dutch gravitational reference level that is approximately the average summer height of the North Sea.

4 DD Name: IENC Code: reflev_4 Date accepted: 2001-05-31

Name: Mean Sea Level

Definition: (MSL) - the average height of the surface of the sea at a tide station for all stages of the tide over a 19-year period, usually determined from hourly height readings measured from a fixed predetermined reference level. (IHO Dictionary, S-32, 5th Edition, 3156)

5 DD Name: IENC Code: reflev_5 Date accepted: 2001-05-31

Name: Other datum

Definition:

Value	Data Dictionary (DD) Reference		
6	DD Name: IENC	Code: reflev_6	Date accepted: 2001-05-31
	Name:	National Geodetic Vertical Datum - NGVD29	
	Definition:	The name, after May 10, 1973, of the Sea Level Datum of 1929.	
7	DD Name: IENC	Code: reflev_7	Date accepted: 2001-05-31
	Name:	North American Vertical Datum - NAVD88	
	Definition:	The vertical control datum established in 1991 by the minimum-constraint adjustment of geodetic leveling observations in Canada, the United States, and Mexico.	
8	DD Name: IENC	Code: reflev_8	Date accepted: 2001-05-31
	Name:	Mean sea level 1912	
	Definition:	A vertical control datum established for vertical control in the United States by the general adjustment of 1912.	
9	DD Name: IENC	Code: reflev_9	Date accepted: 2001-05-31
	Name:	Mean sea level 1929	
	Definition:	A vertical control datum established for vertical control in the United States by the general adjustment of 1929.	
10	DD Name: IENC	Code: reflev_10	Date accepted: 2015-03-09
	Name:	Tweede Algemene Waterpassing (TAW)	
	Definition:	All heights in Belgium are referenced to TAW	

Attribute	Related issue
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Acronym: lg_rel Code: 18008
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31
Definition: Indication of the related legal issue

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: IENC Code: lg_rel_1 Date accepted: 2001-05-31 Name: other Definition:
2	DD Name: IENC Code: lg_rel_2 Date accepted: 2001-05-31 Name: usage of waterway Definition:
3	DD Name: IENC Code: lg_rel_3 Date accepted: 2001-05-31 Name: carriage of equipment Definition:
4	DD Name: IENC Code: lg_rel_4 Date accepted: 2001-05-31 Name: task,operation Definition:

Attribute	Restriction
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Acronym: restrn

Code: 17004

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Restriction

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: restrn_1 Date accepted: 2001-05-31

Name: anchoring prohibited

Definition: an area within which anchoring is not permitted.

2 DD Name: IENC Code: restrn_2 Date accepted: 2001-05-31

Name: anchoring restricted

Definition: a specified area designated by appropriate authority, within which anchoring is restricted in accordance with certain specified conditions.

7 DD Name: IENC Code: restrn_7 Date accepted: 2001-05-31

Name: entry prohibited

Definition: an area within which navigation and/or anchoring is prohibited. (adapted from IHO Dictionary, S-32, 5th Edition, 4044)

8 DD Name: IENC Code: restrn_8 Date accepted: 2001-05-31

Name: entry restricted

Definition: a specified area designated by appropriate authority, within which navigation is restricted in accordance with certain specified conditions. (adapted from IHO Dictionary, S-32, 5th Edition, 4366)

13 DD Name: IENC Code: restrn_13 Date accepted: 2001-05-31

Name: no wake

Definition: mariners must adjust the speed of their vessels to reduce the wave or wash which may cause erosion or disturb moored vessels.

Value	Data Dictionary (DD) Reference		
14	DD Name: IENC	Code: restrn_14	Date accepted: 2001-05-31
	Name: area to be avoided		
	Definition: an IMO designated area to be avoided, defined as a routeing measure. (adapted from IHO Chart Specifications, M-4, 435.7)		
27	DD Name: IENC	Code: restrn_27	Date accepted: 2001-05-31
	Name: speed restricted		
	Definition: an area within which speed is restricted.		
28	DD Name: IENC	Code: restrn_28	Date accepted: 2001-05-31
	Name: overtaking prohibited		
	Definition: a specified area designated by appropriate authority, within which overtaking is generally prohibited		
29	DD Name: IENC	Code: restrn_29	Date accepted: 2001-05-31
	Name: overtaking of convoys by convoys prohibited		
	Definition: a specified area designated by appropriate authority, within which overtaking between convoys prohibited		
30	DD Name: IENC	Code: restrn_30	Date accepted: 2001-05-31
	Name: passing or overtaking prohibited		
	Definition: a specified area designated by appropriate authority, within which passing or overtaking is generally prohibited		
31	DD Name: IENC	Code: restrn_31	Date accepted: 2001-05-31
	Name: berthing prohibited		
	Definition: a specified area designated by appropriate authority, within which vessels, assemblies of floating material or floating establishments may not berth.		
32	DD Name: IENC	Code: restrn_32	Date accepted: 2001-05-31
	Name: berthing restricted		
	Definition: a specified area designated by appropriate authority, within which berthing is restricted		
33	DD Name: IENC	Code: restrn_33	Date accepted: 2001-05-31
	Name: making fast prohibited		
	Definition: a specified area designated by appropriate authority, within which vessels, assemblies of floating material or floating establishments may not make fast to the bank.		
34	DD Name: IENC	Code: restrn_34	Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	making fast restricted	
	Definition:	a specified area designated by appropriate authority, within which making fast to the bank is restricted	
35	DD Name: IENC	Code: restrn_35	Date accepted: 2001-05-31
	Name:	turning prohibited	
	Definition:	a specified area designated by appropriate authority, within which all turning is generally prohibited	
36	DD Name: IENC	Code: restrn_36	Date accepted: 2001-05-31
	Name:	restricted fairway depth	
	Definition:	an area within which the fairway depth is restricted.	
37	DD Name: IENC	Code: restrn_37	Date accepted: 2001-05-31
	Name:	restricted fairway width	
	Definition:	an area within which the fairway width is restricted.	
38	DD Name: IENC	Code: restrn_38	Date accepted: 2014-11-26
	Name:	use of spuds prohibited	
	Definition:	the use of anchoring spuds (telescopic piles) is prohibited	
40	DD Name: IENC	Code: restrn_40	Date accepted: 2017-11-06
	Name:	SOx emission restricted	
	Definition:	An area within which the emission of Sox is restricted.	
41	DD Name: IENC	Code: restrn_41	Date accepted: 2017-11-06
	Name:	NOx emission restricted	
	Definition:	AN area within which the emission of Nox is restricted.	

Attribute	Sounding datum reference level value
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Acronym: sdrval

Code: 18049

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2019-11-13

Definition: Local value of the sounding datum reference level.

Attribute	Speed reference
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Acronym: lg_spr Code: 18002
Use Type: F
Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31
Definition: Indicates the type of speed measurement

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: IENC Code: lg_spr_1 Date accepted: 2001-05-31 Name: other Definition:
2	DD Name: IENC Code: lg_spr_2 Date accepted: 2001-05-31 Name: speed over ground Definition: The vessel's actual speed, determined by dividing the distance between successive fixes by the time between the fixes
3	DD Name: IENC Code: lg_spr_3 Date accepted: 2001-05-31 Name: speed through water Definition: The vessel's actual speed, determined by subtracting the speed over ground by the current speed

Attribute	Time Schedule Reference
-----------	-------------------------

Acronym: schref

Code: 17093

Use Type: F

Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The string encodes the file name of an external file

Attribute	Transshipping goods
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Acronym: trshgd

Code: 17076

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: List of goods, which can be transshipped

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: trshgd_1 Date accepted: 2001-05-31

Name: containers

Definition: boxes for cargo transport with standardized dimensions.

2 DD Name: IENC Code: trshgd_2 Date accepted: 2001-05-31

Name: bulk goods

Definition: unpacked bulk cargo in the same or a similar kind of nature (homogeneous).

3 DD Name: IENC Code: trshgd_3 Date accepted: 2001-05-31

Name: oil

Definition: Mineral oil or liquid petrolatum, a by-product in the distillation of petroleum to produce gasoline.

4 DD Name: IENC Code: trshgd_4 Date accepted: 2001-05-31

Name: fuel

Definition: liquid fuel, e.g. gasoline, diesel.

5 DD Name: IENC Code: trshgd_5 Date accepted: 2001-05-31

Name: chemicals

Definition: any material used in or obtained by a process in chemistry.

6 DD Name: IENC Code: trshgd_6 Date accepted: 2001-05-31

Name: liquid goods

Value	Data Dictionary (DD) Reference		
	Definition: fluids whose shape is usually determined by the container it fills.		
7	DD Name: IENC	Code: trshgd_7	Date accepted: 2001-05-31
	Name: explosive goods		
	Definition: goods that undergoes decomposition or combustion with great rapidity, evolving much heat and producing a large volume of gas.		
8	DD Name: IENC	Code: trshgd_8	Date accepted: 2001-05-31
	Name: fish		
	Definition: marine animals		
9	DD Name: IENC	Code: trshgd_9	Date accepted: 2001-05-31
	Name: cars		
	Definition: wheeled vehicles		
10	DD Name: IENC	Code: trshgd_10	Date accepted: 2001-05-31
	Name: general cargo		
	Definition: general cargo		

Attribute	Type of AtoN
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Acronym: typatn Code: 18047

Use Type: F

Value Type: L

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2019-10-23

Definition: The type of AtoN being referenced

Enumerations:

Value	Data Dictionary (DD) Reference
1	DD Name: IENC Code: typatn_1 Date accepted: 2019-10-23 Name: Aid to Navigation Definition: Aid to Navigation
2	DD Name: IENC Code: typatn_2 Date accepted: 2019-10-23 Name: Physical AIS Aid to Navigation Definition: An AIS station located on an existing physical aid to navigation
3	DD Name: IENC Code: typatn_3 Date accepted: 2019-10-23 Name: Virtual AIS Aid to Navigation Definition: An electronically charted, but non-existent as a physical aid to navigation

Attribute	Type of Ship
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Acronym: shptyp

Code: 33066

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Type of ship

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: shptyp_1 Date accepted: 2001-05-31

Name: general cargo vessel

Definition: a vessel which is designed for carrying general cargo, e.g. boxes, sacks.

2 DD Name: IENC Code: shptyp_2 Date accepted: 2001-05-31

Name: container vessel

Definition: a vessel which is designed for carrying containers.

3 DD Name: IENC Code: shptyp_3 Date accepted: 2001-05-31

Name: tanker

Definition: a vessel which is designed for carrying liquid goods, e.g. oil or water.

4 DD Name: IENC Code: shptyp_4 Date accepted: 2001-05-31

Name: sailing vessel

Definition: a vessel that is powered by the wind; often having several masts.

5 DD Name: IENC Code: shptyp_5 Date accepted: 2001-05-31

Name: fishing vessel

Definition: a vessel that is used and equipped for the fishing of living aquatic resources.

6 DD Name: IENC Code: shptyp_6 Date accepted: 2001-05-31

Name: special purpose vessel.

Value	Data Dictionary (DD) Reference			
	Definition:	a vessel that fulfills special purposes e.g. hovercrafts, pilot boats		
7	DD Name: IENC	Code: shptyp_7	Date accepted:	2001-05-31
	Name:	man of War		
	Definition:	armed naval vessel.		
8	DD Name: IENC	Code: shptyp_8	Date accepted:	2001-05-31
	Name:	submarine		
	Definition:	a vessel that is capable of operating for an extended period of time underwater.		
9	DD Name: IENC	Code: shptyp_9	Date accepted:	2001-05-31
	Name:	high speed craft		
	Definition:			
10	DD Name: IENC	Code: shptyp_10	Date accepted:	2001-05-31
	Name:	bulk carrier		
	Definition:	a vessel which is designed for carrying bulk goods, e.g. coal, ore or grain.		
11	DD Name: IENC	Code: shptyp_11	Date accepted:	2001-05-31
	Name:	seaplane		
	Definition:	airplane designed to take off from and alight on water.		
12	DD Name: IENC	Code: shptyp_12	Date accepted:	2001-05-31
	Name:	tugboat		
	Definition:	a powerful small boat designed to pull or push larger ships or powerless barges.		
13	DD Name: IENC	Code: shptyp_13	Date accepted:	2001-05-31
	Name:	passenger vessel		
	Definition:	a vessel which is designed for carrying passengers and which serves mainly as cruise vessel.		
14	DD Name: IENC	Code: shptyp_14	Date accepted:	2001-05-31
	Name:	ferry		
	Definition:	a vessel which is designed for carrying passengers, and sometimes their vehicles, on scheduled services.		
15	DD Name: IENC	Code: shptyp_15	Date accepted:	2001-05-31
	Name:	boat		
	Definition:	a small vessel.		

Attribute	UN location code
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Acronym: unlocd Code: 17077
Use Type: F
Value Type: T

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The attribute unlocd should be used to encode the UN Location Code (<http://www.unece.org/cefact/locode/service/main.htm>) or - in Europe - the Inland Ship Reporting Standard (ISRS) Code

Acronym:	useshp	Code:	17094
Use Type:	F		
Value Type:	E		

DD Name: IENC Date accepted: 2001-05-31

Definition: Indication of the way the ship is used

Value	Data Dictionary (DD) Reference
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1	DD Name: IENC	Code: useshp_1	Date accepted: 2001-05-31
	Name: liner trade		
	Definition: ship is used to carry goods on a scheduled service		
2	DD Name: IENC	Code: useshp_2	Date accepted: 2001-05-31
	Name: occasional professional shipping		
	Definition: ship is occasional used for professional shipping		
3	DD Name: IENC	Code: useshp_3	Date accepted: 2001-05-31
	Name: leisure		
	Definition: ship is used for leisure activities		

Attribute	Value at other locally relevant water level
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Acronym: othwat

Code: 17086

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Value at waterway gauge in case of a specific water level, which is locally of importance or of interest for navigation

Attribute	Value at relevant high water level
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Acronym: higwat

Code: 17080

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Value at waterway gauge in case of exact high water level (according to official regulations at the specific section of waterway)

Attribute	Value at relevant low water level
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Acronym: lowwat

Code: 17082

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Value at waterway gauge in case of exact low water level (according to official regulations at the specific section of waterway)

Attribute	Value at relevant mean water level
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Acronym: meawat

Code: 17084

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Value at waterway gauge in case of exact mean water level (according to official regulations at the specific section of waterway)

Attribute	Vertical datum
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Acronym: verdat

Code: 17005

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Vertical datum

Enumerations:

Value Data Dictionary (DD) Reference

12 DD Name: IENC Code: verdat_12 Date accepted: 2001-05-31

Name: Mean lower low water

Definition: (MLLW) - the average height of the lower low waters at a place over a 19-year period. (IHO Dictionary, S-32, 5th Edition, 3145)

31 DD Name: IENC Code: verdat_31 Date accepted: 2001-05-31

Name: Local low water reference level

Definition: low water reference level of the local area

32 DD Name: IENC Code: verdat_32 Date accepted: 2001-05-31

Name: Local high water reference level

Definition: high water reference level of the local area

33 DD Name: IENC Code: verdat_33 Date accepted: 2001-05-31

Name: Local mean water reference level

Definition: mean water reference level of the local area

34 DD Name: IENC Code: verdat_34 Date accepted: 2001-05-31

Name: Equivalent height of water (German GIW)

Definition: A low water level which is the result of a defined low water discharge - called "equivalent discharge".

35 DD Name: IENC Code: verdat_35 Date accepted: 2001-05-31

Value	Data Dictionary (DD) Reference		
	Name:	Highest Shipping Height of Water (German HSW)	
	Definition:	upper limit of water levels where navigation is allowed	
36	DD Name: IENC	Code: verdat_36	Date accepted: 2001-05-31
	Name:	Reference low water level according to Danube Commission	
	Definition:	The water level at a discharge, which is exceeded 94 % of the year within a period of 30 years.	
37	DD Name: IENC	Code: verdat_37	Date accepted: 2001-05-31
	Name:	Highest shipping height of water according to Danube Commission	
	Definition:	The water level at a discharge, which is exceeded 1 % of the year within a period of 30 years.	
38	DD Name: IENC	Code: verdat_38	Date accepted: 2001-05-31
	Name:	Dutch river low water reference level (OLR)	
	Definition:	The water level at a discharge, which is exceeded 95 % of the year within a period of 20 years.	
39	DD Name: IENC	Code: verdat_39	Date accepted: 2001-05-31
	Name:	Russian project water level	
	Definition:	Conditional low water level with established probability (Hydrographic Terminology Dictionary, HDNO, 1984).	
40	DD Name: IENC	Code: verdat_40	Date accepted: 2001-05-31
	Name:	Russian normal backwater level	
	Definition:	Highest water level derived from the upper backwater stream in watercourse or reservoir under the normal operational conditions. (Hydrographic Terminology Dictionary, HDNO, 1984).	
41	DD Name: IENC	Code: verdat_41	Date accepted: 2001-05-31
	Name:	Ohio River Datum	
	Definition:		
42	DD Name: IENC	Code: verdat_42	Date accepted: 2015-02-23
	Name:	Approximate LAT	
	Definition:		
43	DD Name: IENC	Code: verdat_43	Date accepted: 2015-02-23
	Name:	Dutch High Water Reference Level (MHW)	
	Definition:		
24	DD Name: HYDRO	Code: VERDAT_24	Date accepted: 2000-11-01

Value	Data Dictionary (DD) Reference		
	Name:	Local datum	
	Definition:	an arbitrary datum defined by a local harbour authority, from which leves and tidal heights are measured by this authority.	
23	DD Name: HYDRO	Code: VERDAT_23	Date accepted: 2000-11-01
	Name:	Lowest astronomical tide	
	Definition:	(LAT) - the lowest tide level which can be predicted to occur under average meterological conditions and under any combination of astronomical conditions. (IHO Dictionary, S-32, 5th Edition, 2936)	
30	DD Name: HYDRO	Code: VERDAT_30	Date accepted: 2000-11-01
	Name:	Highest astronomical tide	
	Definition:	(HAT) - the highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions. (IHO Dictionary, S-32, 5th Edition, 2244).	
45	DD Name: IENC	Code: verdat_45	Date accepted: 2019-10-08
	Name:	Dutch estuary low water reference level (OLW)	
	Definition:		

Attribute	Vertical river datum reference level value
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Acronym: vcrval

Code: 18048

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2019-11-13

Definition: Local value of the vertical clearance reference level.

Attribute	Water displacement unit
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Acronym: lg_wdu

Code: 18007

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Units of measure for water displacement

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: lg_wdu_1 Date accepted: 2001-05-31

Name: other

Definition:

2 DD Name: IENC Code: lg_wdu_2 Date accepted: 2001-05-31

Name: cubic meters

Definition:

3 DD Name: IENC Code: lg_wdu_3 Date accepted: 2001-05-31

Name: tonnes

Definition:

Attribute	Water level effect
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Acronym: watlev

Code: 17104

Use Type: F

Value Type: E

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: Effect of water level

Enumerations:

Value Data Dictionary (DD) Reference

1 DD Name: IENC Code: watlev_1 Date accepted: 2001-05-31

Name: partly submerged at high water

Definition: partially covered and partially dry at high water.

2 DD Name: IENC Code: watlev_2 Date accepted: 2001-05-31

Name: always dry

Definition: not covered at high water under average meteorological conditions.

3 DD Name: IENC Code: watlev_3 Date accepted: 2001-05-31

Name: always under water/submerged

Definition: remains covered by water at all times under average meteorological conditions.

4 DD Name: IENC Code: watlev_4 Date accepted: 2001-05-31

Name: covers and uncovers

Definition: expression intended to indicate an area of a reef or other projection from the bottom of a body of water which periodically extends above and is submerged below the surface. Also referred to as dries or uncovers. (IHO Dictionary, S-32, 5th Edition, 1111)

8 DD Name: IENC Code: watlev_8 Date accepted: 2001-05-31

Name: above mean water level

Definition: above a water level called "mean water" which is the arithmetic mean value of all water levels within a certain period of time

9 DD Name: IENC Code: watlev_9 Date accepted: 2001-05-31

Value Data Dictionary (DD) Reference

Name: below mean water level

Definition: below a water level called "mean water" which is the arithmetic mean value of all water levels within a certain period of time

Attribute	Waterway distance
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Acronym: wtwdis

Code: 17064

Use Type: F

Value Type: F

Data Dictionary (DD) Reference:

DD Name: IENC Date accepted: 2001-05-31

Definition: The distance measured from an origin of a river or canal